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Curtin University – Department of Computing

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Unit name:	Data Structures and Algorithms	Unit ID:	COMP1002
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Date of submission:	28/05/26	Which assignment?	ZipRide I (Leave blank if the unit has only one assignment.)

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Contents

Overview 5

Module 1: Graph-Based Route Planning 7

graph.py 7

linked_list.py 9

min.heap.py 10

stack_queue.py 11

Module 2: Hash-Based Passenger and Driver Lookup 12

hash_table.py 12

Module 3: Heap-Based Pickup Scheduling 13

max_heap.py 13

Module 4: Sorting Pickup Requests 15

advanced_sorting.py 15

conditions_module4.py 16

UML DIAGRAM 16

User Guide 19

main.py 19

Module 1 33

Module 2 37

Test case 60

Module 3 63

Module 4 67

Dataset of 100 pickup requests: 68

Dataset of 500 pickup requests: 74

Dataset of 1000 pickup requests: 102

Runtime Analysis of module 4 160

Assumptions 162

Limitations 162

Conclusion 162

References 163

Figure 1: Data Structures 5
Figure 2 19
Figure 3 20
Figure 4 21
Figure 5 22
Figure 6 23
Figure 7 24
Figure 8 25
Figure 9 26
Figure 10 26
Figure 11 27
Figure 12 27
Figure 13 28
Figure 14 29
Figure 15 30
Figure 16 30
Figure 17 31
Figure 18 31
Figure 19 32
Figure 20 32
Figure 21 33
Figure 22: Route Network 34
Figure 23 35
Figure 24 36
Figure 25 37
Figure 26 38
Figure 27 39
Figure 28 40
Figure 29 41
Figure 30 42
Figure 31 43
Figure 32 44
Figure 33 45
Figure 34 46
Figure 35 47
Figure 36 48
Figure 37 49
Figure 38 50
Figure 39 51
Figure 40 52
Figure 41 53
Figure 42 53
Figure 43 54

Figure 44 54
Figure 45 55
Figure 46 55
Figure 47 56
Figure 48 56
Figure 49 57
Figure 50 57
Figure 51 58
Figure 52 58
Figure 53 59
Figure 54 59
Figure 55 66
Figure 56 66
Figure 57 66
Figure 58 67
Figure 59: Dataset of 100 pickup requests in random order 68
Figure 60: Dataset of 100 pickup requests in nearly sorted order 68
Figure 61: Dataset of 100 pickup requests in reverse order 69
Figure 62: Random order sorted 70
Figure 63: Nearly sorted list sorted 70
Figure 64: Reversed list sorted 71
Figure 65: Short Reflection 71
Figure 66: Random order sorted 72
Figure 67: Nearly sorted list sorted 72
Figure 68: Reverse list sorted 73
Figure 69: Short reflection on using Quick Sort for 100 pickup requests 73
Figure 70 74
Figure 71 74
Figure 72 75
Figure 73 75
Figure 74: Short reflection on using Merge Sort on different lists of 500 pickup requests 92
Figure 75: Short Reflection of using Quick Sort on different 500 pickup request lists 102
Figure 76: Short Reflection on using Merge Sort to sort different 1000 pickup request list 140
Figure 77: Short reflection on using Quick Sort to sort different 1000 pickup lists 159
Figure 78: Timing Table 160

Overview

Nowadays, ride-hailing companies face many challenges. For example, they have rising operating costs caused by surging fuel prices and maintenance costs, so their drivers are encouraged to choose the shortest routes, which ensures that they take the least time possible to pick up clients and drop them to their destination ('What Is a Taxi Dispatch System?', n.d.). Moreover, the companies have the obligation of keeping logs and documenting all of their rides to aid performance analysis ('What Is a Taxi Dispatch System?', n.d.). All of these problems are answered in ZipRide Dispatch System.

ZipRide Dispatch System is a backend system that pairs passengers with the closest driver available, updates the data of driver and passenger, dispatches high-priority pickups and provides dispatch reports using efficient data structures and algorithms such as Graph, Hash Table, Heap, and Sorting.

In other words, ZipRide Dispatch System simulates the operation of a ride-hailing company.

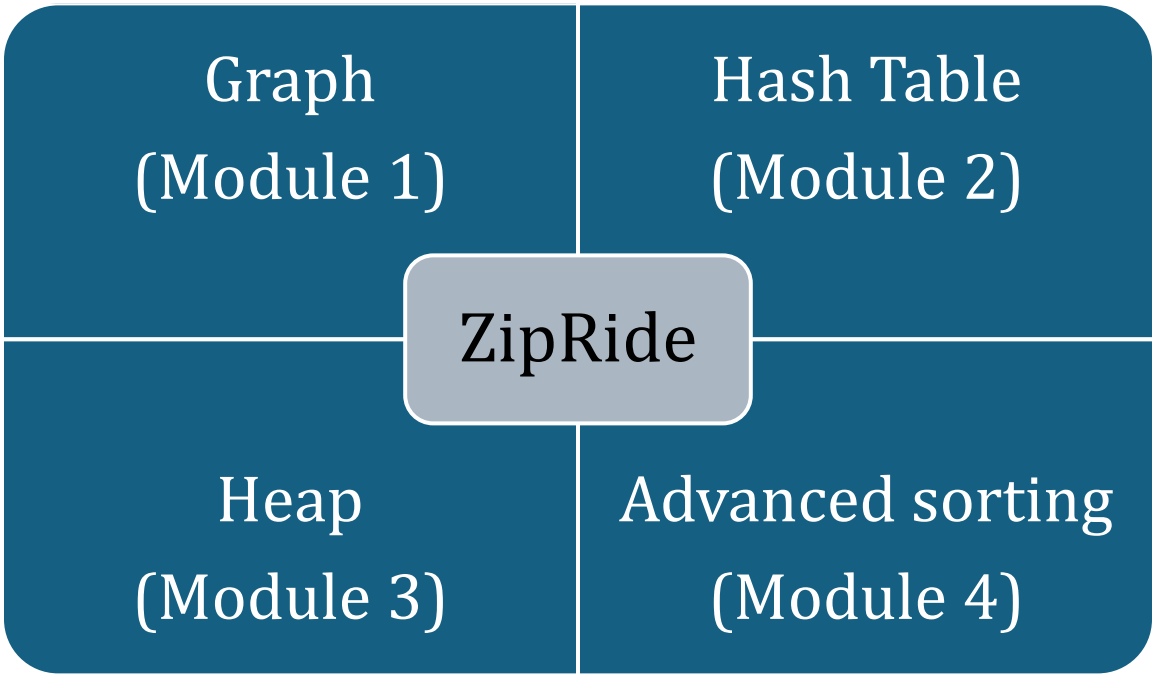


Figure 1: Data Structures

1. **Graph**

   In Module 1, a weighted undirected graph is used to represent the road network in which the nodes represent the following locations: Mall, Suburbs, Park, School, Supermarket, University, Church, Zoo and Harbor inside of an urban region. Furthermore, the edge weights represent the driving time, in minutes, between two edges. The Dijkstra algorithm provides the shortest path from the driver's current location to the passenger's current location. It also gives the minimum waiting time for the passenger in minutes. Harbor is the only isolated node and there are three cycles.

2. **HashTable**

   In Module 2, a hash table is used to store and access both passenger and driver data. Passenger data comprises of Passenger ID, Passenger Name, Passenger Current Location and Membership Tier while Driver data has the following fields: Driver ID, Driver Name, Driver Current Location and Availability Status. The HashTable allows fast retrieval of either passenger or driver records. In case of collision when inserting records, chaining was used.

   In order to avoid clustering, the table size given is twice the size of the total passenger and driver records, and is changed to its nearest odd number. This additional space provides ample space for the modulo-based hash function implemented to provide an index at which each record is stored. When the actual table size is a prime number, there is a higher possibility of keys being distributed uniformly across the hash table.

3. **Heap**

   In Module 3, a max heap is used to schedule highest priority pickups on top. The calculation of priority is based on passenger importance and driver proximity, combining the use of Module 1 (Graph) and Module 2 (HashTable). The driver is selected if they are available and if their estimated pickup time calculated using the Dijkstra algorithm from Module 1, is the shortest compared to other available drivers. Once dispatched, that is, extracted from the heap, the driver is given the status of "Busy". Thus, they are removed from the pool of available drivers.

4. **Sorting**

   In Module 4, a sorting algorithm, both Merge Sort and Quick sort, are used to sort the pickup requests before the pickup requests are added onto the heap. It optimises the performance of the heap.

Module 1: Graph-Based Route Planning

graph.py

This file contains all classes and modules which were used to implement the Module 1 instead of using functions from built-in libraries to build the routes used by the drivers of ZipRide.

1) **class DSAGraphNode**

   This class is used to create a single node which represents a location.

   Attributes:

   - label (str) – name of vertex (location)
   - value (int) – value of vertex
   - adj_list (linked list) – stores adjacency list of a certain vertex
   - visited (bool) – indicates if a vertex (location) has been visited

Module	Description
get_label	Returns label of node
get_value	Returns value of node
get_adjacent_node	Displays all the neighbours of node
get_visited	Checks if node was visited or not by returning True or False
check_adjacent	checks if this node is already connected to node
add_edge_into	Creates connection between two nodes by adding them into each other's adjacency list
delete_edge_from	Deletes connection between two nodes by removing them from each other's adjacency list
set_visited	Sets node as visited
clear_visited	Sets node as unvisited

2) **class DSAGraphEdge**

   This class is used to create an edge by storing a node and its weight.

   Attributes:

   - node – node object (location object containing label and value)
   - weight(int) – driving time

Module	Description
get_node	Returns node
get_weight	Returns the weight of the edge

3) class DSAGraph

This class is used to build the road network using the modules found inside it.
Attributes:

- vertices (linked list) - linked list containing all vertices added into graph
- _next_node(object) – next node of a certain node

Module	Description
get_vertex	traverses the vertices list to check if a certain vertex exists already
has_vertex	Checks if vertex exists or not
add_vertex	Adds vertex in graph
delete_vertex	Deletes vertex from graph
add_edge	connects 2 vertices together
delete_edge	deconnects 2 vertices together
get_vertex_count	Returns total number of vertices inside graph
get_edge_count	Returns total number of edges inside graph
is_adjacent	Checks if the two vertices are neighbours
get_adjacent	returns adjacency list of a vertex
display_as_list	Displays adjacency list
display_as_matrix	Displays adjacency matrix
degree	counts how many neighbours a single vertex has
has_value	checks if any vertex has a certain value
next_vertex sort	returns only next vertex sort vertices in alphabetical or numeric ascending order
sort_adjacent	sort neighbours in alphabetical or numeric ascending order
breadth_first_search	Displays all reachable vertices level by level starting from first vertex in alphabetical order
get_unvisited_adjacent	Returns neighbour which was not visited
depth_first_search	Explores routes as deeply as possible
display_visually	Displays graph visually
dijkstra	Displays shortest route and time to drive from one location to another
bfs	Displays all reachable vertices level by level starting from given vertex
dfs	Displays cycle path if cycle is detected
get_cycle_path	Returns the cycle path

linked_list.py

This python file contains modules used inside some classes of graph. Linked lists were used to store all the vertices, neighbours and edges created inside the graph. They were used because of their flexibility to add and remove nodes compared to queues which used arrays.

1) **DSAListNode**

   This class is used to create a linked list node which is used later inside the class DSALinkedList.

   Attributes:

   - value (object) – stores object inside node (stores location)
   - next (object) – points to object next to the current one
   - prev (object) - points to object before the current one

Module	Description
getValue	Returns value inside of list node
setValue	Sets value inside of list node
getNext	Returns the node next to the current one
getPrev	Returns the node previous to the current one
setNext	Sets value inside of next list node
setPrev	Sets value inside of previous list node

2) **DSALinkedList**

   This class is used to create linked lists using class DSAListNode to create a node.

   Attributes:

   - head(obj) – points to node in first position
   - tail (obj) - points to node in last position
   - count (int) – number of nodes inside of linked list

Module	Description
insertFirst	creates a new node with a given value in linked list and inserts it in front
insertLast	creates a new node with a given value in linked list and inserts it at the back
isEmpty	Checks if linked list is empty
peekFirst	Returns the value found in front
peekLast	Returns the value found at the back
removeFirst	Removes the value found in front

removeLast	Removes the value found at the back
remove	Removes any value by first traversing the linked list to find it
get_count	Returns count of nodes inside of linked list
display	Displays linked list

min.heap.py

The class DSAMINHeap() was used to create a priority queue which was used to implement the djikstra algorithm inside graph.py. The priority queue adds all pickup times and trickles up the shortest one.

1) **class DSAHeapEntry**

   This class is used to store a single Heap Entry object.

   Attributes:

   - priority (int) – priority level calculated by adding driving time from source location to destination
   - value (str) – label of node (location)

Module	Description
get_priority	Returns priority
set_priority	Sets priority
get_value	Returns value
set_value	Sets value

2) **class DSAMINHeap**

   This class is used to store Heap Entry objects by trickling up objects with least priorities.

   Attributes:

   - size (int) – total number of slots inside of min heap
   - heap (list) – list to store all items inside heap (acts as min heap)
   - count (int) – number of items inside of heap

Module	Description
heap_full	Checks if heap is full
heap_empty	Checks if heap is empty
add	Adds an entry. It is trickled up if its priority is smaller than those above it.
trickle_up	Moves index if its entry's priority is smaller than those above it.

remove	Removes an entry and trickles it down
trickle_down	Moves index down and swaps a small index in its place
display_heap	Displays min heap
heapify	Converts a random heap array into a min heap
swap	swaps max element with first element
heap_sort	Sorts heap into ascending order
read_csv	reads csv file and adds each line as an entry
save_csv	displays all heap items inside a csv file

stack_queue.py

This file contains classes for queue and stack.

1) class DSAStack

This class contains all modules to use stack. Stack was used in implementing depth_first_search (returns all nodes explored in depth) and dfs (returns a list of path if a cycle is detected).

Attributes:

- size (int) – capacity of stack
- count (int) – number of item present inside of stack
- stack (array) – array which contains content of stack

Module	Description
get_count	Returns count of items inside of stack
isEmpty	Checks if stack is empty by comparing count to 0
isFull	Checks if stack is full by comparing count to length of stack array
push	insert an item
top	Returns item on top of stack
pop	Removes item on top of stack
display_stack	Displays contents of stack

2) class DSAQueue

This class contains all modules required to use a queue.

Attributes:

- size (int) – capacity of queue
- count (int) – number of item present inside of queue
- queue (array) – array which contains content of queue

- front (int) - index of first item
- rear – index of last item

Module	Description
get_count	Returns count of items inside of queue
isEmpty	Checks if stack is empty by comparing count to 0
isFull	Checks if stack is full by comparing count to length of queue array
enqueue	insert an item from the back
dequeue	Remove item from the front
top	Returns item in first position
display_queue	displays the queue

Module 2: Hash-Based Passenger and Driver Lookup

hash_table.py

This file contains all modules to create and use hash tables to store passenger and driver records. These modules are to be used instead of built-in functions to implement a hash table.

class DSAHashEntry contains all the details which either make up passenger or driver record.

Attributes:

- key (str)– Passenger ID / Driver ID
- type_record (str) – Passenger / Driver
- name (str)– Passenger name / Driver name
- location (str) – Current Passenger / Driver location
- status (Object) – Membership Tier/ Availability Status

class DSAHashTable

This class is used to allocated a key to items before they are stored inside the hash table (array). It uses chaining in case of collision.

Attributes:

- actual_size (int) – hash table size as a prime number
- hash_array (list) – listwhich has a linked list at each index
- count (int) – number of records inserted inside of hash table

Module	Description
get	returns entire entry
has_key	Checks if we have the key (gives True/False)
_find_next_prime	converts number into prime number
_hash	Converts a key into an index
put	put a record in the table
_find	returns hash index if key found
remove	deletes a record
get_load_factor	calculates the percentage of occupied space in hash table
_resize	Risizes hash table if limit of 70% is exceeded or if less than 30% of hash table is occupied
read_csv	reads csv file and adds each line as a hash entry
save_csv	displays hash table entries inside a csv file

Module 3: Heap-Based Pickup Scheduling

max_heap.py

This file contains the classes used to implement Heap which is used to store pickup requests and dispatch drivers to the current location of passengers.

class PickupRequest

This class stores the full details of pickup requests that is, passenger and driver details.

Attributes:

- id (str) – Passenger ID
- name (str) – Passenger name
- location (str) – current location of passenger
- tier (object) – membership tier of passenger
- driver_name (str) – Driver name
- pickup_time (int) – Estimated pickup time calculated using Dijkstra algorithm
- driver_location (str) – current location of driver
- driver_status (str) – Availability status of driver
- shortest_path (list) – list of nodes from driver current location to passenger current location

Module	Description
get_id	Returns passenger ID
get_name	Returns passenger name
get_location	Returns passenger location

get_tier	Returns membership tier of passenger
get_driver_name	Returns driver name
get_pickup_time	Returns pickup time
get_driver_status	Returns driver availability status

class DSAHeapEntry

This class stores the contents of a max Heap entry. It consists of priority and pickup_request. Heap entries are stored inside the Heap.

Attributes:

- priority (int) – priority level calculated using the following formula: Priority = (6 – M) + 1000 / T
- pickup_request (object) – stores the object created by class PickupRequest (full details of both passenger and driver)

Module	Description
get_priority	Returns priority
get_pickup_request	Returns pickup_request object

class Heap

This class stores all modules used to create a heap. The class Heap enables pickup requests to be scheduled and dispatched after passengers get an available driver.

Attributes:

- size (int) – number of Heap Entries which Heap (list) can store
- heap (list) – list storing Heap Entries
- count (int) – number of Heap Entries added to Heap

Module	Description
get_count	Returns count of total Heap Entries added to Heap
set_priority	Returns priority calculated using the following formula: Priority = (6 – M) + 1000 / T
heap_full	Checks if heap is full by comparing count with size. Returns True or False
heap_empty	Checks if heap is empty by comparing count with 0. . Returns True or False
add	Adds Heap Entry from Heap
trickle_up	Moves Heap Entry up if its priority is higher than those of its parents
remove	Removes Heap Entry from Heap
peek	Returns top Heap Entry in Heap
trickle_down	Moves Heap Entry down if its priority is less than those of its parents
display_heap	Displays contents of Heap
heapify	Rearranges an array into a heap

swap	swaps last Heap Entry with first Heap Entry
heap_sort	sorts the heap
read_csv	reads csv file and adds its contents into Heap
save_csv	displays all heap items inside a csv file

Module 4: Sorting Pickup Requests

advanced_sorting.py

This file contains the class which stores the following sorting algorithms: Merge Sort and Quick Sort. Top-down Merge Sort was implemented as it is efficient in sorting large datasets especially ones like 500 and 1000 pickup requests required in the assignment. Moreover, its average time complexity is O(log n) meaning that it is relatively fast. It is stable as it ensures that equal elements keeps the order inside of the input list (GeeksforGeeks 06:51:07+00:00).

Moreover, median of three Quick Sort was implemented as it is also efficient in sorting large datasets. Also, since no temporary arrays are used to sort the datasets, it is cache friendly ('Quick Sort - GeeksforGeeks', n.d.).

class sorting

This class stores all modules of sorting algorithms used to sort pickup requests before adding them to Heap.

Top-down Merge Sort for pickup request object

Module	Description
bubbleSort	Used to sort list storing pickup requests after they obtained the estimated pickup time
mergeSort	Sorts the priority and pickup_request object by estimated pickup time
mergeSortRecurse	Divides input list into two parts recursively
merge	Puts the two lists together

Top-down Merge Sort for pickup request list containing only passenger name and estimated pickup time in tuple form

Module	Description
mergeSort _list	Sorts pickup list by estimated pickup time
mergeSortRecurse list	Divides input list into two parts recursively
merge list	Puts the two lists together

Median of Three Quick Sort for pickup request object

Module	Description
QuickSortMedian3	Sorts the priority and pickup_request object by estimated pickup time
doPartitioning	Sorts list using pivot index
quickSortMedian3Recurse	Recursively sorts left and right partition of input list
median_of_3	Returns the middle index in list

Median of Three Quick Sort for pickup request list containing only passenger name and estimated pickup time in tuple form

Module	Description
QuickSortMedian3 list	Sorts the pickup_request by estimated pickup time
doPartitioning list	Sorts list using pivot index
quickSortMedian3Recurse list	Recursively sorts left and right partition of input list
median_of_3 list	Returns the middle index in list

conditions_module4.py

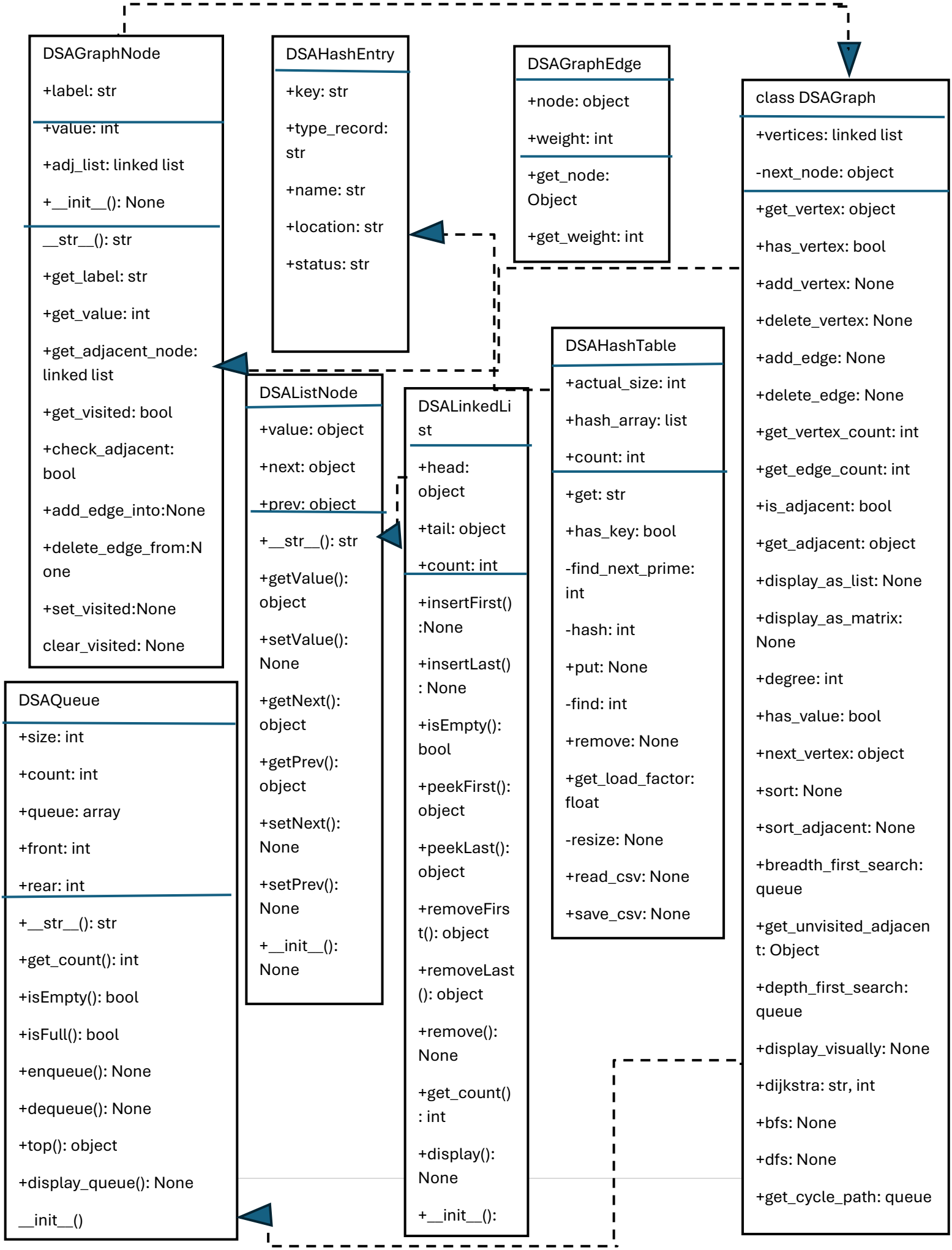
class Test_conditions

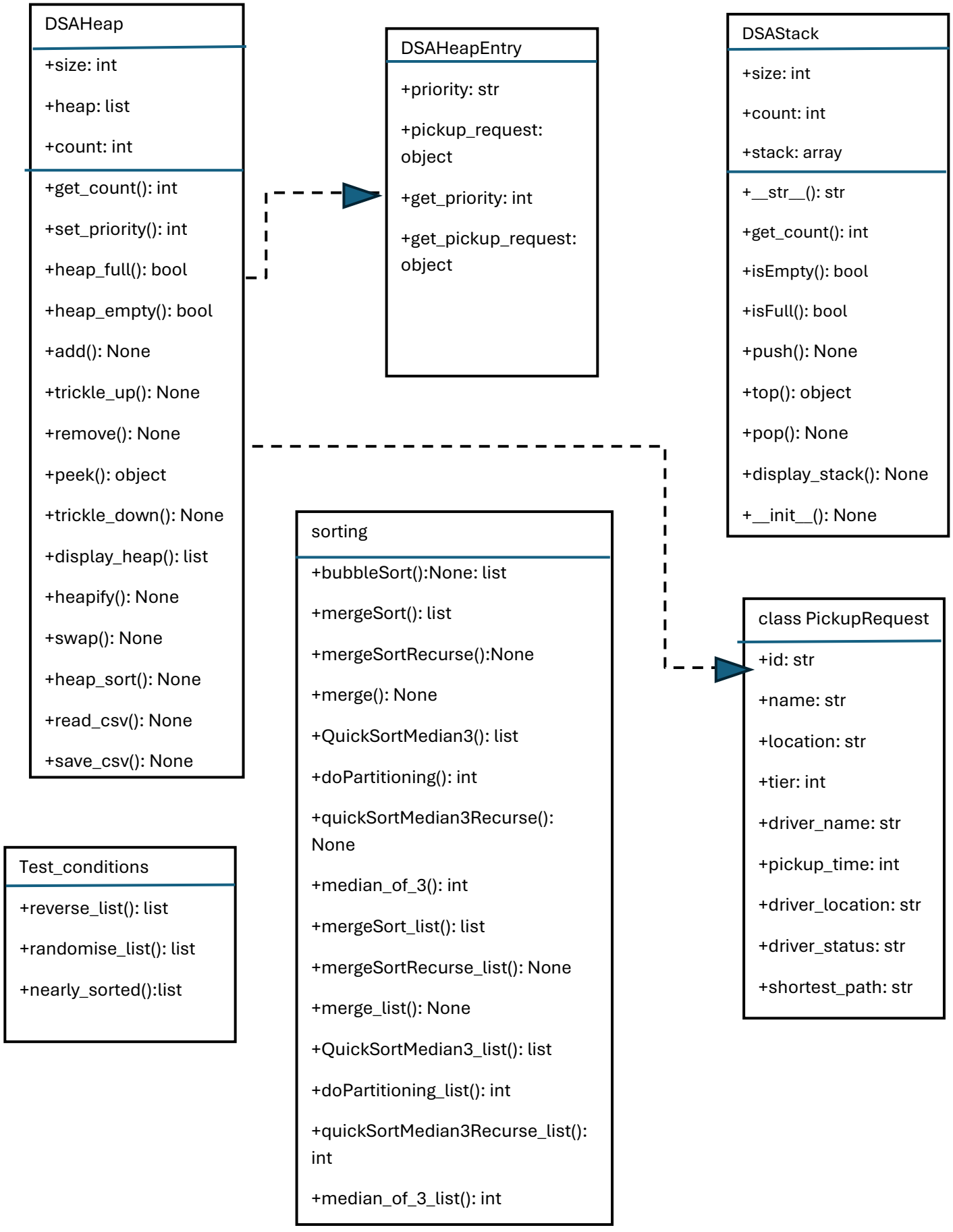
This class contains all modules used to create random, near sorted and reverse lists to get test datasets of 100, 500 and 1000 pickup requests.

Module	Description
reverse_list	reverses a list
randomise_list	shuffles the contents of a list randomly
nearly_sorted	makes the list nearly sorted (shuffles only 10% of list)

UML DIAGRAM

The UML Diagram below allows a clear understanding of how the classes work together to enable the ZipRide Dispatch System to work. As it can be seen class DSAGraph depends on class DSAGraphNode which in turn depends on DSAGraphEdge. Another example of dependency is that DSAHashTable depends on DSAHashEntry. Those are denoted by the dotted line.


DSAGraphNode
+label: str
+value: int
+adj_list: linked list
+__init__(): None
__str__(): str
+get_label: str
+get_value: int
+get_adjacent_node: linked list
+get_visited: bool
+check_adjacent: bool
+add_edge_into:None
+delete_edge_from:None
+set_visited:None
clear_visited: None
DSAHashEntry
+key: str
+type_record: str
+name: str
+location: str
+status: str
DSAGraphEdge
+node: object
+weight: int
+get_node: Object
+get_weight: int
class DSAGraph
+vertices: linked list
-next_node: object
+get_vertex: object
+has_vertex: bool
+add_vertex: None
+delete_vertex: None
+add_edge: None
+delete_edge: None
+get_vertex_count: int
+get_edge_count: int
+is_adjacent: bool
+get_adjacent: object
+display_as_list: None
+display_as_matrix: None
+degree: int
+has_value: bool
+next_vertex: object
+sort: None
+sort_adjacent: None
+breadth_first_search: queue
+get_unvisited_adjacent: Object
+depth_first_search: queue
+display_visually: None
+dijkstra: str, int
+bfs: None
+dfs: None
+get_cycle_path: queue
DSAHashTable
+actual_size: int
+hash_array: list
+count: int
+get: str
+has_key: bool
-find_next_prime: int
-hash: int
+put: None
-find: int
+remove: None
+get_load_factor: float
-resize: None
+read_csv: None
+save_csv: None
DSAListNode
+value: object
+next: object
+prev: object
+__str__(): str
+getValue(): object
+setValue(): None
+getNext(): object
+getPrev(): object
+setNext(): None
+setPrev(): None
+__init__(): None
DSALinkedList
+head: object
+tail: object
+count: int
+insertFirst() :None
+insertLast() : None
+isEmpty(): bool
+peekFirst(): object
+peekLast(): object
+removeFirst(): object
+removeLast(): object
+remove(): None
+get_count() : int
+display(): None
+__init__():
DSAQueue
+size: int
+count: int
+queue: array
+front: int
+rear: int
+__str__(): str
+get_count(): int
+isEmpty(): bool
+isFull(): bool
+enqueue(): None
+dequeue(): None
+top(): object
+display_queue(): None
__init__()


DSAHeap
+size: int
+heap: list
+count: int
+get_count(): int
+set_priority(): int
+heap_full(): bool
+heap_empty(): bool
+add(): None
+trickle_up(): None
+remove(): None
+peek(): object
+trickle_down(): None
+display_heap(): list
+heapify(): None
+swap(): None
+heap_sort(): None
+read_csv(): None
+save_csv(): None
DSAHeapEntry
+priority: str
+pickup_request: object
+get_priority: int
+get_pickup_request: object
DSAStack
+size: int
+count: int
+stack: array
+__str__(): str
+get_count(): int
+isEmpty(): bool
+isFull(): bool
+push(): None
+top(): object
+pop(): None
+display_stack(): None
+__init__(): None
sorting
+bubbleSort():None: list
+mergeSort(): list
+mergeSortRecurse():None
+merge(): None
+QuickSortMedian3(): list
+doPartitioning(): int
+quickSortMedian3Recurse(): None
+median_of_3(): int
+mergeSort_list(): list
+mergeSortRecurse_list(): None
+merge_list(): None
+QuickSortMedian3_list(): list
+doPartitioning_list(): int
+quickSortMedian3Recurse_list(): int
+median_of_3_list(): int
class PickupRequest
+id: str
+name: str
+location: str
+tier: int
+driver_name: str
+pickup_time: int
+driver_location: str
+driver_status: str
+shortest_path: str
Test_conditions
+reverse_list(): list
+randomise_list(): list
+nearly_sorted():list

main.py

This python file combines all of the 4 modules to simulate how ZipRide Dispatch System works.

To execute main.py, run the following command on terminal:

python3 main.py

```
********************************************************************************
Module 1: Graph-Based Route Planning
********************************************************************************
********************************************************************************
Module 2: Hash-Based Passenger and Driver Lookup
********************************************************************************

INSERTION at index 32
New Entry: 14495655  Passenger   Sofia Bonfiglio  Mall  4


UPDATED at index 32
New Entry: 14495655  Passenger   Sofia Bonfiglio  Mall  1


INSERTION at index 75
New Entry: 14224671  Passenger   Ashlee Capellan  School  5


UPDATED at index 75
New Entry: 14224671  Passenger   Ashlee Capellan  School  3
```

Figure 2

```
INSERTION at index 83
New Entry: 14707100  Passenger   Dona Mcinnes  Suburbs  3


INSERTION at index 21
New Entry: 14644633  Passenger   Maricela Landreneau  Park  1


INSERTION at index 62
New Entry: 14147356  Passenger   Elinor Raco  Supermarket  2


INSERTION at index 43
New Entry: 14393910  Passenger   Cody Mcmartin  Suburbs  3


INSERTION at index 88
New Entry: 14799737  Passenger   Katy Vacek  Suburbs  2


INSERTION at index 52
New Entry: 14431660  Passenger   Mathew Mercedes  Suburbs  1

The load factor every 10 inserts is 8.25%
```

Figure 3

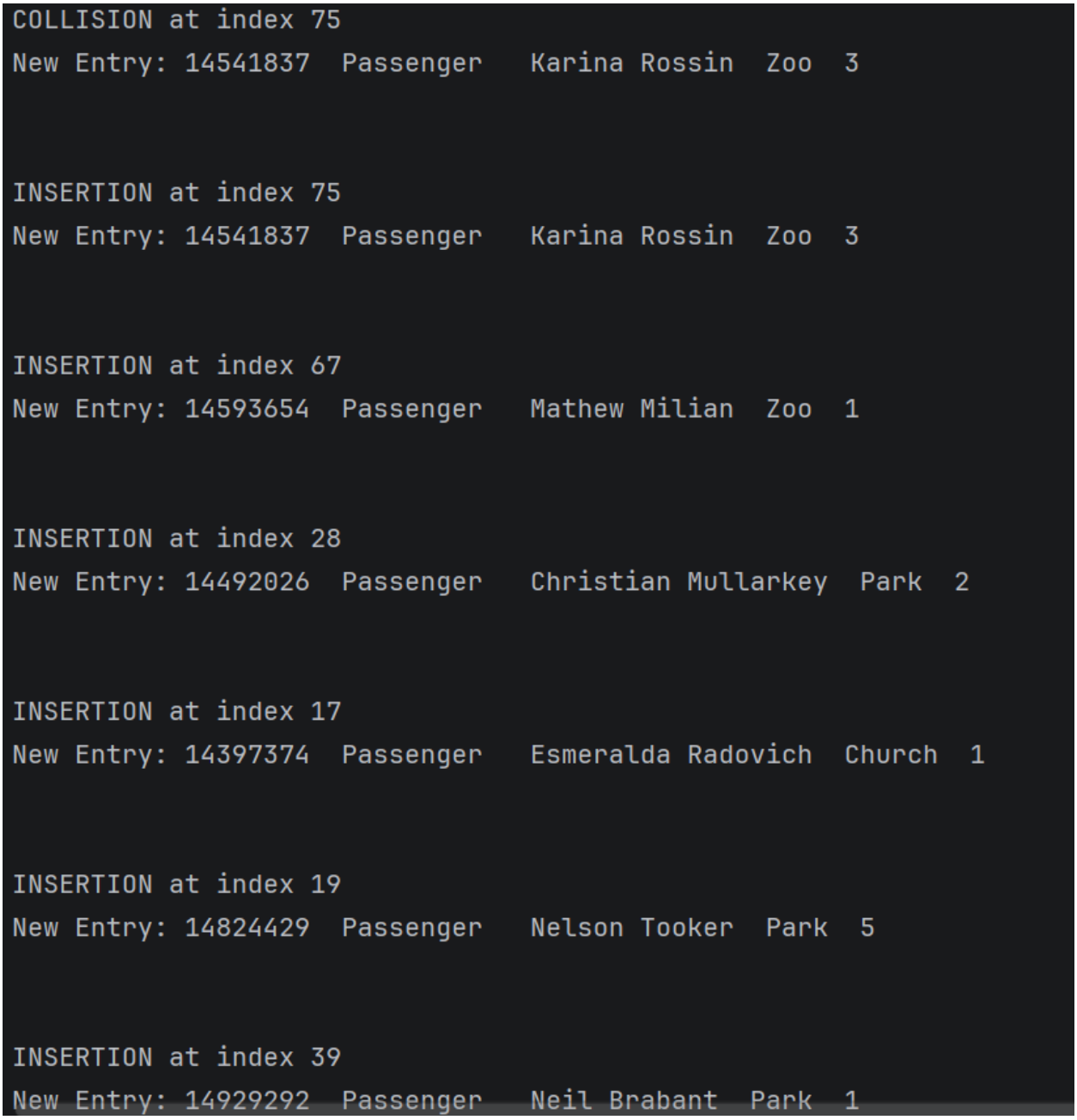

COLLISION at index 75
New Entry: 14541837 Passenger Karina Rossin Zoo 3
INSERTION at index 75
New Entry: 14541837 Passenger Karina Rossin Zoo 3
INSERTION at index 67
New Entry: 14593654 Passenger Mathew Milian Zoo 1
INSERTION at index 28
New Entry: 14492026 Passenger Christian Mullarkey Park 2
INSERTION at index 17
New Entry: 14397374 Passenger Esmeralda Radovich Church 1
INSERTION at index 19
New Entry: 14824429 Passenger Nelson Tooker Park 5
INSERTION at index 39
New Entry: 14929292 Passenger Neil Brabant Park 1

Figure 4

```
INSERTION at index 13
New Entry: 14786148  Passenger   Clinton Blumer  Church  2


INSERTION at index 69
New Entry: 14973358  Passenger   Julio Degroat  Suburbs  5


INSERTION at index 55
New Entry: 14557200  Passenger   Elinor Greig  Suburbs  4


INSERTION at index 49
New Entry: 14472268  Passenger   Malinda Dodgen  University  1

The load factor every 10 inserts is 18.56%

INSERTION at index 81
New Entry: 14359796  Passenger   Mathew Eslick  Mall  4


INSERTION at index 42
New Entry: 14770506  Passenger   Christian Durall  Church  1
```

Figure 5

```
INSERTION at index 1
New Entry: 14778665  Passenger   Cornell Uhri  School  2


INSERTION at index 3
New Entry: 14481609  Passenger   Amal Moznett  Mall  4


INSERTION at index 12
New Entry: 14354152  Driver   Max Plain  Church  Available


INSERTION at index 7
New Entry: 14338567  Driver   Earnestine Croswell  Park  Busy


INSERTION at index 95
New Entry: 14277290  Driver   Clayton Ricard  Suburbs  Offline


INSERTION at index 42
New Entry: 14476919  Driver   Ted Tallmadge  Park  Available
```

Figure 6

```
INSERTION at index 24
New Entry: 14032266  Driver   Kelly Cheatam  Mall  Available


INSERTION at index 3
New Entry: 14094113  Driver   Jessie Leandro  School  Available


INSERTION at index 69
New Entry: 14490422  Driver   Guy Kadel  Mall  Busy


INSERTION at index 39
New Entry: 14650254  Driver   Althea Warring  University  Busy


INSERTION at index 74
New Entry: 14355490  Driver   Tania Mogensen  Suburbs  Offline


INSERTION at index 86
New Entry: 14478629  Driver   Ted Makuch  School  Available

The load factor every 10 inserts is 10.31%
```

Figure 7

```
INSERTION at index 61
New Entry: 14827960  Driver   Katy Callier  Park  Offline


INSERTION at index 36
New Entry: 14607868  Driver   Jamie Seager  Church  Available


INSERTION at index 49
New Entry: 14080040  Driver   Guy Loredo  Supermarket  Available


INSERTION at index 32
New Entry: 14286103  Driver   Dollie Dignan  Mall  Available


INSERTION at index 22
New Entry: 14030930  Driver   Jessie Mayville  School  Available


INSERTION at index 23
New Entry: 14973623  Driver   Christian Busbee  Supermarket  Offline


COLLISION at index 7
New Entry: 14158835  Driver   Jamie Shake  Zoo  Offline
```

Figure 8

```
INSERTION at index 7
New Entry: 14158835  Driver   Jamie Shake  Zoo  Offline


INSERTION at index 28
New Entry: 14129722  Driver   Edwina Esters  University  Available


COLLISION at index 42
New Entry: 14207712  Driver   Harriett Zollinger  Zoo  Busy


INSERTION at index 42
New Entry: 14207712  Driver   Harriett Zollinger  Zoo  Busy


INSERTION at index 71
New Entry: 14151577  Driver   Nannie Mcgown  Zoo  Available

The load factor every 10 inserts is 20.62%

INSERTION at index 90
New Entry: 14905261  Driver   Yuriko Stimer  Supermarket  Available
```

Figure 9

```
INSERTION at index 88
New Entry: 14088484  Driver   Khalilah Dittrick  Zoo  Offline


INSERTION at index 65
New Entry: 14210394  Driver   Lonnie Traxson  Park  Busy
```

Figure 10

```
*********************************************************************************
Module 4: Sorting Pickup Requests
*********************************************************************************
Request from Mathew Mercedes has been rejected due to unavailable drivers
Request from Elinor Greig has been rejected due to unavailable drivers
Request from Elinor Raco has been rejected due to unavailable drivers
Request from Mathew Milian has been rejected due to unavailable drivers
Request from Julio Degroat has been rejected due to unavailable drivers
Request from Karina Rossin has been rejected due to unavailable drivers
Request from Ashlee Capellan has been rejected due to unavailable drivers
Request from Mathew Eslick has been rejected due to unavailable drivers
Request from Dona Mcinnes has been rejected due to unavailable drivers
Request from Katy Vacek has been rejected due to unavailable drivers
```

Figure 11

```
*********************************************************************************
Module 3: Heap-Based Pickup Scheduling
*********************************************************************************
Inserted into Heap = Priority: 1004.0 - Pickup request: 14778665    Cornell Uhri    School  2
Assigned Driver: Jessie Leandro Pickup time: 0 mins Driver Current Location: School
Shortest Path from School to School - School
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2

Inserted into Heap = Priority: 1002.0 - Pickup request: 14481609    Amal Moznett    Mall    4
Assigned Driver: Kelly Cheatam  Pickup time: 0 mins Driver Current Location: Mall
Shortest Path from Mall to Mall - Mall
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4

Inserted into Heap = Priority: 1004.0 - Pickup request: 14786148    Clinton Blumer  Church  2
Assigned Driver: Max Plain  Pickup time: 0 mins Driver Current Location: Church
Shortest Path from Church to Church - Church
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
```

Figure 12

```
Inserted into Heap = Priority: 1005.0 - Pickup request: 14397374    Esmeralda Radovich  Church  1
Assigned Driver: Jamie Seager   Pickup time: 0 mins Driver Current Location: Church
Shortest Path from Church to Church - Church
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4

Inserted into Heap = Priority: 1001.0 - Pickup request: 14824429    Nelson Tooker   Park    5
Assigned Driver: Ted Tallmadge  Pickup time: 0 mins Driver Current Location: Park
Shortest Path from Park to Park - Park
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5

Inserted into Heap = Priority: 65.5 - Pickup request: 14393910  Cody Mcmartin   Suburbs 3
Assigned Driver: Guy Loredo Pickup time: 16 mins    Driver Current Location: Supermarket
Shortest Path from Supermarket to Suburbs - ['Supermarket', 'Suburbs']
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
```

Figure 13

```
Inserted into Heap = Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park    1
Assigned Driver: Edwina Esters  Pickup time: 19 mins    Driver Current Location: University
Shortest Path from University to Park - ['University', 'Park']
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1

Inserted into Heap = Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park    2
Assigned Driver: Dollie Dignan  Pickup time: 24 mins    Driver Current Location: Mall
Shortest Path from Mall to Park - ['Mall', 'Park']
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
```

Figure 14

```
Inserted into Heap = Priority: 38.33 - Pickup request: 14929292 Neil Brabant    Park    1
Assigned Driver: Ted Makuch Pickup time: 30 mins    Driver Current Location: School
Shortest Path from School to Park - ['School', 'Park']
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1

Inserted into Heap = Priority: 32.78 - Pickup request: 14770506 Christian Durall    Church  1
Assigned Driver: Nannie Mcgown  Pickup time: 36 mins    Driver Current Location: Zoo
Shortest Path from Zoo to Church - ['Zoo', 'Suburbs', 'Church']
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1
Priority: 32.78 - Pickup request: 14770506 Christian Durall Church 1
```

Figure 15

```
Inserted into Heap = Priority: 23.52 - Pickup request: 14495655 Sofia Bonfiglio Mall    1
Assigned Driver: Jessie Mayville    Pickup time: 54 mins    Driver Current Location: School
Shortest Path from School to Mall - ['School', 'Park', 'Mall']
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1
Priority: 32.78 - Pickup request: 14770506 Christian Durall Church 1
Priority: 23.52 - Pickup request: 14495655 Sofia Bonfiglio Mall 1
```

Figure 16

```
Inserted into Heap = Priority: 20.15 - Pickup request: 14472268 Malinda Dodgen  University  1
Assigned Driver: Yuriko Stimer  Pickup time: 66 mins    Driver Current Location: Supermarket
Shortest Path from Supermarket to University - ['Supermarket', 'School', 'Park', 'University']
Priority: 1005.0 - Pickup request: 14397374 Esmeralda Radovich Church 1
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1
Priority: 32.78 - Pickup request: 14770506 Christian Durall Church 1
Priority: 23.52 - Pickup request: 14495655 Sofia Bonfiglio Mall 1
Priority: 20.15 - Pickup request: 14472268 Malinda Dodgen University 1
```

Figure 17

```
Extracted from Heap = Priority: 1005.0 - Pickup request: 14397374   Esmeralda Radovich  Church  1
Assigned Driver: Jamie Seager   Pickup time: 0 mins Driver status: Busy Driver Location: Church
Shortest Path from Church to Church - Church
Priority: 1004.0 - Pickup request: 14778665 Cornell Uhri School 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 20.15 - Pickup request: 14472268 Malinda Dodgen University 1
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1
Priority: 32.78 - Pickup request: 14770506 Christian Durall Church 1
Priority: 23.52 - Pickup request: 14495655 Sofia Bonfiglio Mall 1
```

Figure 18

```
Extracted from Heap = Priority: 1004.0 - Pickup request: 14778665   Cornell Uhri    School  2
Assigned Driver: Jessie Leandro Pickup time: 0 mins Driver status: Busy Driver Location: School
Shortest Path from School to School - School
Priority: 1004.0 - Pickup request: 14786148 Clinton Blumer Church 2
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 23.52 - Pickup request: 14495655 Sofia Bonfiglio Mall 1
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 20.15 - Pickup request: 14472268 Malinda Dodgen University 1
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1
Priority: 32.78 - Pickup request: 14770506 Christian Durall Church 1
```

Figure 19

```
Extracted from Heap = Priority: 1004.0 - Pickup request: 14786148   Clinton Blumer  Church  2
Assigned Driver: Max Plain  Pickup time: 0 mins Driver status: Busy Driver Location: Church
Shortest Path from Church to Church - Church
Priority: 1002.0 - Pickup request: 14481609 Amal Moznett Mall 4
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 32.78 - Pickup request: 14770506 Christian Durall Church 1
Priority: 23.52 - Pickup request: 14495655 Sofia Bonfiglio Mall 1
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 20.15 - Pickup request: 14472268 Malinda Dodgen University 1
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1

Extracted from Heap = Priority: 1002.0 - Pickup request: 14481609   Amal Moznett    Mall    4
Assigned Driver: Kelly Cheatam  Pickup time: 0 mins Driver status: Busy Driver Location: Mall
Shortest Path from Mall to Mall - Mall
Priority: 1001.0 - Pickup request: 14824429 Nelson Tooker Park 5
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1
Priority: 32.78 - Pickup request: 14770506 Christian Durall Church 1
Priority: 23.52 - Pickup request: 14495655 Sofia Bonfiglio Mall 1
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 20.15 - Pickup request: 14472268 Malinda Dodgen University 1
```

Figure 20

```
Extracted from Heap = Priority: 1001.0 - Pickup request: 14824429   Nelson Tooker   Park    5
Assigned Driver: Ted Tallmadge  Pickup time: 0 mins Driver status: Busy Driver Location: Park
Shortest Path from Park to Park - Park
Priority: 65.5 - Pickup request: 14393910 Cody Mcmartin Suburbs 3
Priority: 45.67 - Pickup request: 14492026 Christian Mullarkey Park 2
Priority: 57.63 - Pickup request: 14644633 Maricela Landreneau Park 1
Priority: 38.33 - Pickup request: 14929292 Neil Brabant Park 1
Priority: 32.78 - Pickup request: 14770506 Christian Durall Church 1
Priority: 23.52 - Pickup request: 14495655 Sofia Bonfiglio Mall 1
Priority: 20.15 - Pickup request: 14472268 Malinda Dodgen University 1

Item on top of heap: 65.5 - Pickup request: 14393910    Cody Mcmartin   Suburbs 3
Assigned Driver: Guy Loredo Pickup time: 16 mins    Driver status: Busy Driver Location: Supermarket

Count of rejected requests: 10
Count of accepted requests: 12
```

Figure 21

Module 1

In order to test whether module 1 was well implemented, run the following commands on the terminal:

pip install networkx

python3 test_module1.py

networkx is a python package which was only used to display the graph visually.

The images below are the output generated:

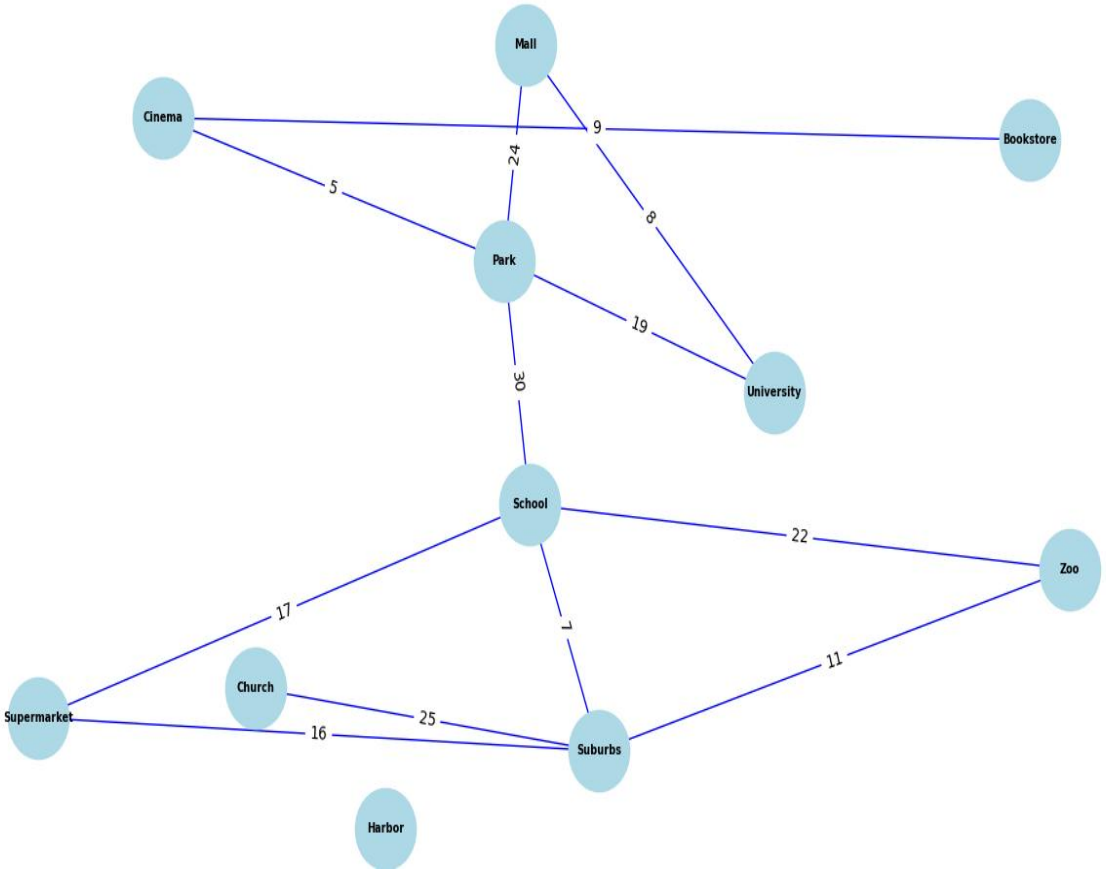


Figure 22: Route Network

```
********************************************************************************
Module 1: Graph-Based Route Planning
********************************************************************************
BFS (Bookstore):
Level 0: ['Bookstore']

Level 1: ['Cinema']

Level 2: ['Park']

Level 3: ['School', 'Mall', 'University']

Level 4: ['Suburbs', 'Supermarket', 'Zoo']

Level 5: ['Church']

DFS Cycle Detection
Cycle path: ['Mall' 'University' 'Park' 'Mall']
None

Dijkstra:
Shortest Path: ['Cinema', 'Park', 'School', 'Supermarket']
Shortest Driving Time: 52 minutes
```

Figure 23

```
Shortest Path: No path to Harbor
Shortest Driving Time: No shortest time

Count num of edges: 12

Count num of vertex: 11


Adjacency list:

Bookstore: Cinema(9)
Church: Suburbs(25)
Cinema: Bookstore(9) Park(5)
Harbor:
Mall: University(8) Park(24)
Park: Cinema(5) School(30) Mall(24) University(19)
School: Suburbs(7) Park(30) Supermarket(17) Zoo(22)
Suburbs: Church(25) Zoo(11) School(7) Supermarket(16)
Supermarket: School(17) Suburbs(16)
University: Mall(8) Park(19)
Zoo: Suburbs(11) School(22)
```

Figure 24

As seen in figure 24, there are 11 vertices and 12 edges which exceed the minimum number required by the assignment. Moreover, the graph was visually displayed. To illustrate that BFS was implemented correctly, “Bookstore” node was used as source location and all its reachable locations were grouped level by level in figure 23. The DFS module was given “Mall” as input and as a result, the locations involved in the cycle were

displayed as shown in figure 23. Furthermore, as shown in figure 23, the djikstra algorithm finds the shortest path between “Cinema” and “Supermarket”.

Module 2

To test whether module 2 was well implemented, ensure that passenger_driver_records.csv and test_module2.csv are in the same directory with test_module2.py, and run the following command on the terminal:

- python3 test_module2.py

```
********************************************************************************
Module 2: Hash-Based Passenger and Driver Lookup Test
********************************************************************************
INSERTING RECORDS


INSERTION at index 39
New Entry: 14495655  Passenger   Sofia Bonfiglio  Mall  4


INSERTION at index 9
New Entry: 14224671  Passenger   Ashlee Capellan  School  5


INSERTION at index 45
New Entry: 14707100  Passenger   Dona Mcinnes  Suburbs  3


INSERTION at index 1
New Entry: 14644633  Passenger   Maricela Landreneau  Park  1


INSERTION at index 41
New Entry: 14147356  Passenger   Elinor Raco  Supermarket  2
```

Figure 25

```
INSERTION at index 15
New Entry: 14393910  Passenger   Cody Mcmartin  Suburbs  3


INSERTION at index 17
New Entry: 14799737  Passenger   Katy Vacek  Suburbs  2


INSERTION at index 31
New Entry: 14431660  Passenger   Mathew Mercedes  Suburbs  1


INSERTION at index 33
New Entry: 14541837  Passenger   Karina Rossin  Zoo  3


INSERTION at index 42
New Entry: 14593654  Passenger   Mathew Milian  Zoo  1

The load factor every 10 inserts is 21.28%

INSERTION at index 35
New Entry: 14492026  Passenger   Christian Mullarkey  Park  2
```

Figure 26

```
INSERTION at index 23
New Entry: 14397374  Passenger   Esmeralda Radovich  Church  1


INSERTION at index 34
New Entry: 14824429  Passenger   Nelson Tooker  Park  5


INSERTION at index 32
New Entry: 14929292  Passenger   Neil Brabant  Park  1


INSERTION at index 5
New Entry: 14786148  Passenger   Clinton Blumer  Church  2


COLLISION at index 9
New Entry: 14973358  Passenger   Julio Degroat  Suburbs  5


INSERTION at index 9
New Entry: 14973358  Passenger   Julio Degroat  Suburbs  5
```

Figure 27

```
COLLISION at index 31
New Entry: 14557200  Passenger   Elinor Greig  Suburbs  4


INSERTION at index 31
New Entry: 14557200  Passenger   Elinor Greig  Suburbs  4


COLLISION at index 9
New Entry: 14472268  Passenger   Malinda Dodgen  University  1


INSERTION at index 9
New Entry: 14472268  Passenger   Malinda Dodgen  University  1


INSERTION at index 37
New Entry: 14359796  Passenger   Mathew Eslick  Mall  4


INSERTION at index 36
New Entry: 14770506  Passenger   Christian Durall  Church  1

The load factor every 10 inserts is 42.55%
```

Figure 28

```
COLLISION at index 15
New Entry: 14354152  Driver   Max Plain  Church  Available


INSERTION at index 15
New Entry: 14354152  Driver   Max Plain  Church  Available


COLLISION at index 34
New Entry: 14338567  Driver   Earnestine Croswell  Park  Busy


INSERTION at index 34
New Entry: 14338567  Driver   Earnestine Croswell  Park  Busy


INSERTION at index 0
New Entry: 14277290  Driver   Clayton Ricard  Suburbs  Offline


COLLISION at index 33
New Entry: 14476919  Driver   Ted Tallmadge  Park  Available
```

Figure 29

```
INSERTION at index 33
New Entry: 14476919  Driver   Ted Tallmadge  Park  Available


INSERTION at index 8
New Entry: 14032266  Driver   Kelly Cheatam  Mall  Available


COLLISION at index 0
New Entry: 14094113  Driver   Jessie Leandro  School  Available


INSERTION at index 0
New Entry: 14094113  Driver   Jessie Leandro  School  Available


INSERTION at index 24
New Entry: 14490422  Driver   Guy Kadel  Mall  Busy


COLLISION at index 8
New Entry: 14650254  Driver   Althea Warring  University  Busy
```

Figure 30

```
INSERTION at index 8
New Entry: 14650254  Driver   Althea Warring  University  Busy


COLLISION at index 41
New Entry: 14355490  Driver   Tania Mogensen  Suburbs  Offline


INSERTION at index 41
New Entry: 14355490  Driver   Tania Mogensen  Suburbs  Offline


INSERTION at index 44
New Entry: 14478629  Driver   Ted Makuch  School  Available

The load factor every 10 inserts is 63.83%

COLLISION at index 17
New Entry: 14827960  Driver   Katy Callier  Park  Offline


INSERTION at index 17
New Entry: 14827960  Driver   Katy Callier  Park  Offline
```

Figure 31

```
COLLISION at index 17
New Entry: 14607868  Driver   Jamie Seager  Church  Available


INSERTION at index 17
New Entry: 14607868  Driver   Jamie Seager  Church  Available


COLLISION at index 44
New Entry: 14080040  Driver   Guy Loredo  Supermarket  Available


INSERTION at index 44
New Entry: 14080040  Driver   Guy Loredo  Supermarket  Available


INSERTION at index 3
New Entry: 14094113  Driver   Jessie Leandro  School  Available


INSERTION at index 95
New Entry: 14277290  Driver   Clayton Ricard  Suburbs  Offline


INSERTION at index 21
New Entry: 14644633  Passenger   Maricela Landreneau  Park  1
```

Figure 32

```
INSERTION at index 13
New Entry: 14786148  Passenger   Clinton Blumer  Church  2


INSERTION at index 39
New Entry: 14650254  Driver   Althea Warring  University  Busy


INSERTION at index 24
New Entry: 14032266  Driver   Kelly Cheatam  Mall  Available


INSERTION at index 49
New Entry: 14472268  Passenger   Malinda Dodgen  University  1


INSERTION at index 69
New Entry: 14973358  Passenger   Julio Degroat  Suburbs  5


INSERTION at index 75
New Entry: 14224671  Passenger   Ashlee Capellan  School  5
```

Figure 33

```
INSERTION at index 12
New Entry: 14354152  Driver   Max Plain  Church  Available


INSERTION at index 43
New Entry: 14393910  Passenger   Cody Mcmartin  Suburbs  3


INSERTION at index 36
New Entry: 14607868  Driver   Jamie Seager  Church  Available


INSERTION at index 61
New Entry: 14827960  Driver   Katy Callier  Park  Offline


INSERTION at index 88
New Entry: 14799737  Passenger   Katy Vacek  Suburbs  2


INSERTION at index 17
New Entry: 14397374  Passenger   Esmeralda Radovich  Church  1
```

Figure 34

```
COLLISION at index 69
New Entry: 14490422  Driver   Guy Kadel  Mall  Busy


INSERTION at index 69
New Entry: 14490422  Driver   Guy Kadel  Mall  Busy


INSERTION at index 55
New Entry: 14557200  Passenger   Elinor Greig  Suburbs  4


INSERTION at index 52
New Entry: 14431660  Passenger   Mathew Mercedes  Suburbs  1


COLLISION at index 39
New Entry: 14929292  Passenger   Neil Brabant  Park  1


INSERTION at index 39
New Entry: 14929292  Passenger   Neil Brabant  Park  1
```

Figure 35

```
INSERTION at index 42
New Entry: 14476919  Driver   Ted Tallmadge  Park  Available


COLLISION at index 75
New Entry: 14541837  Passenger   Karina Rossin  Zoo  3


INSERTION at index 75
New Entry: 14541837  Passenger   Karina Rossin  Zoo  3


INSERTION at index 7
New Entry: 14338567  Driver   Earnestine Croswell  Park  Busy


INSERTION at index 19
New Entry: 14824429  Passenger   Nelson Tooker  Park  5


INSERTION at index 28
New Entry: 14492026  Passenger   Christian Mullarkey  Park  2
```

Figure 36

```
COLLISION at index 42
New Entry: 14770506  Passenger   Christian Durall  Church  1


INSERTION at index 42
New Entry: 14770506  Passenger   Christian Durall  Church  1


INSERTION at index 81
New Entry: 14359796  Passenger   Mathew Eslick  Mall  4


INSERTION at index 32
New Entry: 14495655  Passenger   Sofia Bonfiglio  Mall  4


INSERTION at index 74
New Entry: 14355490  Driver   Tania Mogensen  Suburbs  Offline


INSERTION at index 62
New Entry: 14147356  Passenger   Elinor Raco  Supermarket  2
```

Figure 37

```
INSERTION at index 67
New Entry: 14593654  Passenger   Mathew Milian  Zoo  1


COLLISION at index 49
New Entry: 14080040  Driver   Guy Loredo  Supermarket  Available


INSERTION at index 49
New Entry: 14080040  Driver   Guy Loredo  Supermarket  Available


INSERTION at index 86
New Entry: 14478629  Driver   Ted Makuch  School  Available


INSERTION at index 83
New Entry: 14707100  Passenger   Dona Mcinnes  Suburbs  3


COLLISION at index 32
New Entry: 14286103  Driver   Dollie Dignan  Mall  Available
```

Figure 38

```
INSERTION at index 32
New Entry: 14286103  Driver   Dollie Dignan  Mall  Available


INSERTION at index 22
New Entry: 14030930  Driver   Jessie Mayville  School  Available


INSERTION at index 23
New Entry: 14973623  Driver   Christian Busbee  Supermarket  Offline


COLLISION at index 7
New Entry: 14158835  Driver   Jamie Shake  Zoo  Offline


INSERTION at index 7
New Entry: 14158835  Driver   Jamie Shake  Zoo  Offline


COLLISION at index 28
New Entry: 14129722  Driver   Edwina Esters  University  Offline
```

Figure 39

```
INSERTION at index 28
New Entry: 14129722  Driver   Edwina Esters  University  Offline


COLLISION at index 42
New Entry: 14207712  Driver   Harriett Zollinger  Zoo  Busy


INSERTION at index 42
New Entry: 14207712  Driver   Harriett Zollinger  Zoo  Busy


INSERTION at index 71
New Entry: 14151577  Driver   Nannie Mcgown  Zoo  Busy

The load factor every 10 inserts is 41.24%

INSERTION at index 90
New Entry: 14905261  Driver   Yuriko Stimer  Supermarket  Available


INSERTION at index 1
New Entry: 14778665  Passenger   Cornell Uhri  School  2
```

Figure 40

```
COLLISION at index 3
New Entry: 14481609  Passenger   Amal Moznett  Mall  4


INSERTION at index 3
New Entry: 14481609  Passenger   Amal Moznett  Mall  4


COLLISION at index 88
New Entry: 14088484  Driver   Khalilah Dittrick  Zoo  Offline


INSERTION at index 88
New Entry: 14088484  Driver   Khalilah Dittrick  Zoo  Offline


INSERTION at index 65
New Entry: 14210394  Driver   Lonnie Traxson  Park  Busy
```

Figure 41

```
UPDATED at index 32
New Entry: 14495655  Passenger   Sofia Bonfiglio  Mall  4
```

Figure 42

```
Time taken to insert: 0.009879 seconds
DELETING RECORDS

Deleted entry at index 32: 14495655  Passenger   Sofia Bonfiglio  Mall  4
Deleted entry at index 65: 14210394  Driver   Lonnie Traxson  Park  Busy
Key not found
Time taken to delete: 5.24999e-05 seconds

SEARCHING RECORDS

GOT ENTRY: 14147356  Passenger Elinor Raco   Supermarket 2
GOT ENTRY: 14770506  Passenger Christian Durall   Church 1
Key not found
Key not found
Time taken to search: 4.33e-05 seconds
```

Figure 43

```
Wrong record type
```

Figure 44

```
********************************************************************************
Performance Analysis
********************************************************************************

INSERTING RECORDS

INSERTING 10 RECORDS


INSERTION at index 22
New Entry: 14495655  Passenger   Sofia Bonfiglio  Mall  4


INSERTION at index 8
New Entry: 14224671  Passenger   Ashlee Capellan  School  5


INSERTION at index 19
New Entry: 14707100  Passenger   Dona Mcinnes  Suburbs  3


INSERTION at index 7
New Entry: 14644633  Passenger   Maricela Landreneau  Park  1
```

Figure 45

```
INSERTION at index 65
New Entry: 14210394  Driver   Lonnie Traxson  Park  Busy

Time taken to insert 10 records: 0.0001553 seconds
Time taken to insert 30 records: 0.0042397999 seconds
Time taken to insert 45 records: 0.0085557 seconds
```

Figure 46

```
DELETING RECORDS
DELETING 5 RECORDS

Deleted entry at index 22: 14495655  Passenger   Sofia Bonfiglio  Mall  4
Deleted entry at index 8: 14224671  Passenger   Ashlee Capellan  School  5
Deleted entry at index 19: 14707100  Passenger   Dona Mcinnes  Suburbs  3
Deleted entry at index 7: 14644633  Passenger   Maricela Landreneau  Park  1
Deleted entry at index 6: 14147356  Passenger   Elinor Raco  Supermarket  2
```

Figure 47

```
DELETING 10 RECORDS

Deleted entry at index 46: 14210394  Driver   Lonnie Traxson  Park  Busy
Deleted entry at index 12: 14088484  Driver   Khalilah Dittrick  Zoo  Offline
Deleted entry at index 41: 14481609  Passenger   Amal Moznett  Mall  4
Deleted entry at index 32: 14778665  Passenger   Cornell Uhri  School  2
Deleted entry at index 10: 14905261  Driver   Yuriko Stimer  Supermarket  Available
Deleted entry at index 6: 14393910  Passenger   Cody Mcmartin  Suburbs  3
Deleted entry at index 49: 14799737  Passenger   Katy Vacek  Suburbs  2
Deleted entry at index 21: 14431660  Passenger   Mathew Mercedes  Suburbs  1
Deleted entry at index 34: 14541837  Passenger   Karina Rossin  Zoo  3
Deleted entry at index 14: 14593654  Passenger   Mathew Milian  Zoo  1
DELETING 20 RECORDS
```

Figure 48

```
DELETING 20 RECORDS

Deleted entry at index 17: 14397374  Passenger   Esmeralda Radovich  Church  1
Deleted entry at index 19: 14824429  Passenger   Nelson Tooker  Park  5
Deleted entry at index 39: 14929292  Passenger   Neil Brabant  Park  1
Deleted entry at index 13: 14786148  Passenger   Clinton Blumer  Church  2
Deleted entry at index 69: 14973358  Passenger   Julio Degroat  Suburbs  5
Deleted entry at index 55: 14557200  Passenger   Elinor Greig  Suburbs  4
Deleted entry at index 49: 14472268  Passenger   Malinda Dodgen  University  1
Deleted entry at index 81: 14359796  Passenger   Mathew Eslick  Mall  4
Deleted entry at index 42: 14770506  Passenger   Christian Durall  Church  1
Deleted entry at index 12: 14354152  Driver   Max Plain  Church  Available
Deleted entry at index 7: 14338567  Driver   Earnestine Croswell  Park  Busy
Deleted entry at index 95: 14277290  Driver   Clayton Ricard  Suburbs  Offline
Deleted entry at index 42: 14476919  Driver   Ted Tallmadge  Park  Available
Deleted entry at index 24: 14032266  Driver   Kelly Cheatam  Mall  Available
Deleted entry at index 3: 14094113  Driver   Jessie Leandro  School  Available
Deleted entry at index 69: 14490422  Driver   Guy Kadel  Mall  Busy
Deleted entry at index 39: 14650254  Driver   Althea Warring  University  Busy
Deleted entry at index 74: 14355490  Driver   Tania Mogensen  Suburbs  Offline
Deleted entry at index 86: 14478629  Driver   Ted Makuch  School  Available
Deleted entry at index 61: 14827960  Driver   Katy Callier  Park  Offline
```

Figure 49

```
Time taken to delete 5 records: 7.38001e-05 seconds
Time taken to delete 10 records: 0.0001109 seconds
Time taken to delete 20 records: 0.0002557 seconds

Time Complexity for deletion is O(1) since the time taken are almost the same
```

Figure 50

```
SEARCHING RECORDS
SEARCHING 5 RECORDS

GOT ENTRY: 14210394  Driver Lonnie Traxson   Park Busy
GOT ENTRY: 14088484  Driver Khalilah Dittrick   Zoo Offline
GOT ENTRY: 14481609  Passenger Amal Moznett   Mall 4
GOT ENTRY: 14778665  Passenger Cornell Uhri   School 2
GOT ENTRY: 14905261  Driver Yuriko Stimer   Supermarket Available
```

Figure 51

```
 SEARCHING 10 RECORDS

 GOT ENTRY: 14492026  Passenger Christian Mullarkey   Park 2
 GOT ENTRY: 14397374  Passenger Esmeralda Radovich   Church 1
 GOT ENTRY: 14824429  Passenger Nelson Tooker   Park 5
 GOT ENTRY: 14929292  Passenger Neil Brabant   Park 1
 GOT ENTRY: 14786148  Passenger Clinton Blumer   Church 2
 GOT ENTRY: 14495655  Passenger Sofia Bonfiglio   Mall 4
 GOT ENTRY: 14224671  Passenger Ashlee Capellan   School 5
 GOT ENTRY: 14707100  Passenger Dona Mcinnes   Suburbs 3
 GOT ENTRY: 14644633  Passenger Maricela Landreneau   Park 1
 GOT ENTRY: 14147356  Passenger Elinor Raco   Supermarket 2
```

Figure 52

```
SEARCHING 20 RECORDS

GOT ENTRY: 14495655  Passenger Sofia Bonfiglio   Mall 4
GOT ENTRY: 14224671  Passenger Ashlee Capellan   School 5
GOT ENTRY: 14707100  Passenger Dona Mcinnes   Suburbs 3
GOT ENTRY: 14644633  Passenger Maricela Landreneau   Park 1
GOT ENTRY: 14147356  Passenger Elinor Raco   Supermarket 2
GOT ENTRY: 14210394  Driver Lonnie Traxson   Park Busy
GOT ENTRY: 14088484  Driver Khalilah Dittrick   Zoo Offline
GOT ENTRY: 14481609  Passenger Amal Moznett   Mall 4
GOT ENTRY: 14778665  Passenger Cornell Uhri   School 2
GOT ENTRY: 14905261  Driver Yuriko Stimer   Supermarket Available
GOT ENTRY: 14607868  Driver Jamie Seager   Church Available
GOT ENTRY: 14080040  Driver Guy Loredo   Supermarket Available
GOT ENTRY: 14286103  Driver Dollie Dignan   Mall Available
GOT ENTRY: 14030930  Driver Jessie Mayville   School Available
GOT ENTRY: 14973623  Driver Christian Busbee   Supermarket Offline
GOT ENTRY: 14158835  Driver Jamie Shake   Zoo Offline
GOT ENTRY: 14129722  Driver Edwina Esters   University Offline
GOT ENTRY: 14207712  Driver Harriett Zollinger   Zoo Busy
GOT ENTRY: 14151577  Driver Nannie Mcgown   Zoo Busy
GOT ENTRY: 14393910  Passenger Cody Mcmartin   Suburbs 3
```

Figure 53

```
Time taken to search 5 records: 0.0001228 seconds
Time taken to search 10 records: 9.29e-05 seconds
Time taken to search 20 records: 0.0002075001 seconds

Time Complexity for searching is O(1) since the time taken are almost the same
```

Figure 54

For hash table, a modulo-based hash function was implemented.

As seen above, 45 records (22 passenger and 23 driver records) were inserted inside the hash table from figure 1 to figure 41. Each time a record is inserted or deleted or found or another record collides with it, a message appears on the terminal. When the user adds a duplicate record, that is, an already existing record, the hash table updates it as seen in figure 42. An error message appears when user tries to delete or find non existing record from hash table as shown in figure 43. Also, the load factor is printed on terminal for every 10 insert. If the load factor exceeds 70% or is less than 30%, the hash table is resized.

For collision-handling, chaining was used. In order to prevent worst case scenarios such as many records being stored in same array slot, the hash tables are given a size which is twice the number of records inserted. The worst-case scenario of having too many records stored at the same index in a linked list, can cause the time complexity of searching a record to be O(n).

Test case

Test Case	Test Case Description	Input	Expected Result	Program Output
1	Validate Input	14210394, Driver,Lonnie Traxson,Park,Busy	The record is properly inserted	INSERTION at index 65 New Entry: 14210394 Driver Lonnie Traxson Park Busy (Refer to figure 46)
2	Invalid input	11234087, "Seller", "Chloe Cheng", "Harbor", 1	An error message is displayed	Wrong record type (refer to figure 44)
3	Collision Handling	14481609,Passenger ,Amal Moznett,Mall,4	Displays at which index the record collided	COLLISION at index 3 New Entry: 14481609 Passenger Amal Moznett Mall 4 INSERTION at index 3 New Entry: 14481609 Passenger Amal Moznett Mall 4 (refer to figure 41)
4	Duplicate record handling	14495655, "Passenger", "Sofia Bonfiglio", "Mall", 4	Displays an update message	UPDATED at index 32 New Entry: 14495655 Passenger Sofia Bonfiglio Mall 4 (refer to figure 42)

5	Tracking and displaying load factor every 10 inserts	14495655,Passenger,Sofia Bonfiglio,Mall,4 14224671,Passenger,Ashlee Capellan,School,5 14707100,Passenger,Dona Mcinnes,Suburbs,3 14644633,Passenger,Maricela Landreneau,Park,1 14147356,Passenger,Elinor Raco,Supermarket,2 14393910,Passenger,Cody Mcmartin,Suburbs,3 14799737,Passenger,Katy Vacek,Suburbs,2 14431660,Passenger,Mathew Mercedes,Suburbs,1 14541837,Passenger,Karina Rossin,Zoo,3 14593654,Passenger,Mathew Milian,Zoo,1	The insertion or collision are displayed. Also the load factor is displayed	INSERTION at index 39 New Entry: 14495655 Passenger Sofia Bonfiglio Mall 4 INSERTION at index 9 New Entry: 14224671 Passenger Ashlee Capellan School 5 INSERTION at index 45 New Entry: 14707100 Passenger Dona Mcinnes Suburbs 3 INSERTION at index 1 New Entry: 14644633 Passenger Maricela Landreneau Park 1 INSERTION at index 41 New Entry: 14147356 Passenger Elinor Raco Supermarket 2 INSERTION at index 15 New Entry: 14393910 Passenger Cody Mcmartin Suburbs 3 INSERTION at index 17 New Entry: 14799737 Passenger Katy Vacek Suburbs 2 INSERTION at index 31

				New Entry: 14431660 Passenger Mathew Mercedes Suburbs 1 INSERTION at index 33 New Entry: 14541837 Passenger Karina Rossin Zoo 3 INSERTION at index 42 New Entry: 14593654 Passenger Mathew Milian Zoo 1 The load factor every 10 inserts is 21.28% (refer to figure 25 and figure 26)
6	Deleting existing record	14495655	The passenger or driver record deleted is displayed	Deleted entry at index 32: 14495655 Passenger Sofia Bonfiglio Mall 4 (refer to figure 47)
7	Deleting non existent record	39390000	Error message is displayed	Key not found (refer to figure 43)
8	Searching existing record	14147356	The record details are displayed	GOT ENTRY: 14147356 Passenger Elinor Raco Supermarket 2 (refer to figure 43)
9	Searching non existent record	9475	Error message is displayed	Key not found (refer to figure 43)
10	Searching deleted record	14495655	Error message is displayed	Key not found (refer to figure 43)

A summary of performance analysis which starts as from figure 45 is shown below:

Insertion

Num of Records	10	30	45
Time Taken (in seconds)	0.0001665	0.0004991	0.0023235

Time Complexity for insertion is O(1) since the time taken are almost the same.

Deletion

Num of Records	5	10	20
Time Taken (in seconds)	6.32e-05	0.0001518	0.0002099

Time Complexity for deletion is O(1) since the time taken are almost the same.

Searching

Num of Records	5	10	20
Time Taken (in seconds)	3.86e-05	0.0001213	0.0001985

Time Complexity for searching is O(1) since the time taken are almost the same.

Module 3

To test whether module 3 was well implemented, ensure that passengers.csv, drivers.csv, passenger_hash_table.csv and driver_hash_table.csv are in the same directory with test_module3.py, and run the following command on the terminal:

- python3 test_module3.py

Test Case	Test Case Description	Input	Expected Result	Program Output
1	Membership tier has changed	14495655,Passenger,Sofia Bonfiglio,Mall,4 14495655,Passenger,Sofia Bonfiglio,Mall,1	Update message is displayed	INSERTION at index 18 New Entry: 14495655 Passenger Sofia Bonfiglio Mall 4

				UPDATED at index 18 New Entry: 14495655 Passenger Sofia Bonfiglio Mall 1 (refer to figure 55)
2	No available drivers	14431660,Passenger,Mathew Mercedes,Suburbs,1	Pickup request rejection message is displayed	Request from Mathew Mercedes has been rejected due to unavailable drivers (refer to figure 56)
3	Insert into heap	1003.0, 14541837,Passenger,Karina Rossin,Zoo,3	Insertion message is printed	Inserted into Heap = Priority: 1003.0 - Pickup request: 14541837 Karina Rossin Zoo 3 Assigned Driver: Nannie Mcgown Pickup time: 0 mins Driver Current Location: Zoo Shortest Path from Zoo to Zoo - Zoo Priority: 1003.0 - Pickup request: 14541837 Karina Rossin Zoo 3 (refer to figure 57)
4	Extract from heap	None	Extracted pickup request is displayed	Extracted from Heap = Priority: 29.39 - Pickup request: 14397374 Esmeralda Radovich Church 1 Assigned Driver: Guy Loredo

				Pickup time: 41 mins Driver status: Busy Driver Location: Supermarket Shortest Path from Supermarket to Church - ['Supermarket', 'Suburbs', 'Church'] Priority: 25.41 - Pickup request: 14472268 Malinda Dodgen University 1 (refer to figure 58)
5	Peek at top item on heap	None	Top item is displayed	Item on top of heap: 1005.0 - Pickup request: 14929292 Neil Brabant Park 1 Assigned Driver: Ted Tallmadge Pickup time: 0 mins Driver status: Busy Driver Location: Park

All the requirements and edge cases were considered for module 3. As required, 12 insert operations and 12 extractions were demonstrated. Moreover, all edge cases were considered. For example, when a passenger gets an available driver, the driver is given the "Busy" status once inserted into heap. This prevents other passengers from getting the same driver. If passengers do not get a driver, their pickup request is rejected with a message. Invalid inputs are handled using Error handling inside the class DSAHeap modules. When the output of test_module3.py is displayed, the pickup requests and heap array are displayed after insertion or extraction from heap. It ensures that the user understands the log displayed.

```
INSERTION at index 18
New Entry: 14495655  Passenger   Sofia Bonfiglio  Mall  4


UPDATED at index 18
New Entry: 14495655  Passenger   Sofia Bonfiglio  Mall  1
```

Figure 55

```
Request from Julio Degroat has been rejected due to unavailable drivers
Request from Mathew Mercedes has been rejected due to unavailable drivers
Request from Ashlee Capellan has been rejected due to unavailable drivers
Request from Katy Vacek has been rejected due to unavailable drivers
Request from Cornell Uhri has been rejected due to unavailable drivers
Request from Elinor Raco has been rejected due to unavailable drivers
Request from Dona Mcinnes has been rejected due to unavailable drivers
Request from Mathew Eslick has been rejected due to unavailable drivers
```

Figure 56

```
Item on top of heap: 1005.0 - Pickup request: 14929292  Neil Brabant    Park    1
Assigned Driver: Ted Tallmadge  Pickup time: 0 mins Driver status: Busy Driver Location: Park
```

Figure 57

```
Inserted into Heap = Priority: 1003.0 - Pickup request: 14541837    Karina Rossin   Zoo 3
Assigned Driver: Nannie Mcgown  Pickup time: 0 mins Driver Current Location: Zoo
Shortest Path from Zoo to Zoo - Zoo
Priority: 1003.0 - Pickup request: 14541837 Karina Rossin Zoo 3
```

Figure 58

```
Count - 10
Extracted from Heap = Priority: 29.39 - Pickup request: 14397374    Esmeralda Radovich  Church  1
Assigned Driver: Guy Loredo Pickup time: 41 mins    Driver status: Busy Driver Location: Supermarket
Shortest Path from Supermarket to Church - ['Supermarket', 'Suburbs', 'Church']
Priority: 25.41 - Pickup request: 14472268 Malinda Dodgen University 1
```

Figure 59

Module 4

To test whether module 4 was well implemented, ensure that 2000Pickup(module4).csv, module4_driver_records.csv and passenger_hash_table.csv are in the same directory with test_module4.py, and run the following command on the terminal:

- python4 test_module3.py

This file is used to test how Merge Sort and Quick Sort handle 100, 500 and 1000 datasets which are generated by adding the scheduled pickups, driver details, shortest path, pickup time calculated using Dijkstra and priority calculated using the formula given in the assignment to a list.

The list is then sorted into ascending order for each dataset using BubbleSort algorithm. Then the passenger's name and pickup time are extracted for each pickup request and added to another list. This simple list is then sorted in three orders: random, nearly sorted and reverse using the modules found in class Test_Conditions inside conditions_module4.py. Three new test lists are created for each dataset of 100, 500, and 1,000 pickup requests.

After that, both Merge Sort and Quick Sort are used to sort the datasets. The diagrams below show the output obtained:

Dataset of 100 pickup requests:

```
*********************************************************************************
Sorting of 100 Pickup Requests
*********************************************************************************

Test List of 100 pickup requests in random order
[('Vickie Giffin', 51), ('Curtis Thrower', 61), ('Annie Markel', 0), ('Sean Schubert', 0), ('Orville Gillaspie', 8), ('Phyllis Beene', 42),
 ('Allen Mckenna', 24), ('Karen Pena', 8), ('Cody Basta', 0), ('Nelson Guitierrez', 38), ('Bessie Minton', 0), ('Kelly Cheatam', 24),
 ('Wayne Merkle', 51), ('Martin Harbin', 0), ('Clarence Keyser', 44), ('Chad Doucet', 42), ('Jose Gregg', 42), ('Gary Thorson', 0), ('Jorge
 Ventimiglia', 0), ('Peter Whitman', 0), ('Nelson Tooker', 0), ('Marilyn Mickens', 0), ('Yvette Dowdy', 0), ('Noreen Durfey', 0), ('Shawn
 Rodriquez', 0), ('Eileen Bunch', 0), ('Joseph Favors', 0), ('Keith Dykes', 0), ('Luz Coomer', 0), ('Amy Bedford', 0), ('Gary Burmeister',
 7), ('Chris Rahman', 0), ('Bruce Dailey', 24), ('Eva Pursley', 0), ('Russell Ashmore', 0), ('Curtis Cordell', 35), ('Jeffrey Keenan', 0),
 ('Dean Buchanon', 42), ('Theresa Townsley', 0), ('Vernon Moultrie', 7), ('Dustin Gadsden', 67), ('Deanna Arndt', 0), ('Timothy Congdon',
 7), ('Harold Seaman', 30), ('Norman Styles', 0), ('Harold Hamill', 11), ('Hazel Justice', 24), ('Timothy Vetter', 38), ('Mike Rodarte',
 51), ('Cassandra Creed', 24), ('Stephen Fitz', 0), ('Jacob Meeker', 0), ('Denise Herrington', 38), ('Olivia Fall', 29), ('Jessica
 Villegas', 41), ('Eileen Haddock', 33), ('Manuel Hammack', 30), ('Tony Brogan', 67), ('Sherri Kinser', 76), ('Antonio Braddock', 44),
 ('Jocelyn Lynde', 35), ('Chad Steffens', 29), ('Arlene Bergmann', 0), ('Sherry Winfrey', 62), ('Jeremy Hyman', 0), ('Larry Chapa', 35),
 ('Debbie Ruch', 62), ('Ronald Henning', 33), ('Jodi Holm', 0), ('Erik Weinman', 52), ('Philip Hammonds', 0), ('Jessie Efaw', 42), ('Richard
  Melvin', 35), ('Paul Albers', 52), ('Teresa Booth', 52), ('Jonathan Wagoner', 47), ('Elaine Elias', 48), ('Wayne Crabb', 37), ('Jeff
 Singletary', 0), ('Raymond Hynes', 51), ('Gregory Despain', 0), ('William Backus', 51), ('Benjamin Prado', 7), ('Jack Kolb', 29), ('Michael
  Michel', 0), ('Adam Harden', 53), ('Dale Ormsby', 0), ('Opal Tobias', 53), ('Howard Storm', 0), ('Gabriela Keaney', 62), ('Erma Creech',
 61), ('Jeremy Catalano', 62), ('Larry Cantu', 24), ('Chad Burbank', 62), ('Travis Llamas', 0), ('Emily Chadwick', 0), ('Vivian Milton', 0),
  ('Scott Luevano', 53), ('Christopher Merwin', 76), ('Fred Meeker', 76)]
```

Figure 60: Dataset of 100 pickup requests in random order

```
Test List of 100 pickup requests in nearly sorted order
[('Jeff Singletary', 0), ('Vickie Giffin', 51), ('Jacob Meeker', 0), ('Jorge Ventimiglia', 0), ('Vivian Milton', 0), ('Stephen Fitz', 0),
 ('Travis Llamas', 0), ('Eva Pursley', 0), ('Howard Storm', 0), ('Gary Thorson', 0), ('Jodi Holm', 0), ('Deanna Arndt', 0), ('Gregory
 Despain', 0), ('Martin Harbin', 0), ('Bessie Minton', 0), ('Dustin Gadsden', 67), ('Nelson Guitierrez', 38), ('Annie Markel', 0), ('Jeremy
 Hyman', 0), ('Peter Whitman', 0), ('William Backus', 51), ('Marilyn Mickens', 0), ('Yvette Dowdy', 0), ('Noreen Durfey', 0), ('Shawn
 Rodriquez', 0), ('Eileen Bunch', 0), ('Amy Bedford', 0), ('Keith Dykes', 0), ('Luz Coomer', 0), ('Joseph Favors', 0), ('Emily Chadwick',
 0), ('Chris Rahman', 0), ('Philip Hammonds', 0), ('Norman Styles', 0), ('Russell Ashmore', 0), ('Sean Schubert', 0), ('Jeffrey Keenan', 0),
  ('Dale Ormsby', 0), ('Theresa Townsley', 0), ('Vernon Moultrie', 7), ('Timothy Congdon', 7), ('Benjamin Prado', 7), ('Gary Burmeister',
 7), ('Orville Gillaspie', 8), ('Karen Pena', 8), ('Harold Hamill', 11), ('Hazel Justice', 24), ('Kelly Cheatam', 24), ('Allen Mckenna',
 24), ('Cassandra Creed', 24), ('Bruce Dailey', 24), ('Larry Cantu', 24), ('Mike Rodarte', 51), ('Olivia Fall', 29), ('Jack Kolb', 29),
 ('Harold Seaman', 30), ('Manuel Hammack', 30), ('Eileen Haddock', 33), ('Ronald Henning', 33), ('Curtis Cordell', 35), ('Jocelyn Lynde',
 35), ('Larry Chapa', 35), ('Richard Melvin', 35), ('Wayne Crabb', 37), ('Michael Michel', 0), ('Denise Herrington', 38), ('Timothy Vetter',
  38), ('Jessica Villegas', 41), ('Dean Buchanon', 42), ('Jose Gregg', 42), ('Phyllis Beene', 42), ('Raymond Hynes', 51), ('Chad Doucet',
 42), ('Clarence Keyser', 44), ('Antonio Braddock', 44), ('Jonathan Wagoner', 47), ('Elaine Elias', 48), ('Chad Steffens', 29), ('Cody
 Basta', 0), ('Jessie Efaw', 42), ('Wayne Merkle', 51), ('Nelson Tooker', 0), ('Teresa Booth', 52), ('Paul Albers', 52), ('Erik Weinman',
 52), ('Adam Harden', 53), ('Scott Luevano', 53), ('Opal Tobias', 53), ('Curtis Thrower', 61), ('Erma Creech', 61), ('Debbie Ruch', 62),
 ('Jeremy Catalano', 62), ('Gabriela Keaney', 62), ('Chad Burbank', 62), ('Sherry Winfrey', 62), ('Arlene Bergmann', 0), ('Tony Brogan',
 67), ('Sherri Kinser', 76), ('Christopher Merwin', 76), ('Fred Meeker', 76)]
```

Figure 61: Dataset of 100 pickup requests in nearly sorted order

```
Test List of 100 pickup requests in reversed order
[('Fred Meeker', 76), ('Christopher Merwin', 76), ('Sherri Kinser', 76), ('Tony Brogan', 67), ('Dustin Gadsden', 67), ('Sherry Winfrey',
62), ('Chad Burbank', 62), ('Gabriela Keaney', 62), ('Jeremy Catalano', 62), ('Debbie Ruch', 62), ('Erma Creech', 61), ('Curtis Thrower',
61), ('Opal Tobias', 53), ('Scott Luevano', 53), ('Adam Harden', 53), ('Erik Weinman', 52), ('Paul Albers', 52), ('Teresa Booth', 52),
('William Backus', 51), ('Wayne Merkle', 51), ('Raymond Hynes', 51), ('Vickie Giffin', 51), ('Mike Rodarte', 51), ('Elaine Elias', 48),
('Jonathan Wagoner', 47), ('Antonio Braddock', 44), ('Clarence Keyser', 44), ('Chad Doucet', 42), ('Jessie Efaw', 42), ('Phyllis Beene',
42), ('Jose Gregg', 42), ('Dean Buchanon', 42), ('Jessica Villegas', 41), ('Timothy Vetter', 38), ('Denise Herrington', 38), ('Nelson
Guitierrez', 38), ('Wayne Crabb', 37), ('Richard Melvin', 35), ('Larry Chapa', 35), ('Jocelyn Lynde', 35), ('Curtis Cordell', 35), ('Ronald
Henning', 33), ('Eileen Haddock', 33), ('Manuel Hammack', 30), ('Harold Seaman', 30), ('Jack Kolb', 29), ('Olivia Fall', 29), ('Chad
Steffens', 29), ('Larry Cantu', 24), ('Bruce Dailey', 24), ('Cassandra Creed', 24), ('Allen Mckenna', 24), ('Kelly Cheatam', 24), ('Hazel
Justice', 24), ('Harold Hamill', 11), ('Karen Pena', 8), ('Orville Gillaspie', 8), ('Gary Burmeister', 7), ('Benjamin Prado', 7), ('Timothy
Congdon', 7), ('Vernon Moultrie', 7), ('Theresa Townsley', 0), ('Dale Ormsby', 0), ('Jeffrey Keenan', 0), ('Sean Schubert', 0), ('Russell
Ashmore', 0), ('Norman Styles', 0), ('Philip Hammonds', 0), ('Chris Rahman', 0), ('Emily Chadwick', 0), ('Joseph Favors', 0), ('Luz
Coomer', 0), ('Keith Dykes', 0), ('Amy Bedford', 0), ('Eileen Bunch', 0), ('Shawn Rodriquez', 0), ('Noreen Durfey', 0), ('Yvette Dowdy',
0), ('Marilyn Mickens', 0), ('Nelson Tooker', 0), ('Peter Whitman', 0), ('Jeremy Hyman', 0), ('Annie Markel', 0), ('Michael Michel', 0),
('Arlene Bergmann', 0), ('Bessie Minton', 0), ('Martin Harbin', 0), ('Gregory Despain', 0), ('Deanna Arndt', 0), ('Jodi Holm', 0), ('Gary
Thorson', 0), ('Howard Storm', 0), ('Eva Pursley', 0), ('Travis Llamas', 0), ('Stephen Fitz', 0), ('Vivian Milton', 0), ('Jorge
Ventimiglia', 0), ('Jacob Meeker', 0), ('Cody Basta', 0), ('Jeff Singletary', 0)]
```

Figure 62: Dataset of 100 pickup requests in reverse order

Merge Sort for 100 pickup requests:

```
********************************************************************************
Sorting of 100 Random Pickup Requests using Mergesort
********************************************************************************
Note: The first names are not the same for the different test scenarios as the lists are sorted using only the pickup times not the names.

[('Annie Markel', 0), ('Sean Schubert', 0), ('Cody Basta', 0), ('Bessie Minton', 0), ('Martin Harbin', 0), ('Gary Thorson', 0), ('Jorge
 Ventimiglia', 0), ('Peter Whitman', 0), ('Nelson Tooker', 0), ('Marilyn Mickens', 0), ('Yvette Dowdy', 0), ('Noreen Durfey', 0), ('Shawn
 Rodriquez', 0), ('Eileen Bunch', 0), ('Joseph Favors', 0), ('Keith Dykes', 0), ('Luz Coomer', 0), ('Amy Bedford', 0), ('Chris Rahman', 0),
 ('Eva Pursley', 0), ('Russell Ashmore', 0), ('Jeffrey Keenan', 0), ('Theresa Townsley', 0), ('Deanna Arndt', 0), ('Norman Styles', 0),
 ('Stephen Fitz', 0), ('Jacob Meeker', 0), ('Arlene Bergmann', 0), ('Jeremy Hyman', 0), ('Jodi Holm', 0), ('Philip Hammonds', 0), ('Jeff
 Singletary', 0), ('Gregory Despain', 0), ('Michael Michel', 0), ('Dale Ormsby', 0), ('Howard Storm', 0), ('Travis Llamas', 0), ('Emily
 Chadwick', 0), ('Vivian Milton', 0), ('Gary Burmeister', 7), ('Vernon Moultrie', 7), ('Timothy Congdon', 7), ('Benjamin Prado', 7),
 ('Orville Gillaspie', 8), ('Karen Pena', 8), ('Harold Hamill', 11), ('Allen Mckenna', 24), ('Kelly Cheatam', 24), ('Bruce Dailey', 24),
 ('Hazel Justice', 24), ('Cassandra Creed', 24), ('Larry Cantu', 24), ('Olivia Fall', 29), ('Chad Steffens', 29), ('Jack Kolb', 29),
 ('Harold Seaman', 30), ('Manuel Hammack', 30), ('Eileen Haddock', 33), ('Ronald Henning', 33), ('Curtis Cordell', 35), ('Jocelyn Lynde',
 35), ('Larry Chapa', 35), ('Richard Melvin', 35), ('Wayne Crabb', 37), ('Nelson Guitierrez', 38), ('Timothy Vetter', 38), ('Denise
 Herrington', 38), ('Jessica Villegas', 41), ('Phyllis Beene', 42), ('Chad Doucet', 42), ('Jose Gregg', 42), ('Dean Buchanon', 42), ('Jessie
  Efaw', 42), ('Clarence Keyser', 44), ('Antonio Braddock', 44), ('Jonathan Wagoner', 47), ('Elaine Elias', 48), ('Vickie Giffin', 51),
 ('Wayne Merkle', 51), ('Mike Rodarte', 51), ('Raymond Hynes', 51), ('William Backus', 51), ('Erik Weinman', 52), ('Paul Albers', 52),
 ('Teresa Booth', 52), ('Adam Harden', 53), ('Opal Tobias', 53), ('Scott Luevano', 53), ('Curtis Thrower', 61), ('Erma Creech', 61),
 ('Sherry Winfrey', 62), ('Debbie Ruch', 62), ('Gabriela Keaney', 62), ('Jeremy Catalano', 62), ('Chad Burbank', 62), ('Dustin Gadsden',
 67), ('Tony Brogan', 67), ('Sherri Kinser', 76), ('Christopher Merwin', 76), ('Fred Meeker', 76)]
Duration taken by mergesort to sort 100 pickup requests in random order: 0.000325 seconds
```

Figure 63: Random order sorted

```
********************************************************************************
Sorting of 100 Nearly Sorted Pickup Requests using Mergesort
********************************************************************************

[('Jeff Singletary', 0), ('Jacob Meeker', 0), ('Jorge Ventimiglia', 0), ('Vivian Milton', 0), ('Stephen Fitz', 0), ('Travis Llamas', 0),
 ('Eva Pursley', 0), ('Howard Storm', 0), ('Gary Thorson', 0), ('Jodi Holm', 0), ('Deanna Arndt', 0), ('Gregory Despain', 0), ('Martin
 Harbin', 0), ('Bessie Minton', 0), ('Annie Markel', 0), ('Jeremy Hyman', 0), ('Peter Whitman', 0), ('Marilyn Mickens', 0), ('Yvette Dowdy',
  0), ('Noreen Durfey', 0), ('Shawn Rodriquez', 0), ('Eileen Bunch', 0), ('Amy Bedford', 0), ('Keith Dykes', 0), ('Luz Coomer', 0), ('Joseph
  Favors', 0), ('Emily Chadwick', 0), ('Chris Rahman', 0), ('Philip Hammonds', 0), ('Norman Styles', 0), ('Russell Ashmore', 0), ('Sean
 Schubert', 0), ('Jeffrey Keenan', 0), ('Dale Ormsby', 0), ('Theresa Townsley', 0), ('Michael Michel', 0), ('Cody Basta', 0), ('Nelson
 Tooker', 0), ('Arlene Bergmann', 0), ('Vernon Moultrie', 7), ('Timothy Congdon', 7), ('Benjamin Prado', 7), ('Gary Burmeister', 7),
 ('Orville Gillaspie', 8), ('Karen Pena', 8), ('Harold Hamill', 11), ('Hazel Justice', 24), ('Kelly Cheatam', 24), ('Allen Mckenna', 24),
 ('Cassandra Creed', 24), ('Bruce Dailey', 24), ('Larry Cantu', 24), ('Olivia Fall', 29), ('Jack Kolb', 29), ('Chad Steffens', 29), ('Harold
  Seaman', 30), ('Manuel Hammack', 30), ('Eileen Haddock', 33), ('Ronald Henning', 33), ('Curtis Cordell', 35), ('Jocelyn Lynde', 35),
 ('Larry Chapa', 35), ('Richard Melvin', 35), ('Wayne Crabb', 37), ('Nelson Guitierrez', 38), ('Denise Herrington', 38), ('Timothy Vetter',
 38), ('Jessica Villegas', 41), ('Dean Buchanon', 42), ('Jose Gregg', 42), ('Phyllis Beene', 42), ('Chad Doucet', 42), ('Jessie Efaw', 42),
 ('Clarence Keyser', 44), ('Antonio Braddock', 44), ('Jonathan Wagoner', 47), ('Elaine Elias', 48), ('Vickie Giffin', 51), ('William
 Backus', 51), ('Mike Rodarte', 51), ('Raymond Hynes', 51), ('Wayne Merkle', 51), ('Teresa Booth', 52), ('Paul Albers', 52), ('Erik
 Weinman', 52), ('Adam Harden', 53), ('Scott Luevano', 53), ('Opal Tobias', 53), ('Curtis Thrower', 61), ('Erma Creech', 61), ('Debbie
 Ruch', 62), ('Jeremy Catalano', 62), ('Gabriela Keaney', 62), ('Chad Burbank', 62), ('Sherry Winfrey', 62), ('Dustin Gadsden', 67), ('Tony
 Brogan', 67), ('Sherri Kinser', 76), ('Christopher Merwin', 76), ('Fred Meeker', 76)]
Duration taken by mergesort to sort 100 pickup requests in nearly sorted order: 0.000287 seconds
```

Figure 64: Nearly sorted list sorted

```
*************************************************************************
Sorting of 100 Reversed Pickup Requests using Mergesort
*************************************************************************

[('Theresa Townsley', 0), ('Dale Ormsby', 0), ('Jeffrey Keenan', 0), ('Sean Schubert', 0), ('Russell Ashmore', 0), ('Norman Styles', 0),
 ('Philip Hammonds', 0), ('Chris Rahman', 0), ('Emily Chadwick', 0), ('Joseph Favors', 0), ('Luz Coomer', 0), ('Keith Dykes', 0), ('Amy
 Bedford', 0), ('Eileen Bunch', 0), ('Shawn Rodriquez', 0), ('Noreen Durfey', 0), ('Yvette Dowdy', 0), ('Marilyn Mickens', 0), ('Nelson
 Tooker', 0), ('Peter Whitman', 0), ('Jeremy Hyman', 0), ('Annie Markel', 0), ('Michael Michel', 0), ('Arlene Bergmann', 0), ('Bessie
 Minton', 0), ('Martin Harbin', 0), ('Gregory Despain', 0), ('Deanna Arndt', 0), ('Jodi Holm', 0), ('Gary Thorson', 0), ('Howard Storm', 0),
  ('Eva Pursley', 0), ('Travis Llamas', 0), ('Stephen Fitz', 0), ('Vivian Milton', 0), ('Jorge Ventimiglia', 0), ('Jacob Meeker', 0), ('Cody
  Basta', 0), ('Jeff Singletary', 0), ('Gary Burmeister', 7), ('Benjamin Prado', 7), ('Timothy Congdon', 7), ('Vernon Moultrie', 7), ('Karen
  Pena', 8), ('Orville Gillaspie', 8), ('Harold Hamill', 11), ('Larry Cantu', 24), ('Bruce Dailey', 24), ('Cassandra Creed', 24), ('Allen
 Mckenna', 24), ('Kelly Cheatam', 24), ('Hazel Justice', 24), ('Jack Kolb', 29), ('Olivia Fall', 29), ('Chad Steffens', 29), ('Manuel
 Hammack', 30), ('Harold Seaman', 30), ('Ronald Henning', 33), ('Eileen Haddock', 33), ('Richard Melvin', 35), ('Larry Chapa', 35),
 ('Jocelyn Lynde', 35), ('Curtis Cordell', 35), ('Wayne Crabb', 37), ('Timothy Vetter', 38), ('Denise Herrington', 38), ('Nelson
 Guitierrez', 38), ('Jessica Villegas', 41), ('Chad Doucet', 42), ('Jessie Efaw', 42), ('Phyllis Beene', 42), ('Jose Gregg', 42), ('Dean
 Buchanon', 42), ('Antonio Braddock', 44), ('Clarence Keyser', 44), ('Jonathan Wagoner', 47), ('Elaine Elias', 48), ('William Backus', 51),
 ('Wayne Merkle', 51), ('Raymond Hynes', 51), ('Vickie Giffin', 51), ('Mike Rodarte', 51), ('Erik Weinman', 52), ('Paul Albers', 52),
 ('Teresa Booth', 52), ('Opal Tobias', 53), ('Scott Luevano', 53), ('Adam Harden', 53), ('Erma Creech', 61), ('Curtis Thrower', 61),
 ('Sherry Winfrey', 62), ('Chad Burbank', 62), ('Gabriela Keaney', 62), ('Jeremy Catalano', 62), ('Debbie Ruch', 62), ('Tony Brogan', 67),
 ('Dustin Gadsden', 67), ('Fred Meeker', 76), ('Christopher Merwin', 76), ('Sherri Kinser', 76)]
Duration taken by mergesort to sort 100 pickup requests in reverse order: 0.000298 seconds
```

Figure 65: Reversed list sorted

```
The Top Down MergeSort algorithm Reflection for different scenarios of 100 pickup request:
Duration to sort 100 pickup requests in random order: 0.000325 seconds
Duration to sort 100 pickup requests in nearly sorted order: 0.000287 seconds
Duration to sort 100 pickup requests in reverse order: 0.000298 seconds

Looking at the durations for each scenario of 100 pickup request, MergeSort takes less time to
    sort the nearly_sorted list while more time is taken to sort the random list
Its time complexity is O(n log n) and its space complexity is O(n) as it needs temporary lists to enable merging.
```

Figure 66: Short Reflection

Quick Sort for 100 pickup requests:

```
********************************************************************************
Sorting of 100 Random Pickup Requests using Quicksort
********************************************************************************
Note: The first names are not the same for the different test scenarios as the lists are sorted using only the pickup times not the names.

[('Theresa Townsley', 0), ('Deanna Arndt', 0), ('Annie Markel', 0), ('Norman Styles', 0), ('Cody Basta', 0), ('Stephen Fitz', 0), ('Martin
 Harbin', 0), ('Jacob Meeker', 0), ('Jorge Ventimiglia', 0), ('Arlene Bergmann', 0), ('Nelson Tooker', 0), ('Jeremy Hyman', 0), ('Yvette
 Dowdy', 0), ('Jodi Holm', 0), ('Shawn Rodriquez', 0), ('Philip Hammonds', 0), ('Joseph Favors', 0), ('Jeff Singletary', 0), ('Luz Coomer',
 0), ('Gregory Despain', 0), ('Chris Rahman', 0), ('Michael Michel', 0), ('Russell Ashmore', 0), ('Dale Ormsby', 0), ('Bessie Minton', 0),
 ('Howard Storm', 0), ('Sean Schubert', 0), ('Travis Llamas', 0), ('Peter Whitman', 0), ('Emily Chadwick', 0), ('Noreen Durfey', 0),
 ('Vivian Milton', 0), ('Keith Dykes', 0), ('Jeffrey Keenan', 0), ('Eileen Bunch', 0), ('Eva Pursley', 0), ('Amy Bedford', 0), ('Marilyn
 Mickens', 0), ('Gary Thorson', 0), ('Gary Burmeister', 7), ('Benjamin Prado', 7), ('Timothy Congdon', 7), ('Vernon Moultrie', 7), ('Orville
  Gillaspie', 8), ('Karen Pena', 8), ('Harold Hamill', 11), ('Hazel Justice', 24), ('Kelly Cheatam', 24), ('Bruce Dailey', 24), ('Cassandra
 Creed', 24), ('Larry Cantu', 24), ('Allen Mckenna', 24), ('Olivia Fall', 29), ('Jack Kolb', 29), ('Chad Steffens', 29), ('Harold Seaman',
 30), ('Manuel Hammack', 30), ('Eileen Haddock', 33), ('Ronald Henning', 33), ('Curtis Cordell', 35), ('Larry Chapa', 35), ('Richard
 Melvin', 35), ('Jocelyn Lynde', 35), ('Wayne Crabb', 37), ('Timothy Vetter', 38), ('Denise Herrington', 38), ('Nelson Guitierrez', 38),
 ('Jessica Villegas', 41), ('Dean Buchanon', 42), ('Phyllis Beene', 42), ('Jessie Efaw', 42), ('Jose Gregg', 42), ('Chad Doucet', 42),
 ('Antonio Braddock', 44), ('Clarence Keyser', 44), ('Jonathan Wagoner', 47), ('Elaine Elias', 48), ('William Backus', 51), ('Mike Rodarte',
  51), ('Raymond Hynes', 51), ('Vickie Giffin', 51), ('Wayne Merkle', 51), ('Paul Albers', 52), ('Teresa Booth', 52), ('Erik Weinman', 52),
 ('Opal Tobias', 53), ('Scott Luevano', 53), ('Adam Harden', 53), ('Erma Creech', 61), ('Curtis Thrower', 61), ('Gabriela Keaney', 62),
 ('Debbie Ruch', 62), ('Chad Burbank', 62), ('Sherry Winfrey', 62), ('Jeremy Catalano', 62), ('Dustin Gadsden', 67), ('Tony Brogan', 67),
 ('Sherri Kinser', 76), ('Christopher Merwin', 76), ('Fred Meeker', 76)]

Duration taken by QuickSort to sort 100 pickup requests in random order: 0.000167 seconds
```

Figure 67: Random order sorted

```
********************************************************************************
Sorting of 100 Nearly Sorted Pickup Requests using Quicksort
********************************************************************************

[('Amy Bedford', 0), ('Keith Dykes', 0), ('Jeff Singletary', 0), ('Luz Coomer', 0), ('Jorge Ventimiglia', 0), ('Joseph Favors', 0),
 ('Stephen Fitz', 0), ('Emily Chadwick', 0), ('Eva Pursley', 0), ('Chris Rahman', 0), ('Gary Thorson', 0), ('Philip Hammonds', 0), ('Deanna
 Arndt', 0), ('Norman Styles', 0), ('Martin Harbin', 0), ('Russell Ashmore', 0), ('Annie Markel', 0), ('Sean Schubert', 0), ('Peter
 Whitman', 0), ('Jeffrey Keenan', 0), ('Yvette Dowdy', 0), ('Dale Ormsby', 0), ('Shawn Rodriquez', 0), ('Theresa Townsley', 0), ('Vivian
 Milton', 0), ('Michael Michel', 0), ('Jacob Meeker', 0), ('Cody Basta', 0), ('Howard Storm', 0), ('Nelson Tooker', 0), ('Gregory Despain',
 0), ('Arlene Bergmann', 0), ('Jeremy Hyman', 0), ('Eileen Bunch', 0), ('Bessie Minton', 0), ('Noreen Durfey', 0), ('Marilyn Mickens', 0),
 ('Jodi Holm', 0), ('Travis Llamas', 0), ('Vernon Moultrie', 7), ('Gary Burmeister', 7), ('Benjamin Prado', 7), ('Timothy Congdon', 7),
 ('Orville Gillaspie', 8), ('Karen Pena', 8), ('Harold Hamill', 11), ('Cassandra Creed', 24), ('Kelly Cheatam', 24), ('Allen Mckenna', 24),
 ('Bruce Dailey', 24), ('Larry Cantu', 24), ('Hazel Justice', 24), ('Olivia Fall', 29), ('Chad Steffens', 29), ('Jack Kolb', 29), ('Harold
 Seaman', 30), ('Manuel Hammack', 30), ('Eileen Haddock', 33), ('Ronald Henning', 33), ('Curtis Cordell', 35), ('Larry Chapa', 35),
 ('Richard Melvin', 35), ('Jocelyn Lynde', 35), ('Wayne Crabb', 37), ('Denise Herrington', 38), ('Timothy Vetter', 38), ('Nelson
 Guitierrez', 38), ('Jessica Villegas', 41), ('Chad Doucet', 42), ('Dean Buchanon', 42), ('Jessie Efaw', 42), ('Phyllis Beene', 42), ('Jose
 Gregg', 42), ('Antonio Braddock', 44), ('Clarence Keyser', 44), ('Jonathan Wagoner', 47), ('Elaine Elias', 48), ('Wayne Merkle', 51),
 ('Mike Rodarte', 51), ('Raymond Hynes', 51), ('Vickie Giffin', 51), ('William Backus', 51), ('Paul Albers', 52), ('Erik Weinman', 52),
 ('Teresa Booth', 52), ('Scott Luevano', 53), ('Opal Tobias', 53), ('Adam Harden', 53), ('Erma Creech', 61), ('Curtis Thrower', 61),
 ('Gabriela Keaney', 62), ('Jeremy Catalano', 62), ('Sherry Winfrey', 62), ('Debbie Ruch', 62), ('Chad Burbank', 62), ('Dustin Gadsden',
 67), ('Tony Brogan', 67), ('Sherri Kinser', 76), ('Christopher Merwin', 76), ('Fred Meeker', 76)]
Duration taken by QuickSort to sort 100 pickup requests in nearly sorted order: 0.000254 seconds
```

Figure 68: Nearly sorted list sorted

```
*************************************************************************
Sorting of 100 Reverse Pickup Requests using Quicksort
*************************************************************************

[('Michael Michel', 0), ('Arlene Bergmann', 0), ('Theresa Townsley', 0), ('Bessie Minton', 0), ('Jeffrey Keenan', 0), ('Martin Harbin', 0),
 ('Russell Ashmore', 0), ('Gregory Despain', 0), ('Philip Hammonds', 0), ('Deanna Arndt', 0), ('Emily Chadwick', 0), ('Jodi Holm', 0), ('Luz
  Coomer', 0), ('Gary Thorson', 0), ('Amy Bedford', 0), ('Howard Storm', 0), ('Shawn Rodriquez', 0), ('Eva Pursley', 0), ('Yvette Dowdy',
 0), ('Travis Llamas', 0), ('Nelson Tooker', 0), ('Stephen Fitz', 0), ('Jeremy Hyman', 0), ('Vivian Milton', 0), ('Sean Schubert', 0),
 ('Jorge Ventimiglia', 0), ('Dale Ormsby', 0), ('Jacob Meeker', 0), ('Chris Rahman', 0), ('Cody Basta', 0), ('Keith Dykes', 0), ('Jeff
 Singletary', 0), ('Noreen Durfey', 0), ('Annie Markel', 0), ('Eileen Bunch', 0), ('Peter Whitman', 0), ('Marilyn Mickens', 0), ('Joseph
 Favors', 0), ('Norman Styles', 0), ('Gary Burmeister', 7), ('Vernon Moultrie', 7), ('Timothy Congdon', 7), ('Benjamin Prado', 7), ('Karen
 Pena', 8), ('Orville Gillaspie', 8), ('Harold Hamill', 11), ('Allen Mckenna', 24), ('Bruce Dailey', 24), ('Cassandra Creed', 24), ('Kelly
 Cheatam', 24), ('Hazel Justice', 24), ('Larry Cantu', 24), ('Jack Kolb', 29), ('Chad Steffens', 29), ('Olivia Fall', 29), ('Manuel
 Hammack', 30), ('Harold Seaman', 30), ('Ronald Henning', 33), ('Eileen Haddock', 33), ('Richard Melvin', 35), ('Jocelyn Lynde', 35),
 ('Curtis Cordell', 35), ('Larry Chapa', 35), ('Wayne Crabb', 37), ('Denise Herrington', 38), ('Nelson Guitierrez', 38), ('Timothy Vetter',
 38), ('Jessica Villegas', 41), ('Jose Gregg', 42), ('Chad Doucet', 42), ('Dean Buchanon', 42), ('Phyllis Beene', 42), ('Jessie Efaw', 42),
 ('Clarence Keyser', 44), ('Antonio Braddock', 44), ('Jonathan Wagoner', 47), ('Elaine Elias', 48), ('Mike Rodarte', 51), ('Raymond Hynes',
 51), ('Vickie Giffin', 51), ('William Backus', 51), ('Wayne Merkle', 51), ('Paul Albers', 52), ('Teresa Booth', 52), ('Erik Weinman', 52),
 ('Scott Luevano', 53), ('Adam Harden', 53), ('Opal Tobias', 53), ('Curtis Thrower', 61), ('Erma Creech', 61), ('Gabriela Keaney', 62),
 ('Chad Burbank', 62), ('Debbie Ruch', 62), ('Sherry Winfrey', 62), ('Jeremy Catalano', 62), ('Tony Brogan', 67), ('Dustin Gadsden', 67),
 ('Fred Meeker', 76), ('Christopher Merwin', 76), ('Sherri Kinser', 76)]

Duration taken by QuickSort to sort 100 pickup requests in reverse order: 0.000250 seconds
```

Figure 69: Reverse list sorted

```
The Median of 3 QuickSort algorithm Reflection for different scenarios for 100 pickup requests:
Duration to sort 100 pickup requests in random order: 0.000167 seconds
Duration to sort 100 pickup requests in nearly sorted order: 0.000254 seconds
Duration to sort 100 pickup requests in reverse order: 0.000250 seconds

Looking at the durations for each scenario of 100 pickup request, QuickSort Median of 3 takes less time to
    sort the random list while more time is taken to sort the nearly_sorted list
Its time complexity is O(n log n) and its space complexity is Olog(n).
```

Figure 70: Short reflection on using Quick Sort for 100 pickup requests

Dataset of 500 pickup requests:

500 Pickup request list in Random order:

```
Test List of 500 pickup requests in random order
[('Dale Charlton', 51), ('Carmella Sheffey', 5), ('Paul Gowen', 11), ('Dorcas Abbitt', 0), ('Carolyn Kay', 0), ('Elaine Elias', 0), ('Dianne
  Mixon', 24), ('Joshua Spooner', 0), ('Jorge Madry', 62), ('James Urrutia', 0), ('Maribel Earnest', 0), ('Kurt Prowse', 0), ('Jeffery
 Ruiz', 17), ('Corey Devane', 24), ('Wilfred Mourning', 0), ('Deanna Arndt', 0), ('Shawn Leigh', 53), ('Jesse Horn', 17), ('Chris Reiner',
 76), ('Walter Roussel', 33), ('Michael Lefevre', 0), ('Heather Laureano', 0), ('Danielle Tincher', 0), ('Gary Poole', 35), ('Mabel Gist',
 0), ('Tyrone Ottman', 33), ('Joseph Linville', 0), ('Javier Weatherwax', 62), ('Larry Cantu', 0), ('Harold Hamill', 0), ('Rosemary
 Mcculley', 0), ('Eileen Bunch', 0), ('Kevin Schreiber', 0), ('Opal Tobias', 0), ('Clayton Salls', 0), ('Lillie Frisby', 0), ('Judith
 Lemay', 0), ('Kenneth Lewellen', 61), ('Luis Lehmann', 0), ('Louis Marino', 51), ('Willie Pence', 62), ('Dustin Gadsden', 0), ('Clara
 Wainwright', 51), ('Lester Gallivan', 29), ('Dennis Housley', 0), ('Keith Hyder', 5), ('Candice Swisher', 0), ('Dennis Gault', 0), ('Cindy
 Penney', 0), ('Sherry Winfrey', 0), ('Denise Herrington', 0), ('Rita Witkowski', 53), ('Elizabeth Dismuke', 29), ('Douglas Sedillo', 24),
 ('Teresa Samaniego', 33), ('Anthony Pires', 24), ('Stanley Littleton', 0), ('Russell Ashmore', 0), ('Ethel Reitz', 0), ('Nelson Tooker',
 0), ('Dale Ormsby', 0), ('Philip Hammonds', 0), ('Matthew Massa', 0), ('Tony Brogan', 0), ('Katy Tressler', 33), ('Bryan Corcoran', 51),
 ('Kelli Schulze', 0), ('Louis Schreiber', 14), ('Nicole Ellingson', 14), ('Jacob Dandrea', 76), ('Zachary Falgoust', 0), ('Scott Simons',
 35), ('Wilma Rey', 0), ('Chad Pardo', 0), ('Michael Michel', 0), ('Debbie Ruch', 0), ('Daniel Trimble', 0), ('Vernon Sermons', 0), ('Elsie
 Sowers', 0), ('Scott Prescott', 42), ('Max Kirkendall', 0), ('Belen Sheerin', 67), ('Shelia Sorrells', 61), ('Steve Ellingson', 38),
 ('Hilda Wilmoth', 53), ('Christopher Merwin', 0), ('Pearlie Storie', 14), ('Michele Ford', 61), ('Edward Reader', 0), ('Vivian Milton', 0),
  ('Arlene Wellington', 61), ('Cassandra Creed', 0), ('Wesley Harm', 0), ('Douglas Dabney', 0), ('Bruce Dailey', 0), ('Nelson Guitierrez',
 0), ('Albert Lockett', 0), ('Michele Valliere', 0), ('Chris Mcmorris', 44), ('Juana Greenleaf', 0), ('Olivia Fall', 0), ('Kristine Cave',
 29), ('Sofia Kelling', 53), ('Allen Velarde', 0), ('Corinne Kocsis', 0), ('Darren Hadaway', 0), ('Ingrid Mutter', 0), ('Roslyn Philipp',
 14), ('Roberto Munroe', 62), ('Becky Zinn', 0), ('Natalie Bynum', 29), ('Mark Tate', 41), ('Edward Scalise', 0), ('Ronald Rushton', 0),
 ('Eva Pursley', 0), ('Tonya Niemi', 0), ('Joshua Kingsbury', 5), ('Ramona Stacy', 0), ('Shelley Scherer', 29), ('Jacob Meeker', 0),
 ('Gerald Cousin', 0), ('Christopher List', 29), ('Donald Parmer', 5), ('Jose Gregg', 0), ('Jimmy Munoz', 53), ('Marjorie Tello', 0),
 ('Andrew Colunga', 41), ('Annie Markel', 0), ('Jack Kolb', 0), ('Gerald Whitesell', 17), ('Veronica Voss', 38), ('Kelly Cheatam', 0),
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Figure 71

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('Andrew Colunga', 41), ('Annie Markel', 0), ('Jack Kolb', 0), ('Gerald Whitesell', 17), ('Veronica Voss', 38), ('Kelly Cheatam', 0),
('Sonia Turgeon', 0), ('Mark Grimm', 5), ('Martin Harbin', 0), ('Joseph Favors', 0), ('Alberta Bradley', 53), ('Shane Dare', 38),
('Harriett Loveall', 0), ('Eunice Barr', 52), ('Matthew Will', 38), ('Gary Thorson', 0), ('Alice Reese', 52), ('Darren Suchan', 0), ('Roger
 Cool', 53), ('Leon Urbina', 0), ('Justin Hutson', 0), ('Allen Mckenna', 0), ('Steve Rushing', 0), ('Darren Pedrosa', 76), ('Johnny
Turnbull', 0), ('Julio Naccarato', 38), ('Cody Basta', 0), ('Orville Gillaspie', 0), ('Darlene Sandler', 17), ('Mike Rodarte', 0),
('Geraldine Flora', 5), ('Verla Pali', 76), ('Hugo Vogelgesang', 33), ('Roberto Routon', 14), ('Vernon Moultrie', 0), ('George Blow', 17),
('Sally Soule', 0), ('Ingrid Devaul', 33), ('Larry Chapa', 0), ('Keith Dykes', 0), ('Joseph Donohue', 53), ('Bobby Faris', 5), ('Norman
Mcnulty', 0), ('Sonia Bass', 0), ('Stephen Fitz', 0), ('Russell Treat', 0), ('Joseph Audet', 53), ('Lonnie Neveu', 5), ('Ted Yetman', 33),
('Jonathan Gower', 0), ('Stanley Bybee', 38), ('Michael Scheffler', 51), ('Lena Judge', 0), ('Steve Hix', 44), ('Noreen Durfey', 0),
('Larry Mitchel', 5), ('Russell Oshea', 0), ('Charlene Pennington', 35), ('Jacob Ingle', 0), ('Fernando Sobers', 35), ('Steven Ferrari',
0), ('Alma Mcneal', 0), ('Daphne Renz', 0), ('Jessie Mayville', 67), ('Nora Neel', 35), ('Justin Soloman', 0), ('Billy Partridge', 53),
('Travis Trull', 61), ('Yvette Scalise', 62), ('Elinor Raco', 0), ('Jeff Moten', 0), ('Tina Haugen', 5), ('Hector Horning', 0), ('Manuel
Hammack', 0), ('Scott Luevano', 0), ('Guy Winders', 76), ('Javier Hausman', 0), ('Walter Chapa', 5), ('Jamie Sing', 5), ('Elnora Coppedge',
 33), ('Alice Knoll', 0), ('Danny Corrigan', 0), ('Gary Burmeister', 0), ('Adam Bingham', 5), ('Cherie Mcguckin', 42), ('Richard Melvin',
0), ('Norman Styles', 0), ('Sharonda Remsberg', 17), ('Luz Coomer', 0), ('Gabriela Keaney', 0), ('Kayla Paynter', 0), ('Emily Chadwick',
0), ('Timothy Vetter', 0), ('Clarence Batchelder', 67), ('Carl Campos', 0), ('Scott Ring', 0), ('Anthony Deaver', 52), ('Bruce Rickert',
29), ('Christian Klemm', 0), ('Guy Gonsoulin', 38), ('Tanya Pickering', 52), ('Jeanette Herbst', 62), ('Allen Towns', 42), ('Eugene Singh',
 35), ('Patsy Horne', 0), ('Jack Tso', 52), ('Jo Patino', 0), ('Chad Maclean', 67), ('Harry Norfleet', 0), ('Arthur Larrabee', 0),
('Earnestine Croswell', 5), ('Christopher Biles', 0), ('Keith Hurt', 5), ('Lilia Rojero', 29), ('Harriett Curatolo', 0), ('Benjamin Prado',
 0), ('Lonnie Kitzmiller', 35), ('Jonathan Wagoner', 0), ('Anthony Ramey', 35), ('Esther Hale', 33), ('Diana Welker', 29), ('Howard Hake',
24), ('Juan Hermes', 42), ('Louise Lively', 0), ('Inez Yee', 0), ('Javier Kratochvil', 0), ('Amy Bedford', 0), ('Dianne Alarcon', 44),
('Keisha Ragusa', 0), ('Harold Andre', 5), ('Beatrice Rose', 35), ('Mathew Mercedes', 42), ('Theresa Townsley', 0), ('Jeffrey Marcum', 0),
('Vickie Giffin', 0), ('Howard Storm', 0), ('Lester Kealey', 0), ('Alicia Womble', 0), ('Lois Sexton', 11), ('Jesse Sandstrom', 14),
('Jason Word', 5), ('Carmella Vanwagenen', 0), ('Lance Buono', 33), ('Jodi Holm', 0), ('Doris Mays', 14), ('Rick Brathwaite', 0), ('Leonard
 Urbina', 14), ('Max Madamba', 24), ('Wayne Merkle', 0), ('Yvette Dowdy', 0), ('Cindy Motley', 14), ('Todd Huss', 19), ('Velma Burnett',
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Figure 72

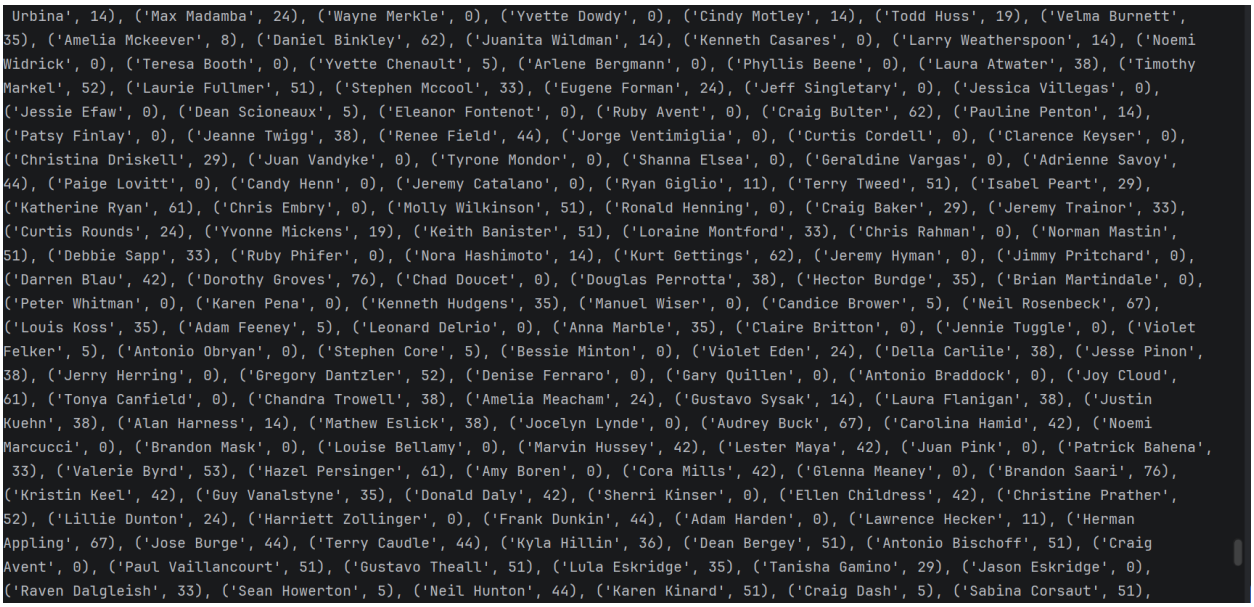


Figure 73



Figure 74

500 pickup request list in nearly sorted order:

[('Chad Pardo', 0), ('Noemi Widrick', 0), ('Keisha Ragusa', 0), ('Cody Basta', 0), ('Valerie Jenson', 67), ('Jorge Ventimiglia', 0), ('Louise Bellamy', 0), ('Vernon Moultrie', 0), ('Asia Gucman', 35), ('Earnestine Croswell', 5), ('Maribel Earnest', 0), ('Howard Storm', 0), ('Ramona Stacy', 0), ('Jodi Holm', 0), ('Lance Buono', 33), ('Gregory Despain', 0), ('Alicia Womble', 0), ('Bessie Minton', 0), ('Paige Lovitt', 0), ('Michael Michel', 0), ('Brandi

Mcauliffe', 0), ('Benjamin Prado', 0), ('Inez Yee', 0), ('Peter Whitman', 0), ('Patsy Finlay', 0), ('Marilyn Mickens', 0), ('Joseph Linville', 0), ('Noreen Durfey', 0), ('Arthur Larrabee', 0), ('Harold Hamill', 0), ('Rosemary Mcculley', 0), ('Eileen Bunch', 0), ('Kevin Schreiber', 0), ('Jessica Villegas', 0), ('Clayton Salls', 0), ('Luz Coomer', 0), ('Wilma Rey', 0), ('Emily Chadwick', 0), ('Luis Lehmann', 0), ('Philip Hammonds', 0), ('Leonard Delrio', 0), ('Russell Ashmore', 0), ('Craig Avent', 0), ('Manuel Hammack', 0), ('Javier Hausman', 0), ('Dale Ormsby', 0), ('Candice Swisher', 0), ('Jonathan Wagoner', 0), ('Scott Ring', 0), ('Teresa Booth', 0), ('Harry Norfleet', 0), ('Jose Gregg', 0), ('Tyrone Mondor', 0), ('Karen Pena', 0), ('Mabel Gist', 0), ('Chad Steffens', 0), ('Louise Lively', 0), ('Dustin Gadsden', 0), ('Ethel Reitz', 0), ('Kelly Cheatam', 0), ('Judith Lemay', 0), ('Scott Luevano', 0), ('Matthew Massa', 0), ('Opal Tobias', 0), ('Jerry Herring', 0), ('Jessie Efaw', 0), ('Kelli Schulze', 0), ('Chad Doucet', 0), ('Johnny Dozier', 0), ('Bruce Dailey', 0), ('Zachary Falgoust', 0), ('Erik Weinman', 0), ('Cindy Penney', 0), ('Olivia Fall', 0), ('Dennis Housley', 0), ('Mark Grimm', 5), ('Daniel Trimble', 0), ('Sherri Kinser', 0), ('Elsie Sowers', 0), ('Vickie Giffin', 0), ('Joy Cloud', 61), ('Jeremy Catalano', 0), ('Wesley Harm', 0), ('Ronald Henning', 0), ('Lester Kealey', 0), ('Christopher Merwin', 0), ('Claire Britton', 0), ('Timothy Vetter', 0), ('Edward Reader', 0), ('Clarence Keyser', 0), ('Juan Vandyke', 0), ('Erma Creech', 0), ('Hector Horning', 0), ('Nelson Guitierrez', 0), ('Nelson Coltrane', 35), ('Ruby Phifer', 0), ('Albert Lockett', 0), ('Michele Valliere', 0), ('Jeff Vanbuskirk', 0), ('Ruby Avent', 0), ('Manuel Wiser', 0), ('Danny Corrigan', 0), ('Travis Trull', 61), ('Allen Velarde', 0), ('Corinne Kocsis', 0), ('Darren Hadaway', 0), ('Ingrid Mutter', 0), ('Rick Brathwaite', 0), ('Noemi Marcucci', 0), ('Brandon Mask', 0), ('Danielle Tincher', 0), ('Amy Boren', 0), ('Edward Scalise', 0), ('Carl Campos', 0), ('Antonio Obryan', 0), ('Tonya Niemi', 0), ('Alice Knoll', 0), ('George Manchester', 0), ('Gary Quillen', 0), ('Jacob Meeker', 0), ('Kenneth Casares', 0), ('Stephen Fitz', 0), ('Lillie Frisby', 0), ('Gary Thorson', 0), ('Jacob Ingle', 0), ('Martin Harbin', 0), ('Alma Mcneal', 0), ('Annie Markel', 0), ('Becky Zinn', 0), ('Nelson Tooker', 0), ('Harriett Curatolo', 0), ('Gary Burmeister', 0), ('Janice Granados', 0), ('Amy Bedford', 0), ('Jeff Moten', 0), ('Joseph Favors', 0), ('Michael Lefevre', 0), ('Norman Styles', 0), ('Harriett Loveall', 0), ('Wayne Crabb', 0), ('Jennie Tuggle', 0), ('Dean Buchanon', 0), ('Juana Greenleaf', 0), ('Theresa Townsley', 0), ('Ellen Millican', 0), ('Adam Harden', 0), ('Justin Hutson', 0), ('Allen Mckenna', 0), ('Adam Bingham', 5), ('Cassandra Creed', 0), ('Johnny Turnbull', 0), ('Jocelyn Lynde', 0), ('Eva Pursley', 0), ('Larry Chapa', 0), ('Chris Embry', 0), ('Mike Rodarte', 0), ('Steve Rushing', 0), ('William Backus', 0), ('Justin Soloman', 0), ('Denise Herrington', 0), ('Jimmy Pritchard', 0), ('Sherry Winfrey', 0), ('Sally Soule', 0), ('Wayne Merkle', 0), ('Leon Urbina', 0), ('Harriett Zollinger', 0), ('Jessie Reta', 0), ('Stanley Littleton', 0), ('Norman Mcnulty', 0), ('Sonia Bass', 0), ('Jason Eskridge', 0), ('Eleanor Fontenot', 0), ('Christopher Biles', 0), ('Douglas Dabney', 0), ('Ronald Rushton', 0), ('Jonathan Gower', 0), ('Carmella Vanwagenen', 0), ('Vivian Milton', 0), ('Lena Judge', 0), ('Deanna Arndt', 0), ('Kurt Prowse', 0), ('Jeremy Hyman', 0), ('Russell Oshea', 0), ('Shawn Rodriquez', 0), ('Glenna Meaney', 0), ('Chris Rahman', 0), ('Stephanie Shanks', 0), ('Pearlie Storie', 14), ('Daphne Renz', 0), ('Jose Flannery', 67), ('Joshua Spooner', 0), ('Paul Albers', 0),

('Russell Treat', 0), ('Larry Cantu', 0), ('Sonia Turgeon', 0), ('Debbie Ruch', 0), ('Elinor Raco', 0), ('Raymond Hynes', 0), ('James Urrutia', 0), ('Antonio Braddock', 0), ('Juan Pink', 0), ('Daisy Palladino', 0), ('Joyce Holtz', 0), ('Candy Henn', 0), ('Jack Kolb', 0), ('Jeff Singletary', 0), ('Dennis Gault', 0), ('Arlene Bergmann', 0), ('Steven Ferrari', 0), ('Keith Dykes', 0), ('Christine King', 0), ('Orville Gillaspie', 0), ('Gerald Cousin', 0), ('Curtis Cordell', 0), ('Jeffrey Marcum', 0), ('Gabriela Keaney', 0), ('Denise Ferraro', 0), ('Darren Suchan', 0), ('Earnestine Simo', 0), ('Marjorie Tello', 0), ('Heather Laureano', 0), ('Yvette Dowdy', 0), ('Vernon Sermons', 0), ('Tony Brogan', 0), ('Christian Klemm', 0), ('Fred Meeker', 0), ('Javier Kratochvil', 0), ('Dorcas Abbitt', 0), ('Darren Cotta', 0), ('Gerald Whitesell', 17), ('Patsy Horne', 0), ('Harold Seaman', 0), ('Jo Patino', 0), ('Jamie Sing', 5), ('Jason Word', 5), ('Keith Hyder', 5), ('Keith Hurt', 5), ('Geraldine Flora', 5), ('Timothy Congdon', 0), ('Dean Scioneaux', 5), ('Donald Parmer', 5), ('Tina Haugen', 5), ('Larry Mitchel', 5), ('Joshua Kingsbury', 5), ('Violet Felker', 5), ('Candice Brower', 5), ('Stephen Core', 5), ('Carmella Sheffey', 5), ('Zelma Vallier', 14), ('Arthur Runge', 52), ('Roger Pellerin', 42), ('Bobby Faris', 5), ('Richard Melvin', 0), ('Yvette Chenault', 5), ('Lonnie Neveu', 5), ('Guy Vanalstyne', 35), ('Craig Dash', 5), ('Brian Martindale', 0), ('Larry Lamp', 7), ('Carole Stuckey', 8), ('Amelia Mckeever', 8), ('Paul Gowen', 11), ('Ryan Giglio', 11), ('Lawrence Hecker', 11), ('Lois Sexton', 11), ('Jesse Sandstrom', 14), ('Nora Hashimoto', 14), ('Ingrid Devaul', 33), ('Hillary Tall', 14), ('Adam Feeney', 5), ('Doris Mays', 14), ('Roslyn Philipp', 14), ('Leonard Urbina', 14), ('Pauline Penton', 14), ('Gustavo Sysak', 14), ('Roberto Routon', 14), ('Cindy Motley', 14), ('Norman Wingfield', 14), ('Elaine Elias', 0), ('Alan Harness', 14), ('Clarence Correll', 14), ('Juanita Wildman', 14), ('Louis Schreiber', 14), ('Larry Weatherspoon', 14), ('Darlene Sandler', 17), ('George Blow', 17), ('Shanna Elsea', 0), ('Jesse Horn', 17), ('Sharonda Remsberg', 17), ('Jeffery Ruiz', 17), ('Brandi Keil', 19), ('Todd Huss', 19), ('Yvonne Mickens', 19), ('Eugene Forman', 24), ('Violet Eden', 24), ('Hilda Wilmoth', 53), ('Daniel Deloach', 24), ('Corey Devane', 24), ('Anthony Pires', 24), ('Howard Hake', 24), ('Dianne Mixon', 24), ('Kristy Strom', 24), ('Curtis Rounds', 24), ('Max Madamba', 24), ('Douglas Sedillo', 24), ('Curtis Bowyer', 24), ('Amelia Meacham', 24), ('Kristine Cave', 29), ('Christina Driskell', 29), ('Diana Welker', 29), ('Elizabeth Dismuke', 29), ('Fannie Hesse', 29), ('Craig Baker', 29), ('Lester Gallivan', 29), ('Bruce Rickert', 29), ('Lilia Rojero', 29), ('Christopher List', 29), ('Natalie Bynum', 29), ('Tanisha Gamino', 29), ('Isabel Peart', 29), ('Shelley Scherer', 29), ('Hugo Vogelgesang', 33), ('Raven Dalgleish', 33), ('Esther Hale', 33), ('Tyrone Ottman', 33), ('Jeremy Trainor', 33), ('Teresa Samaniego', 33), ('Ted Yetman', 33), ('Stephen Mccool', 33), ('Loraine Montford', 33), ('Elnora Coppedge', 33), ('Walter Roussel', 33), ('Debbie Sapp', 33), ('Patrick Bahena', 33), ('Katy Tressler', 33), ('Wilfred Mourning', 0), ('Selena Lenart', 33), ('Nicole Ellingson', 14), ('Charlene Pennington', 35), ('Eugene Singh', 35), ('Kayla Paynter', 0), ('Beatrice Rose', 35), ('Hector Burdge', 35), ('Lonnie Kitzmiller', 35), ('Paul Vaillancourt', 51), ('Juan Hermes', 42), ('Kenneth Hudgens', 35), ('Scott Simons', 35), ('Gary Tate', 35), ('Fernando Sobers', 35), ('Louis Koss', 35), ('Harold Andre', 5), ('Tonya Canfield', 0), ('Anna Marble', 35), ('Velma Burnett', 35), ('Nora Neel', 35), ('Anthony Ramey', 35),

('Yvonne Mccune', 35), ('Lula Eskridge', 35), ('Jill Dickerson', 36), ('Kyla Hillin', 36), ('Della Carlile', 38), ('Guy Gonsoulin', 38), ('Matthew Will', 38), ('Laura Atwater', 38), ('Veronica Voss', 38), ('Jeanne Twigg', 38), ('Douglas Perrotta', 38), ('Jesse Pinon', 38), ('Verna Basso', 38), ('Chandra Trowell', 38), ('Shane Dare', 38), ('Stanley Bybee', 38), ('Laura Flanigan', 38), ('Justin Kuehn', 38), ('Christine Prather', 52), ('Mathew Eslick', 38), ('Steve Ellingson', 38), ('Julio Naccarato', 38), ('Mark Tate', 41), ('Andrew Colunga', 41), ('Lester Maya', 42), ('Terry Brush', 42), ('Marvin Hussey', 42), ('Cherie Mcguckin', 42), ('Darren Blau', 42), ('Mathew Mercedes', 42), ('Tara Nissen', 42), ('Scott Prescott', 42), ('Carolina Hamid', 42), ('Cora Mills', 42), ('Roger Arriola', 42), ('Robert Cooks', 35), ('Kristin Keel', 42), ('Sean Howerton', 5), ('Donald Daly', 42), ('Allen Towns', 42), ('Ellen Childress', 42), ('Renee Field', 44), ('Dianne Alarcon', 44), ('Joyce Etter', 44), ('Frank Dunkin', 44), ('Chris Mcmorris', 44), ('Steve Hix', 44), ('Neil Hunton', 44), ('Jose Burge', 44), ('Terry Caudle', 44), ('Adrienne Savoy', 44), ('Dean Bergey', 51), ('Antonio Bischoff', 51), ('Clara Wainwright', 51), ('Gary Poole', 35), ('Gustavo Theall', 51), ('Bryan Corcoran', 51), ('Terry Tweed', 51), ('Dale Charlton', 51), ('Keith Banister', 51), ('Louis Marino', 51), ('Norman Mastin', 51), ('Karen Kinard', 51), ('Molly Wilkinson', 51), ('Sabina Corsaut', 51), ('Marguerite Mccabe', 51), ('Dan Lain', 51), ('Michael Scheffler', 51), ('Laurie Fullmer', 51), ('Timothy Markel', 52), ('Amelia Jarrell', 76), ('Alice Reese', 52), ('Eunice Barr', 52), ('Mike Schumann', 52), ('Anthony Deaver', 52), ('Harry Skeen', 52), ('Tanya Pickering', 52), ('Lorraine Ake', 38), ('Gregory Dantzler', 52), ('Walter Chapa', 5), ('Rita Witkowski', 53), ('Joseph Donohue', 53), ('Shawn Leigh', 53), ('Sofia Kelling', 53), ('Roger Cool', 53), ('Billy Partridge', 53), ('Mark Dillman', 53), ('Joseph Audet', 53), ('Jimmy Munoz', 53), ('Valerie Byrd', 53), ('Alberta Bradley', 53), ('Lillie Dunton', 24), ('Vincent Berkley', 53), ('Luz Hong', 53), ('Max Kirkendall', 0), ('Arlene Wellington', 61), ('Kenneth Lewellen', 61), ('Billy Luu', 61), ('Philip Gauthier', 61), ('Hazel Persinger', 61), ('Darlene Bergman', 61), ('Geraldine Vargas', 0), ('Katherine Ryan', 61), ('Michele Ford', 61), ('Susie Peoples', 61), ('Shelia Sorrells', 61), ('Jeffery Bounds', 61), ('Craig Bulter', 62), ('Kurt Gettings', 62), ('Ruby Traylor', 62), ('Walter Sappington', 62), ('Daniel Binkley', 62), ('Willie Pence', 62), ('Jorge Madry', 62), ('Matthew Beaman', 62), ('Yvette Scalise', 62), ('Javier Weatherwax', 62), ('Rodney Toth', 62), ('Roberto Munroe', 62), ('Jeanette Herbst', 62), ('Brent Curtsinger', 62), ('Pamela French', 67), ('Carolyn Kay', 0), ('Nereida Viscardi', 67), ('Audrey Buck', 67), ('Neil Rosenbeck', 67), ('Chad Maclean', 67), ('Sarah Waterman', 67), ('Jessie Mayville', 67), ('Belen Sheerin', 67), ('Christian Durall', 67), ('Phyllis Beene', 0), ('Clarence Batchelder', 67), ('Herman Appling', 67), ('Dennis Donahue', 76), ('Chris Reiner', 76), ('Dorothy Groves', 76), ('Christa Monsour', 76), ('Guy Winders', 76), ('Eugene Gore', 76), ('Cody Russom', 76), ('Jacob Dandrea', 76), ('Thomas Giffin', 76), ('Verla Pali', 76), ('Brandon Saari', 76), ('Darren Pedrosa', 76), ('Clara Huck', 76), ('Jack Tso', 52)]

500 pickup request list in reverse order:

[('Amelia Jarrell', 76), ('Clara Huck', 76), ('Darren Pedrosa', 76), ('Brandon Saari', 76), ('Verla Pali', 76), ('Thomas Giffin', 76), ('Jacob Dandrea', 76), ('Cody Russom', 76), ('Eugene Gore', 76), ('Guy Winders', 76), ('Christa Monsour', 76), ('Dorothy Groves', 76), ('Chris Reiner', 76), ('Dennis Donahue', 76), ('Herman Appling', 67), ('Clarence Batchelder', 67), ('Jose Flannery', 67), ('Christian Durall', 67), ('Belen Sheerin', 67), ('Jessie Mayville', 67), ('Sarah Waterman', 67), ('Chad Maclean', 67), ('Neil Rosenbeck', 67), ('Audrey Buck', 67), ('Nereida Viscardi', 67), ('Valerie Jenson', 67), ('Pamela French', 67), ('Brent Curtsinger', 62), ('Jeanette Herbst', 62), ('Roberto Munroe', 62), ('Rodney Toth', 62), ('Javier Weatherwax', 62), ('Yvette Scalise', 62), ('Matthew Beaman', 62), ('Jorge Madry', 62), ('Willie Pence', 62), ('Daniel Binkley', 62), ('Walter Sappington', 62), ('Ruby Traylor', 62), ('Kurt Gettings', 62), ('Craig Bulter', 62), ('Jeffery Bounds', 61), ('Shelia Sorrells', 61), ('Susie Peoples', 61), ('Michele Ford', 61), ('Katherine Ryan', 61), ('Travis Trull', 61), ('Darlene Bergman', 61), ('Hazel Persinger', 61), ('Philip Gauthier', 61), ('Billy Luu', 61), ('Kenneth Lewellen', 61), ('Arlene Wellington', 61), ('Joy Cloud', 61), ('Luz Hong', 53), ('Vincent Berkley', 53), ('Hilda Wilmoth', 53), ('Alberta Bradley', 53), ('Valerie Byrd', 53), ('Jimmy Munoz', 53), ('Joseph Audet', 53), ('Mark Dillman', 53), ('Billy Partridge', 53), ('Roger Cool', 53), ('Sofia Kelling', 53), ('Shawn Leigh', 53), ('Joseph Donohue', 53), ('Rita Witkowski', 53), ('Arthur Runge', 52), ('Gregory Dantzler', 52), ('Christine Prather', 52), ('Tanya Pickering', 52), ('Harry Skeen', 52), ('Anthony Deaver', 52), ('Mike Schumann', 52), ('Eunice Barr', 52), ('Alice Reese', 52), ('Jack Tso', 52), ('Timothy Markel', 52), ('Laurie Fullmer', 51), ('Michael Scheffler', 51), ('Dan Lain', 51), ('Marguerite Mccabe', 51), ('Sabina Corsaut', 51), ('Molly Wilkinson', 51), ('Karen Kinard', 51), ('Norman Mastin', 51), ('Louis Marino', 51), ('Keith Banister', 51), ('Dale Charlton', 51), ('Terry Tweed', 51), ('Bryan Corcoran', 51), ('Gustavo Theall', 51), ('Paul Vaillancourt', 51), ('Clara Wainwright', 51), ('Antonio Bischoff', 51), ('Dean Bergey', 51), ('Adrienne Savoy', 44), ('Terry Caudle', 44), ('Jose Burge', 44), ('Neil Hunton', 44), ('Steve Hix', 44), ('Chris Mcmorris', 44), ('Frank Dunkin', 44), ('Joyce Etter', 44), ('Dianne Alarcon', 44), ('Renee Field', 44), ('Ellen Childress', 42), ('Allen Towns', 42), ('Donald Daly', 42), ('Roger Pellerin', 42), ('Kristin Keel', 42), ('Juan Hermes', 42), ('Roger Arriola', 42), ('Cora Mills', 42), ('Carolina Hamid', 42), ('Scott Prescott', 42), ('Tara Nissen', 42), ('Mathew Mercedes', 42), ('Darren Blau', 42), ('Cherie Mcguckin', 42), ('Marvin Hussey', 42), ('Terry Brush', 42), ('Lester Maya', 42), ('Andrew Colunga', 41), ('Mark Tate', 41), ('Julio Naccarato', 38), ('Steve Ellingson', 38), ('Mathew Eslick', 38), ('Lorraine Ake', 38), ('Justin Kuehn', 38), ('Laura Flanigan', 38), ('Stanley Bybee', 38), ('Shane Dare', 38), ('Chandra Trowell', 38), ('Verna Basso', 38), ('Jesse Pinon', 38), ('Douglas Perrotta', 38), ('Jeanne Twigg', 38), ('Veronica Voss', 38), ('Laura Atwater', 38), ('Matthew Will', 38), ('Guy Gonsoulin', 38), ('Della Carlile', 38), ('Kyla Hillin', 36), ('Jill Dickerson', 36), ('Lula Eskridge', 35), ('Yvonne Mccune', 35), ('Anthony Ramey', 35), ('Nora Neel', 35), ('Velma Burnett', 35), ('Anna Marble', 35), ('Nelson Coltrane', 35), ('Guy Vanalstyne', 35), ('Louis Koss', 35), ('Fernando Sobers', 35), ('Gary Tate', 35), ('Scott Simons', 35), ('Kenneth Hudgens', 35), ('Robert Cooks', 35), ('Gary Poole', 35), ('Lonnie Kitzmiller', 35), ('Hector Burdge', 35),

('Beatrice Rose', 35), ('Asia Gucman', 35), ('Eugene Singh', 35), ('Charlene Pennington', 35), ('Ingrid Devaul', 33), ('Selena Lenart', 33), ('Lance Buono', 33), ('Katy Tressler', 33), ('Patrick Bahena', 33), ('Debbie Sapp', 33), ('Walter Roussel', 33), ('Elnora Coppedge', 33), ('Loraine Montford', 33), ('Stephen Mccool', 33), ('Ted Yetman', 33), ('Teresa Samaniego', 33), ('Jeremy Trainor', 33), ('Tyrone Ottman', 33), ('Esther Hale', 33), ('Raven Dalgleish', 33), ('Hugo Vogelgesang', 33), ('Shelley Scherer', 29), ('Isabel Peart', 29), ('Tanisha Gamino', 29), ('Natalie Bynum', 29), ('Christopher List', 29), ('Lilia Rojero', 29), ('Bruce Rickert', 29), ('Lester Gallivan', 29), ('Craig Baker', 29), ('Fannie Hesse', 29), ('Elizabeth Dismuke', 29), ('Diana Welker', 29), ('Christina Driskell', 29), ('Kristine Cave', 29), ('Amelia Meacham', 24), ('Curtis Bowyer', 24), ('Douglas Sedillo', 24), ('Max Madamba', 24), ('Curtis Rounds', 24), ('Kristy Strom', 24), ('Dianne Mixon', 24), ('Howard Hake', 24), ('Anthony Pires', 24), ('Corey Devane', 24), ('Daniel Deloach', 24), ('Lillie Dunton', 24), ('Violet Eden', 24), ('Eugene Forman', 24), ('Yvonne Mickens', 19), ('Todd Huss', 19), ('Brandi Keil', 19), ('Jeffery Ruiz', 17), ('Sharonda Remsberg', 17), ('Jesse Horn', 17), ('Gerald Whitesell', 17), ('George Blow', 17), ('Darlene Sandler', 17), ('Larry Weatherspoon', 14), ('Louis Schreiber', 14), ('Juanita Wildman', 14), ('Clarence Correll', 14), ('Alan Harness', 14), ('Pearlie Storie', 14), ('Norman Wingfield', 14), ('Cindy Motley', 14), ('Roberto Routon', 14), ('Gustavo Sysak', 14), ('Pauline Penton', 14), ('Leonard Urbina', 14), ('Roslyn Philipp', 14), ('Doris Mays', 14), ('Zelma Vallier', 14), ('Hillary Tall', 14), ('Nicole Ellingson', 14), ('Nora Hashimoto', 14), ('Jesse Sandstrom', 14), ('Lois Sexton', 11), ('Lawrence Hecker', 11), ('Ryan Giglio', 11), ('Paul Gowen', 11), ('Amelia Mckeever', 8), ('Carole Stuckey', 8), ('Larry Lamp', 7), ('Adam Bingham', 5), ('Craig Dash', 5), ('Harold Andre', 5), ('Lonnie Neveu', 5), ('Yvette Chenault', 5), ('Mark Grimm', 5), ('Bobby Faris', 5), ('Sean Howerton', 5), ('Walter Chapa', 5), ('Adam Feeney', 5), ('Carmella Sheffey', 5), ('Stephen Core', 5), ('Candice Brower', 5), ('Violet Felker', 5), ('Joshua Kingsbury', 5), ('Larry Mitchel', 5), ('Tina Haugen', 5), ('Donald Parmer', 5), ('Dean Scioneaux', 5), ('Earnestine Croswell', 5), ('Geraldine Flora', 5), ('Keith Hurt', 5), ('Keith Hyder', 5), ('Jason Word', 5), ('Jamie Sing', 5), ('Jo Patino', 0), ('Harold Seaman', 0), ('Patsy Horne', 0), ('Jack Kolb', 0), ('Darren Cotta', 0), ('Eva Pursley', 0), ('Javier Kratochvil', 0), ('Fred Meeker', 0), ('Christian Klemm', 0), ('Tony Brogan', 0), ('Vernon Sermons', 0), ('Yvette Dowdy', 0), ('Heather Laureano', 0), ('Marjorie Tello', 0), ('Earnestine Simo', 0), ('Darren Suchan', 0), ('Denise Ferraro', 0), ('Gabriela Keaney', 0), ('Jeffrey Marcum', 0), ('Curtis Cordell', 0), ('Gerald Cousin', 0), ('Orville Gillaspie', 0), ('Christine King', 0), ('Keith Dykes', 0), ('Steven Ferrari', 0), ('Arlene Bergmann', 0), ('Dennis Gault', 0), ('Jeff Singletary', 0), ('Shanna Elsea', 0), ('Candy Henn', 0), ('Joyce Holtz', 0), ('Daisy Palladino', 0), ('Juan Pink', 0), ('Antonio Braddock', 0), ('James Urrutia', 0), ('Raymond Hynes', 0), ('Elinor Raco', 0), ('Debbie Ruch', 0), ('Sonia Turgeon', 0), ('Larry Cantu', 0), ('Russell Treat', 0), ('Paul Albers', 0), ('Joshua Spooner', 0), ('Phyllis Beene', 0), ('Daphne Renz', 0), ('Elaine Elias', 0), ('Stephanie Shanks', 0), ('Chris Rahman', 0), ('Glenna Meaney', 0), ('Shawn Rodriquez', 0), ('Russell Oshea', 0), ('Jeremy Hyman', 0), ('Kurt Prowse', 0), ('Deanna Arndt', 0), ('Lena Judge', 0), ('Vivian Milton', 0), ('Carmella Vanwagenen', 0), ('Jonathan Gower', 0), ('Ronald

Rushton', 0), ('Douglas Dabney', 0), ('Christopher Biles', 0), ('Eleanor Fontenot', 0), ('Jason Eskridge', 0), ('Sonia Bass', 0), ('Norman Mcnulty', 0), ('Stanley Littleton', 0), ('Jessie Reta', 0), ('Harriett Zollinger', 0), ('Leon Urbina', 0), ('Wayne Merkle', 0), ('Sally Soule', 0), ('Sherry Winfrey', 0), ('Jimmy Pritchard', 0), ('Denise Herrington', 0), ('Justin Soloman', 0), ('Nelson Guitierrez', 0), ('Steve Rushing', 0), ('Mike Rodarte', 0), ('Chris Embry', 0), ('Larry Chapa', 0), ('Dorcas Abbitt', 0), ('Jocelyn Lynde', 0), ('Johnny Turnbull', 0), ('Cassandra Creed', 0), ('Brian Martindale', 0), ('Allen Mckenna', 0), ('Justin Hutson', 0), ('Adam Harden', 0), ('Ellen Millican', 0), ('Theresa Townsley', 0), ('Juana Greenleaf', 0), ('Dean Buchanon', 0), ('Jennie Tuggle', 0), ('Wayne Crabb', 0), ('Harriett Loveall', 0), ('Norman Styles', 0), ('Michael Lefevre', 0), ('Joseph Favors', 0), ('Jeff Moten', 0), ('Amy Bedford', 0), ('Janice Granados', 0), ('Gary Burmeister', 0), ('Harriett Curatolo', 0), ('Nelson Tooker', 0), ('Becky Zinn', 0), ('Annie Markel', 0), ('Alma Mcneal', 0), ('Martin Harbin', 0), ('Jacob Ingle', 0), ('Gary Thorson', 0), ('Lillie Frisby', 0), ('Stephen Fitz', 0), ('Kenneth Casares', 0), ('Jacob Meeker', 0), ('Gary Quillen', 0), ('George Manchester', 0), ('Alice Knoll', 0), ('Tonya Niemi', 0), ('Antonio Obryan', 0), ('Carl Campos', 0), ('Edward Scalise', 0), ('Amy Boren', 0), ('Danielle Tincher', 0), ('Brandon Mask', 0), ('Noemi Marcucci', 0), ('Rick Brathwaite', 0), ('Ingrid Mutter', 0), ('Darren Hadaway', 0), ('Corinne Kocsis', 0), ('Allen Velarde', 0), ('Geraldine Vargas', 0), ('Danny Corrigan', 0), ('Manuel Wiser', 0), ('Ruby Avent', 0), ('Jeff Vanbuskirk', 0), ('Michele Valliere', 0), ('Albert Lockett', 0), ('Ruby Phifer', 0), ('Tonya Canfield', 0), ('William Backus', 0), ('Hector Horning', 0), ('Erma Creech', 0), ('Juan Vandyke', 0), ('Clarence Keyser', 0), ('Edward Reader', 0), ('Timothy Vetter', 0), ('Claire Britton', 0), ('Christopher Merwin', 0), ('Lester Kealey', 0), ('Ronald Henning', 0), ('Wesley Harm', 0), ('Jeremy Catalano', 0), ('Max Kirkendall', 0), ('Vickie Giffin', 0), ('Elsie Sowers', 0), ('Sherri Kinser', 0), ('Daniel Trimble', 0), ('Richard Melvin', 0), ('Dennis Housley', 0), ('Olivia Fall', 0), ('Cindy Penney', 0), ('Erik Weinman', 0), ('Zachary Falgoust', 0), ('Bruce Dailey', 0), ('Johnny Dozier', 0), ('Chad Doucet', 0), ('Kelli Schulze', 0), ('Jessie Efaw', 0), ('Jerry Herring', 0), ('Opal Tobias', 0), ('Matthew Massa', 0), ('Scott Luevano', 0), ('Judith Lemay', 0), ('Kelly Cheatam', 0), ('Ethel Reitz', 0), ('Dustin Gadsden', 0), ('Louise Lively', 0), ('Chad Steffens', 0), ('Mabel Gist', 0), ('Karen Pena', 0), ('Tyrone Mondor', 0), ('Jose Gregg', 0), ('Harry Norfleet', 0), ('Teresa Booth', 0), ('Scott Ring', 0), ('Jonathan Wagoner', 0), ('Candice Swisher', 0), ('Dale Ormsby', 0), ('Javier Hausman', 0), ('Manuel Hammack', 0), ('Craig Avent', 0), ('Russell Ashmore', 0), ('Leonard Delrio', 0), ('Philip Hammonds', 0), ('Luis Lehmann', 0), ('Emily Chadwick', 0), ('Wilma Rey', 0), ('Luz Coomer', 0), ('Clayton Salls', 0), ('Jessica Villegas', 0), ('Kevin Schreiber', 0), ('Eileen Bunch', 0), ('Rosemary Mcculley', 0), ('Harold Hamill', 0), ('Arthur Larrabee', 0), ('Noreen Durfey', 0), ('Joseph Linville', 0), ('Marilyn Mickens', 0), ('Patsy Finlay', 0), ('Peter Whitman', 0), ('Inez Yee', 0), ('Benjamin Prado', 0), ('Kayla Paynter', 0), ('Michael Michel', 0), ('Paige Lovitt', 0), ('Bessie Minton', 0), ('Alicia Womble', 0), ('Gregory Despain', 0), ('Wilfred Mourning', 0), ('Jodi Holm', 0), ('Ramona Stacy', 0), ('Howard Storm', 0), ('Maribel Earnest', 0), ('Timothy Congdon', 0), ('Brandi

Mcauliffe', 0), ('Vernon Moultrie', 0), ('Louise Bellamy', 0), ('Jorge Ventimiglia', 0), ('Carolyn Kay', 0), ('Cody Basta', 0), ('Keisha Ragusa', 0), ('Noemi Widrick', 0), ('Chad Pardo', 0)]

Merge Sort for 500 pickup requests:

Random order sorted

Note: The first names are not the same for the different test scenarios as the lists are sorted using only the pickup times not the names.

[('Dorcas Abbitt', 0), ('Carolyn Kay', 0), ('Elaine Elias', 0), ('Joshua Spooner', 0), ('James Urrutia', 0), ('Maribel Earnest', 0), ('Kurt Prowse', 0), ('Wilfred Mourning', 0), ('Deanna Arndt', 0), ('Michael Lefevre', 0), ('Heather Laureano', 0), ('Danielle Tincher', 0), ('Mabel Gist', 0), ('Joseph Linville', 0), ('Larry Cantu', 0), ('Harold Hamill', 0), ('Rosemary Mcculley', 0), ('Eileen Bunch', 0), ('Kevin Schreiber', 0), ('Opal Tobias', 0), ('Clayton Salls', 0), ('Lillie Frisby', 0), ('Judith Lemay', 0), ('Luis Lehmann', 0), ('Dustin Gadsden', 0), ('Dennis Housley', 0), ('Candice Swisher', 0), ('Dennis Gault', 0), ('Cindy Penney', 0), ('Sherry Winfrey', 0), ('Denise Herrington', 0), ('Stanley Littleton', 0), ('Russell Ashmore', 0), ('Ethel Reitz', 0), ('Nelson Tooker', 0), ('Dale Ormsby', 0), ('Philip Hammonds', 0), ('Matthew Massa', 0), ('Tony Brogan', 0), ('Kelli Schulze', 0), ('Zachary Falgoust', 0), ('Wilma Rey', 0), ('Chad Pardo', 0), ('Michael Michel', 0), ('Debbie Ruch', 0), ('Daniel Trimble', 0), ('Vernon Sermons', 0), ('Elsie Sowers', 0), ('Max Kirkendall', 0), ('Christopher Merwin', 0), ('Edward Reader', 0), ('Vivian Milton', 0), ('Cassandra Creed', 0), ('Wesley Harm', 0), ('Douglas Dabney', 0), ('Bruce Dailey', 0), ('Nelson Guitierrez', 0), ('Albert Lockett', 0), ('Michele Valliere', 0), ('Juana Greenleaf', 0), ('Olivia Fall', 0), ('Allen Velarde', 0), ('Corinne Kocsis', 0), ('Darren Hadaway', 0), ('Ingrid Mutter', 0), ('Becky Zinn', 0), ('Edward Scalise', 0), ('Ronald Rushton', 0), ('Eva Pursley', 0), ('Tonya Niemi', 0), ('Ramona Stacy', 0), ('Jacob Meeker', 0), ('Gerald Cousin', 0), ('Jose Gregg', 0), ('Marjorie Tello', 0), ('Annie Markel', 0), ('Jack Kolb', 0), ('Kelly Cheatam', 0), ('Sonia Turgeon', 0), ('Martin Harbin', 0), ('Joseph Favors', 0), ('Harriett Loveall', 0), ('Gary Thorson', 0), ('Darren Suchan', 0), ('Leon Urbina', 0), ('Justin Hutson', 0), ('Allen Mckenna', 0), ('Steve Rushing', 0), ('Johnny Turnbull', 0), ('Cody Basta', 0), ('Orville Gillaspie', 0), ('Mike Rodarte', 0), ('Vernon Moultrie', 0), ('Sally Soule', 0), ('Larry Chapa', 0), ('Keith Dykes', 0), ('Norman Mcnulty', 0), ('Sonia Bass', 0), ('Stephen Fitz', 0), ('Russell Treat', 0), ('Jonathan Gower', 0), ('Lena Judge', 0), ('Noreen Durfey', 0), ('Russell Oshea', 0), ('Jacob Ingle', 0), ('Steven Ferrari', 0), ('Alma Mcneal', 0), ('Daphne Renz', 0), ('Justin Soloman', 0), ('Elinor Raco', 0), ('Jeff Moten', 0), ('Hector Horning', 0), ('Manuel Hammack', 0), ('Scott Luevano', 0), ('Javier Hausman', 0), ('Alice Knoll', 0), ('Danny Corrigan', 0), ('Gary Burmeister', 0), ('Richard Melvin', 0), ('Norman Styles', 0), ('Luz Coomer', 0), ('Gabriela Keaney', 0), ('Kayla Paynter', 0), ('Emily Chadwick', 0), ('Timothy Vetter', 0), ('Carl Campos',

0), ('Scott Ring', 0), ('Christian Klemm', 0), ('Patsy Horne', 0), ('Jo Patino', 0), ('Harry Norfleet', 0), ('Arthur Larrabee', 0), ('Christopher Biles', 0), ('Harriett Curatolo', 0), ('Benjamin Prado', 0), ('Jonathan Wagoner', 0), ('Louise Lively', 0), ('Inez Yee', 0), ('Javier Kratochvil', 0), ('Amy Bedford', 0), ('Keisha Ragusa', 0), ('Theresa Townsley', 0), ('Jeffrey Marcum', 0), ('Vickie Giffin', 0), ('Howard Storm', 0), ('Lester Kealey', 0), ('Alicia Womble', 0), ('Carmella Vanwagenen', 0), ('Jodi Holm', 0), ('Rick Brathwaite', 0), ('Wayne Merkle', 0), ('Yvette Dowdy', 0), ('Kenneth Casares', 0), ('Noemi Widrick', 0), ('Teresa Booth', 0), ('Arlene Bergmann', 0), ('Phyllis Beene', 0), ('Jeff Singletary', 0), ('Jessica Villegas', 0), ('Jessie Efaw', 0), ('Eleanor Fontenot', 0), ('Ruby Avent', 0), ('Patsy Finlay', 0), ('Jorge Ventimiglia', 0), ('Curtis Cordell', 0), ('Clarence Keyser', 0), ('Juan Vandyke', 0), ('Tyrone Mondor', 0), ('Shanna Elsea', 0), ('Geraldine Vargas', 0), ('Paige Lovitt', 0), ('Candy Henn', 0), ('Jeremy Catalano', 0), ('Chris Embry', 0), ('Ronald Henning', 0), ('Chris Rahman', 0), ('Ruby Phifer', 0), ('Jeremy Hyman', 0), ('Jimmy Pritchard', 0), ('Chad Doucet', 0), ('Brian Martindale', 0), ('Peter Whitman', 0), ('Karen Pena', 0), ('Manuel Wiser', 0), ('Leonard Delrio', 0), ('Claire Britton', 0), ('Jennie Tuggle', 0), ('Antonio Obryan', 0), ('Bessie Minton', 0), ('Jerry Herring', 0), ('Denise Ferraro', 0), ('Gary Quillen', 0), ('Antonio Braddock', 0), ('Tonya Canfield', 0), ('Jocelyn Lynde', 0), ('Noemi Marcucci', 0), ('Brandon Mask', 0), ('Louise Bellamy', 0), ('Juan Pink', 0), ('Amy Boren', 0), ('Glenna Meaney', 0), ('Sherri Kinser', 0), ('Harriett Zollinger', 0), ('Adam Harden', 0), ('Craig Avent', 0), ('Jason Eskridge', 0), ('Joyce Holtz', 0), ('Harold Seaman', 0), ('Erma Creech', 0), ('Wayne Crabb', 0), ('Marilyn Mickens', 0), ('Christine King', 0), ('Paul Albers', 0), ('Erik Weinman', 0), ('George Manchester', 0), ('Dean Buchanon', 0), ('Brandi Mcauliffe', 0), ('Earnestine Simo', 0), ('Johnny Dozier', 0), ('Fred Meeker', 0), ('William Backus', 0), ('Gregory Despain', 0), ('Timothy Congdon', 0), ('Janice Granados', 0), ('Chad Steffens', 0), ('Jessie Reta', 0), ('Shawn Rodriquez', 0), ('Ellen Millican', 0), ('Darren Cotta', 0), ('Stephanie Shanks', 0), ('Daisy Palladino', 0), ('Raymond Hynes', 0), ('Jeff Vanbuskirk', 0), ('Carmella Sheffey', 5), ('Keith Hyder', 5), ('Joshua Kingsbury', 5), ('Donald Parmer', 5), ('Mark Grimm', 5), ('Geraldine Flora', 5), ('Bobby Faris', 5), ('Lonnie Neveu', 5), ('Larry Mitchel', 5), ('Tina Haugen', 5), ('Walter Chapa', 5), ('Jamie Sing', 5), ('Adam Bingham', 5), ('Earnestine Croswell', 5), ('Keith Hurt', 5), ('Harold Andre', 5), ('Jason Word', 5), ('Yvette Chenault', 5), ('Dean Scioneaux', 5), ('Candice Brower', 5), ('Adam Feeney', 5), ('Violet Felker', 5), ('Stephen Core', 5), ('Sean Howerton', 5), ('Craig Dash', 5), ('Larry Lamp', 7), ('Amelia Mckeever', 8), ('Carole Stuckey', 8), ('Paul Gowen', 11), ('Lois Sexton', 11), ('Ryan Giglio', 11), ('Lawrence Hecker', 11), ('Louis Schreiber', 14), ('Nicole Ellingson', 14), ('Pearlie Storie', 14), ('Roslyn Philipp', 14), ('Roberto Routon', 14), ('Jesse Sandstrom', 14), ('Doris Mays', 14), ('Leonard Urbina', 14), ('Cindy Motley', 14), ('Juanita Wildman', 14), ('Larry Weatherspoon', 14), ('Pauline Penton', 14), ('Nora Hashimoto', 14), ('Gustavo Sysak', 14), ('Alan Harness', 14), ('Zelma Vallier', 14), ('Norman Wingfield', 14), ('Clarence Correll', 14), ('Hillary Tall', 14), ('Jeffery Ruiz', 17), ('Jesse Horn', 17), ('Gerald Whitesell', 17), ('Darlene Sandler', 17), ('George Blow', 17), ('Sharonda Remsberg', 17), ('Todd Huss', 19), ('Yvonne Mickens', 19), ('Brandi Keil', 19),

('Dianne Mixon', 24), ('Corey Devane', 24), ('Douglas Sedillo', 24), ('Anthony Pires', 24), ('Howard Hake', 24), ('Max Madamba', 24), ('Eugene Forman', 24), ('Curtis Rounds', 24), ('Violet Eden', 24), ('Amelia Meacham', 24), ('Lillie Dunton', 24), ('Kristy Strom', 24), ('Daniel Deloach', 24), ('Curtis Bowyer', 24), ('Lester Gallivan', 29), ('Elizabeth Dismuke', 29), ('Kristine Cave', 29), ('Natalie Bynum', 29), ('Shelley Scherer', 29), ('Christopher List', 29), ('Bruce Rickert', 29), ('Lilia Rojero', 29), ('Diana Welker', 29), ('Christina Driskell', 29), ('Isabel Peart', 29), ('Craig Baker', 29), ('Tanisha Gamino', 29), ('Fannie Hesse', 29), ('Walter Roussel', 33), ('Tyrone Ottman', 33), ('Teresa Samaniego', 33), ('Katy Tressler', 33), ('Hugo Vogelgesang', 33), ('Ingrid Devaul', 33), ('Ted Yetman', 33), ('Elnora Coppedge', 33), ('Esther Hale', 33), ('Lance Buono', 33), ('Stephen Mccool', 33), ('Jeremy Trainor', 33), ('Loraine Montford', 33), ('Debbie Sapp', 33), ('Patrick Bahena', 33), ('Raven Dalgleish', 33), ('Selena Lenart', 33), ('Gary Poole', 35), ('Scott Simons', 35), ('Charlene Pennington', 35), ('Fernando Sobers', 35), ('Nora Neel', 35), ('Eugene Singh', 35), ('Lonnie Kitzmiller', 35), ('Anthony Ramey', 35), ('Beatrice Rose', 35), ('Velma Burnett', 35), ('Hector Burdge', 35), ('Kenneth Hudgens', 35), ('Louis Koss', 35), ('Anna Marble', 35), ('Guy Vanalstyne', 35), ('Lula Eskridge', 35), ('Gary Tate', 35), ('Yvonne Mccune', 35), ('Robert Cooks', 35), ('Nelson Coltrane', 35), ('Asia Gucman', 35), ('Kyla Hillin', 36), ('Jill Dickerson', 36), ('Steve Ellingson', 38), ('Veronica Voss', 38), ('Shane Dare', 38), ('Matthew Will', 38), ('Julio Naccarato', 38), ('Stanley Bybee', 38), ('Guy Gonsoulin', 38), ('Laura Atwater', 38), ('Jeanne Twigg', 38), ('Douglas Perrotta', 38), ('Della Carlile', 38), ('Jesse Pinon', 38), ('Chandra Trowell', 38), ('Laura Flanigan', 38), ('Justin Kuehn', 38), ('Mathew Eslick', 38), ('Verna Basso', 38), ('Lorraine Ake', 38), ('Mark Tate', 41), ('Andrew Colunga', 41), ('Scott Prescott', 42), ('Cherie Mcguckin', 42), ('Allen Towns', 42), ('Juan Hermes', 42), ('Mathew Mercedes', 42), ('Darren Blau', 42), ('Carolina Hamid', 42), ('Marvin Hussey', 42), ('Lester Maya', 42), ('Cora Mills', 42), ('Kristin Keel', 42), ('Donald Daly', 42), ('Ellen Childress', 42), ('Tara Nissen', 42), ('Roger Arriola', 42), ('Roger Pellerin', 42), ('Terry Brush', 42), ('Chris Mcmorris', 44), ('Steve Hix', 44), ('Dianne Alarcon', 44), ('Renee Field', 44), ('Adrienne Savoy', 44), ('Frank Dunkin', 44), ('Jose Burge', 44), ('Terry Caudle', 44), ('Neil Hunton', 44), ('Joyce Etter', 44), ('Dale Charlton', 51), ('Louis Marino', 51), ('Clara Wainwright', 51), ('Bryan Corcoran', 51), ('Michael Scheffler', 51), ('Laurie Fullmer', 51), ('Terry Tweed', 51), ('Molly Wilkinson', 51), ('Keith Banister', 51), ('Norman Mastin', 51), ('Dean Bergey', 51), ('Antonio Bischoff', 51), ('Paul Vaillancourt', 51), ('Gustavo Theall', 51), ('Karen Kinard', 51), ('Sabina Corsaut', 51), ('Marguerite Mccabe', 51), ('Dan Lain', 51), ('Eunice Barr', 52), ('Alice Reese', 52), ('Anthony Deaver', 52), ('Tanya Pickering', 52), ('Jack Tso', 52), ('Timothy Markel', 52), ('Gregory Dantzler', 52), ('Christine Prather', 52), ('Mike Schumann', 52), ('Arthur Runge', 52), ('Harry Skeen', 52), ('Shawn Leigh', 53), ('Rita Witkowski', 53), ('Hilda Wilmoth', 53), ('Sofia Kelling', 53), ('Jimmy Munoz', 53), ('Alberta Bradley', 53), ('Roger Cool', 53), ('Joseph Donohue', 53), ('Joseph Audet', 53), ('Billy Partridge', 53), ('Valerie Byrd', 53), ('Mark Dillman', 53), ('Luz Hong', 53), ('Vincent Berkley', 53), ('Kenneth Lewellen', 61), ('Shelia Sorrells', 61), ('Michele Ford', 61), ('Arlene

Wellington', 61), ('Travis Trull', 61), ('Katherine Ryan', 61), ('Joy Cloud', 61), ('Hazel Persinger', 61), ('Billy Luu', 61), ('Darlene Bergman', 61), ('Susie Peoples', 61), ('Jeffery Bounds', 61), ('Philip Gauthier', 61), ('Jorge Madry', 62), ('Javier Weatherwax', 62), ('Willie Pence', 62), ('Roberto Munroe', 62), ('Yvette Scalise', 62), ('Jeanette Herbst', 62), ('Daniel Binkley', 62), ('Craig Bulter', 62), ('Kurt Gettings', 62), ('Walter Sappington', 62), ('Ruby Traylor', 62), ('Matthew Beaman', 62), ('Rodney Toth', 62), ('Brent Curtsinger', 62), ('Belen Sheerin', 67), ('Jessie Mayville', 67), ('Clarence Batchelder', 67), ('Chad Maclean', 67), ('Neil Rosenbeck', 67), ('Audrey Buck', 67), ('Herman Appling', 67), ('Jose Flannery', 67), ('Christian Durall', 67), ('Nereida Viscardi', 67), ('Valerie Jenson', 67), ('Pamela French', 67), ('Sarah Waterman', 67), ('Chris Reiner', 76), ('Jacob Dandrea', 76), ('Darren Pedrosa', 76), ('Verla Pali', 76), ('Guy Winders', 76), ('Dorothy Groves', 76), ('Brandon Saari', 76), ('Eugene Gore', 76), ('Dennis Donahue', 76), ('Clara Huck', 76), ('Christa Monsour', 76), ('Cody Russom', 76), ('Thomas Giffin', 76), ('Amelia Jarrell', 76)]

Duration taken by mergesort to sort 500 pickup requests in random order: 0.001276 seconds

Nearly Sorted list sorted

[('Chad Pardo', 0), ('Noemi Widrick', 0), ('Keisha Ragusa', 0), ('Cody Basta', 0), ('Jorge Ventimiglia', 0), ('Louise Bellamy', 0), ('Vernon Moultrie', 0), ('Maribel Earnest', 0), ('Howard Storm', 0), ('Ramona Stacy', 0), ('Jodi Holm', 0), ('Gregory Despain', 0), ('Alicia Womble', 0), ('Bessie Minton', 0), ('Paige Lovitt', 0), ('Michael Michel', 0), ('Brandi Mcauliffe', 0), ('Benjamin Prado', 0), ('Inez Yee', 0), ('Peter Whitman', 0), ('Patsy Finlay', 0), ('Marilyn Mickens', 0), ('Joseph Linville', 0), ('Noreen Durfey', 0), ('Arthur Larrabee', 0), ('Harold Hamill', 0), ('Rosemary Mcculley', 0), ('Eileen Bunch', 0), ('Kevin Schreiber', 0), ('Jessica Villegas', 0), ('Clayton Salls', 0), ('Luz Coomer', 0), ('Wilma Rey', 0), ('Emily Chadwick', 0), ('Luis Lehmann', 0), ('Philip Hammonds', 0), ('Leonard Delrio', 0), ('Russell Ashmore', 0), ('Craig Avent', 0), ('Manuel Hammack', 0), ('Javier Hausman', 0), ('Dale Ormsby', 0), ('Candice Swisher', 0), ('Jonathan Wagoner', 0), ('Scott Ring', 0), ('Teresa Booth', 0), ('Harry Norfleet', 0), ('Jose Gregg', 0), ('Tyrone Mondor', 0), ('Karen Pena', 0), ('Mabel Gist', 0), ('Chad Steffens', 0), ('Louise Lively', 0), ('Dustin Gadsden', 0), ('Ethel Reitz', 0), ('Kelly Cheatam', 0), ('Judith Lemay', 0), ('Scott Luevano', 0), ('Matthew Massa', 0), ('Opal Tobias', 0), ('Jerry Herring', 0), ('Jessie Efaw', 0), ('Kelli Schulze', 0), ('Chad Doucet', 0), ('Johnny Dozier', 0), ('Bruce Dailey', 0), ('Zachary Falgoust', 0), ('Erik Weinman', 0), ('Cindy Penney', 0), ('Olivia Fall', 0), ('Dennis Housley', 0), ('Daniel Trimble', 0), ('Sherri Kinser', 0), ('Elsie Sowers', 0), ('Vickie Giffin', 0), ('Jeremy Catalano', 0), ('Wesley Harm', 0), ('Ronald Henning', 0), ('Lester Kealey', 0), ('Christopher Merwin', 0), ('Claire Britton', 0), ('Timothy Vetter', 0), ('Edward Reader', 0), ('Clarence Keyser', 0), ('Juan Vandyke', 0), ('Erma Creech', 0), ('Hector Horning', 0), ('Nelson Guitierrez', 0), ('Ruby Phifer', 0), ('Albert Lockett', 0),

('Michele Valliere', 0), ('Jeff Vanbuskirk', 0), ('Ruby Avent', 0), ('Manuel Wiser', 0), ('Danny Corrigan', 0), ('Allen Velarde', 0), ('Corinne Kocsis', 0), ('Darren Hadaway', 0), ('Ingrid Mutter', 0), ('Rick Brathwaite', 0), ('Noemi Marcucci', 0), ('Brandon Mask', 0), ('Danielle Tincher', 0), ('Amy Boren', 0), ('Edward Scalise', 0), ('Carl Campos', 0), ('Antonio Obryan', 0), ('Tonya Niemi', 0), ('Alice Knoll', 0), ('George Manchester', 0), ('Gary Quillen', 0), ('Jacob Meeker', 0), ('Kenneth Casares', 0), ('Stephen Fitz', 0), ('Lillie Frisby', 0), ('Gary Thorson', 0), ('Jacob Ingle', 0), ('Martin Harbin', 0), ('Alma Mcneal', 0), ('Annie Markel', 0), ('Becky Zinn', 0), ('Nelson Tooker', 0), ('Harriett Curatolo', 0), ('Gary Burmeister', 0), ('Janice Granados', 0), ('Amy Bedford', 0), ('Jeff Moten', 0), ('Joseph Favors', 0), ('Michael Lefevre', 0), ('Norman Styles', 0), ('Harriett Loveall', 0), ('Wayne Crabb', 0), ('Jennie Tuggle', 0), ('Dean Buchanon', 0), ('Juana Greenleaf', 0), ('Theresa Townsley', 0), ('Ellen Millican', 0), ('Adam Harden', 0), ('Justin Hutson', 0), ('Allen Mckenna', 0), ('Cassandra Creed', 0), ('Johnny Turnbull', 0), ('Jocelyn Lynde', 0), ('Eva Pursley', 0), ('Larry Chapa', 0), ('Chris Embry', 0), ('Mike Rodarte', 0), ('Steve Rushing', 0), ('William Backus', 0), ('Justin Soloman', 0), ('Denise Herrington', 0), ('Jimmy Pritchard', 0), ('Sherry Winfrey', 0), ('Sally Soule', 0), ('Wayne Merkle', 0), ('Leon Urbina', 0), ('Harriett Zollinger', 0), ('Jessie Reta', 0), ('Stanley Littleton', 0), ('Norman Mcnulty', 0), ('Sonia Bass', 0), ('Jason Eskridge', 0), ('Eleanor Fontenot', 0), ('Christopher Biles', 0), ('Douglas Dabney', 0), ('Ronald Rushton', 0), ('Jonathan Gower', 0), ('Carmella Vanwagenen', 0), ('Vivian Milton', 0), ('Lena Judge', 0), ('Deanna Arndt', 0), ('Kurt Prowse', 0), ('Jeremy Hyman', 0), ('Russell Oshea', 0), ('Shawn Rodriquez', 0), ('Glenna Meaney', 0), ('Chris Rahman', 0), ('Stephanie Shanks', 0), ('Daphne Renz', 0), ('Joshua Spooner', 0), ('Paul Albers', 0), ('Russell Treat', 0), ('Larry Cantu', 0), ('Sonia Turgeon', 0), ('Debbie Ruch', 0), ('Elinor Raco', 0), ('Raymond Hynes', 0), ('James Urrutia', 0), ('Antonio Braddock', 0), ('Juan Pink', 0), ('Daisy Palladino', 0), ('Joyce Holtz', 0), ('Candy Henn', 0), ('Jack Kolb', 0), ('Jeff Singletary', 0), ('Dennis Gault', 0), ('Arlene Bergmann', 0), ('Steven Ferrari', 0), ('Keith Dykes', 0), ('Christine King', 0), ('Orville Gillaspie', 0), ('Gerald Cousin', 0), ('Curtis Cordell', 0), ('Jeffrey Marcum', 0), ('Gabriela Keaney', 0), ('Denise Ferraro', 0), ('Darren Suchan', 0), ('Earnestine Simo', 0), ('Marjorie Tello', 0), ('Heather Laureano', 0), ('Yvette Dowdy', 0), ('Vernon Sermons', 0), ('Tony Brogan', 0), ('Christian Klemm', 0), ('Fred Meeker', 0), ('Javier Kratochvil', 0), ('Dorcas Abbitt', 0), ('Darren Cotta', 0), ('Patsy Horne', 0), ('Harold Seaman', 0), ('Jo Patino', 0), ('Timothy Congdon', 0), ('Richard Melvin', 0), ('Brian Martindale', 0), ('Elaine Elias', 0), ('Shanna Elsea', 0), ('Wilfred Mourning', 0), ('Kayla Paynter', 0), ('Tonya Canfield', 0), ('Max Kirkendall', 0), ('Geraldine Vargas', 0), ('Carolyn Kay', 0), ('Phyllis Beene', 0), ('Earnestine Croswell', 5), ('Mark Grimm', 5), ('Adam Bingham', 5), ('Jamie Sing', 5), ('Jason Word', 5), ('Keith Hyder', 5), ('Keith Hurt', 5), ('Geraldine Flora', 5), ('Dean Scioneaux', 5), ('Donald Parmer', 5), ('Tina Haugen', 5), ('Larry Mitchel', 5), ('Joshua Kingsbury', 5), ('Violet Felker', 5), ('Candice Brower', 5), ('Stephen Core', 5), ('Carmella Sheffey', 5), ('Bobby Faris', 5), ('Yvette Chenault', 5), ('Lonnie Neveu', 5), ('Craig Dash', 5), ('Adam Feeney', 5), ('Harold Andre', 5), ('Sean Howerton', 5), ('Walter Chapa', 5), ('Larry Lamp', 7), ('Carole Stuckey', 8), ('Amelia Mckeever', 8), ('Paul Gowen', 11), ('Ryan Giglio',

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('Michael Scheffler', 51), ('Laurie Fullmer', 51), ('Arthur Runge', 52), ('Christine Prather', 52), ('Timothy Markel', 52), ('Alice Reese', 52), ('Eunice Barr', 52), ('Mike Schumann', 52), ('Anthony Deaver', 52), ('Harry Skeen', 52), ('Tanya Pickering', 52), ('Gregory Dantzler', 52), ('Jack Tso', 52), ('Hilda Wilmoth', 53), ('Rita Witkowski', 53), ('Joseph Donohue', 53), ('Shawn Leigh', 53), ('Sofia Kelling', 53), ('Roger Cool', 53), ('Billy Partridge', 53), ('Mark Dillman', 53), ('Joseph Audet', 53), ('Jimmy Munoz', 53), ('Valerie Byrd', 53), ('Alberta Bradley', 53), ('Vincent Berkley', 53), ('Luz Hong', 53), ('Joy Cloud', 61), ('Travis Trull', 61), ('Arlene Wellington', 61), ('Kenneth Lewellen', 61), ('Billy Luu', 61), ('Philip Gauthier', 61), ('Hazel Persinger', 61), ('Darlene Bergman', 61), ('Katherine Ryan', 61), ('Michele Ford', 61), ('Susie Peoples', 61), ('Shelia Sorrells', 61), ('Jeffery Bounds', 61), ('Craig Bulter', 62), ('Kurt Gettings', 62), ('Ruby Traylor', 62), ('Walter Sappington', 62), ('Daniel Binkley', 62), ('Willie Pence', 62), ('Jorge Madry', 62), ('Matthew Beaman', 62), ('Yvette Scalise', 62), ('Javier Weatherwax', 62), ('Rodney Toth', 62), ('Roberto Munroe', 62), ('Jeanette Herbst', 62), ('Brent Curtsinger', 62), ('Valerie Jenson', 67), ('Jose Flannery', 67), ('Pamela French', 67), ('Nereida Viscardi', 67), ('Audrey Buck', 67), ('Neil Rosenbeck', 67), ('Chad Maclean', 67), ('Sarah Waterman', 67), ('Jessie Mayville', 67), ('Belen Sheerin', 67), ('Christian Durall', 67), ('Clarence Batchelder', 67), ('Herman Appling', 67), ('Amelia Jarrell', 76), ('Dennis Donahue', 76), ('Chris Reiner', 76), ('Dorothy Groves', 76), ('Christa Monsour', 76), ('Guy Winders', 76), ('Eugene Gore', 76), ('Cody Russom', 76), ('Jacob Dandrea', 76), ('Thomas Giffin', 76), ('Verla Pali', 76), ('Brandon Saari', 76), ('Darren Pedrosa', 76), ('Clara Huck', 76)]

Duration taken by mergesort to sort 500 pickup requests in nearly sorted order: 0.001970 seconds

Reversed list sorted

[('Jo Patino', 0), ('Harold Seaman', 0), ('Patsy Horne', 0), ('Jack Kolb', 0), ('Darren Cotta', 0), ('Eva Pursley', 0), ('Javier Kratochvil', 0), ('Fred Meeker', 0), ('Christian Klemm', 0), ('Tony Brogan', 0), ('Vernon Sermons', 0), ('Yvette Dowdy', 0), ('Heather Laureano', 0), ('Marjorie Tello', 0), ('Earnestine Simo', 0), ('Darren Suchan', 0), ('Denise Ferraro', 0), ('Gabriela Keaney', 0), ('Jeffrey Marcum', 0), ('Curtis Cordell', 0), ('Gerald Cousin', 0), ('Orville Gillaspie', 0), ('Christine King', 0), ('Keith Dykes', 0), ('Steven Ferrari', 0), ('Arlene Bergmann', 0), ('Dennis Gault', 0), ('Jeff Singletary', 0), ('Shanna Elsea', 0), ('Candy Henn', 0), ('Joyce Holtz', 0), ('Daisy Palladino', 0), ('Juan Pink', 0), ('Antonio Braddock', 0), ('James Urrutia', 0), ('Raymond Hynes', 0), ('Elinor Raco', 0), ('Debbie Ruch', 0), ('Sonia Turgeon', 0), ('Larry Cantu', 0), ('Russell Treat', 0), ('Paul Albers', 0), ('Joshua Spooner', 0), ('Phyllis Beene', 0), ('Daphne Renz', 0), ('Elaine Elias', 0), ('Stephanie Shanks', 0), ('Chris Rahman', 0), ('Glenna Meaney', 0), ('Shawn Rodriquez', 0), ('Russell Oshea', 0), ('Jeremy Hyman', 0), ('Kurt Prowse', 0), ('Deanna Arndt', 0), ('Lena Judge', 0), ('Vivian Milton', 0), ('Carmella

Vanwagenen', 0), ('Jonathan Gower', 0), ('Ronald Rushton', 0), ('Douglas Dabney', 0), ('Christopher Biles', 0), ('Eleanor Fontenot', 0), ('Jason Eskridge', 0), ('Sonia Bass', 0), ('Norman Mcnulty', 0), ('Stanley Littleton', 0), ('Jessie Reta', 0), ('Harriett Zollinger', 0), ('Leon Urbina', 0), ('Wayne Merkle', 0), ('Sally Soule', 0), ('Sherry Winfrey', 0), ('Jimmy Pritchard', 0), ('Denise Herrington', 0), ('Justin Soloman', 0), ('Nelson Guitierrez', 0), ('Steve Rushing', 0), ('Mike Rodarte', 0), ('Chris Embry', 0), ('Larry Chapa', 0), ('Dorcas Abbitt', 0), ('Jocelyn Lynde', 0), ('Johnny Turnbull', 0), ('Cassandra Creed', 0), ('Brian Martindale', 0), ('Allen Mckenna', 0), ('Justin Hutson', 0), ('Adam Harden', 0), ('Ellen Millican', 0), ('Theresa Townsley', 0), ('Juana Greenleaf', 0), ('Dean Buchanon', 0), ('Jennie Tuggle', 0), ('Wayne Crabb', 0), ('Harriett Loveall', 0), ('Norman Styles', 0), ('Michael Lefevre', 0), ('Joseph Favors', 0), ('Jeff Moten', 0), ('Amy Bedford', 0), ('Janice Granados', 0), ('Gary Burmeister', 0), ('Harriett Curatolo', 0), ('Nelson Tooker', 0), ('Becky Zinn', 0), ('Annie Markel', 0), ('Alma Mcneal', 0), ('Martin Harbin', 0), ('Jacob Ingle', 0), ('Gary Thorson', 0), ('Lillie Frisby', 0), ('Stephen Fitz', 0), ('Kenneth Casares', 0), ('Jacob Meeker', 0), ('Gary Quillen', 0), ('George Manchester', 0), ('Alice Knoll', 0), ('Tonya Niemi', 0), ('Antonio Obryan', 0), ('Carl Campos', 0), ('Edward Scalise', 0), ('Amy Boren', 0), ('Danielle Tincher', 0), ('Brandon Mask', 0), ('Noemi Marcucci', 0), ('Rick Brathwaite', 0), ('Ingrid Mutter', 0), ('Darren Hadaway', 0), ('Corinne Kocsis', 0), ('Allen Velarde', 0), ('Geraldine Vargas', 0), ('Danny Corrigan', 0), ('Manuel Wiser', 0), ('Ruby Avent', 0), ('Jeff Vanbuskirk', 0), ('Michele Valliere', 0), ('Albert Lockett', 0), ('Ruby Phifer', 0), ('Tonya Canfield', 0), ('William Backus', 0), ('Hector Horning', 0), ('Erma Creech', 0), ('Juan Vandyke', 0), ('Clarence Keyser', 0), ('Edward Reader', 0), ('Timothy Vetter', 0), ('Claire Britton', 0), ('Christopher Merwin', 0), ('Lester Kealey', 0), ('Ronald Henning', 0), ('Wesley Harm', 0), ('Jeremy Catalano', 0), ('Max Kirkendall', 0), ('Vickie Giffin', 0), ('Elsie Sowers', 0), ('Sherri Kinser', 0), ('Daniel Trimble', 0), ('Richard Melvin', 0), ('Dennis Housley', 0), ('Olivia Fall', 0), ('Cindy Penney', 0), ('Erik Weinman', 0), ('Zachary Falgoust', 0), ('Bruce Dailey', 0), ('Johnny Dozier', 0), ('Chad Doucet', 0), ('Kelli Schulze', 0), ('Jessie Efaw', 0), ('Jerry Herring', 0), ('Opal Tobias', 0), ('Matthew Massa', 0), ('Scott Luevano', 0), ('Judith Lemay', 0), ('Kelly Cheatam', 0), ('Ethel Reitz', 0), ('Dustin Gadsden', 0), ('Louise Lively', 0), ('Chad Steffens', 0), ('Mabel Gist', 0), ('Karen Pena', 0), ('Tyrone Mondor', 0), ('Jose Gregg', 0), ('Harry Norfleet', 0), ('Teresa Booth', 0), ('Scott Ring', 0), ('Jonathan Wagoner', 0), ('Candice Swisher', 0), ('Dale Ormsby', 0), ('Javier Hausman', 0), ('Manuel Hammack', 0), ('Craig Avent', 0), ('Russell Ashmore', 0), ('Leonard Delrio', 0), ('Philip Hammonds', 0), ('Luis Lehmann', 0), ('Emily Chadwick', 0), ('Wilma Rey', 0), ('Luz Coomer', 0), ('Clayton Salls', 0), ('Jessica Villegas', 0), ('Kevin Schreiber', 0), ('Eileen Bunch', 0), ('Rosemary Mcculley', 0), ('Harold Hamill', 0), ('Arthur Larrabee', 0), ('Noreen Durfey', 0), ('Joseph Linville', 0), ('Marilyn Mickens', 0), ('Patsy Finlay', 0), ('Peter Whitman', 0), ('Inez Yee', 0), ('Benjamin Prado', 0), ('Kayla Paynter', 0), ('Michael Michel', 0), ('Paige Lovitt', 0), ('Bessie Minton', 0), ('Alicia Womble', 0), ('Gregory Despain', 0), ('Wilfred Mourning', 0), ('Jodi Holm', 0), ('Ramona Stacy', 0), ('Howard Storm', 0), ('Maribel Earnest', 0), ('Timothy Congdon', 0), ('Brandi Mcauliffe', 0), ('Vernon Moultrie', 0), ('Louise Bellamy',

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Duration taken by mergesort to sort 500 pickup requests in reverse order: 0.002072 seconds

```
The Top Down MergeSort algorithm Reflection for different scenarios of 500 pickup request:
Duration to sort 500 pickup requests in random order: 0.001276 seconds
Duration to sort 500 pickup requests in nearly sorted order: 0.001970 seconds
Duration to sort 500 pickup requests in reverse order: 0.002072 seconds

Looking at the durations for each scenario of 500 pickup request, MergeSort takes less time to
    sort the random list while more time is taken to sort the reverse list
Its time complexity is O(n log n) and its space complexity is O(n) as it needs temporary lists to enable merging.
```

Figure 75: Short reflection on using Merge Sort on different lists of 500 pickup requests

Quick Sort for 500 pickup request lists

500 pickup request list in random order sorted

Note: The first names are not the same for the different test scenarios as the lists are sorted using only the pickup times not the names.

[('Danny Corrigan', 0), ('Jeff Vanbuskirk', 0), ('Gary Burmeister', 0), ('Carolyn Kay', 0), ('Richard Melvin', 0), ('Joshua Spooner', 0), ('Norman Styles', 0), ('Maribel Earnest', 0), ('Luz Coomer', 0), ('Wilfred Mourning', 0), ('Gabriela Keaney', 0), ('Michael Lefevre', 0), ('Kayla Paynter', 0), ('Danielle Tincher', 0), ('Emily Chadwick', 0), ('Joseph Linville', 0), ('Timothy Vetter', 0), ('Harold Hamill', 0), ('Carl Campos', 0), ('Eileen Bunch', 0), ('Scott Ring', 0), ('Opal Tobias', 0), ('Christian Klemm', 0), ('Lillie Frisby', 0), ('Patsy Horne', 0), ('Luis Lehmann', 0), ('Jo Patino', 0), ('Dennis Housley', 0), ('Harry Norfleet', 0), ('Dennis Gault', 0), ('Arthur Larrabee', 0), ('Sherry Winfrey', 0), ('Christopher Biles', 0), ('Stanley Littleton', 0), ('Harriett Curatolo', 0), ('Ethel Reitz', 0), ('Benjamin Prado', 0), ('Dale Ormsby', 0), ('Jonathan Wagoner', 0), ('Matthew Massa', 0), ('Louise Lively', 0), ('Kelli Schulze', 0), ('Inez Yee', 0), ('Wilma Rey', 0), ('Javier Kratochvil', 0), ('Michael Michel', 0), ('Amy Bedford', 0), ('Daniel Trimble', 0), ('Keisha Ragusa', 0), ('Elsie Sowers', 0), ('Theresa Townsley', 0), ('Christopher Merwin', 0), ('Jeffrey Marcum', 0), ('Vivian Milton', 0), ('Vickie Giffin', 0), ('Wesley Harm', 0), ('Howard Storm', 0), ('Bruce Dailey', 0), ('Lester Kealey', 0), ('Albert Lockett', 0), ('Alicia Womble', 0), ('Juana Greenleaf', 0), ('Carmella Vanwagenen', 0), ('Allen Velarde', 0), ('Jodi Holm', 0), ('Darren Hadaway', 0), ('Rick Brathwaite', 0), ('Becky Zinn', 0), ('Wayne Merkle', 0), ('Ronald Rushton', 0), ('Yvette Dowdy', 0), ('Tonya Niemi', 0), ('Kenneth Casares', 0), ('Jacob Meeker', 0), ('Noemi Widrick', 0), ('Jose Gregg', 0), ('Teresa Booth', 0), ('Annie Markel', 0), ('Arlene Bergmann', 0), ('Kelly Cheatam', 0), ('Phyllis Beene', 0), ('Martin Harbin', 0), ('Jeff Singletary', 0), ('Harriett Loveall', 0), ('Jessica Villegas', 0), ('Darren Suchan', 0), ('Jessie Efaw', 0), ('Justin Hutson', 0), ('Eleanor Fontenot', 0), ('Steve

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Marino', 51), ('Bryan Corcoran', 51), ('Laurie Fullmer', 51), ('Terry Tweed', 51), ('Timothy Markel', 52), ('Gregory Dantzler', 52), ('Christine Prather', 52), ('Anthony Deaver', 52), ('Mike Schumann', 52), ('Jack Tso', 52), ('Arthur Runge', 52), ('Alice Reese', 52), ('Harry Skeen', 52), ('Eunice Barr', 52), ('Tanya Pickering', 52), ('Jimmy Munoz', 53), ('Rita Witkowski', 53), ('Hilda Wilmoth', 53), ('Sofia Kelling', 53), ('Alberta Bradley', 53), ('Roger Cool', 53), ('Joseph Donohue', 53), ('Joseph Audet', 53), ('Billy Partridge', 53), ('Valerie Byrd', 53), ('Mark Dillman', 53), ('Luz Hong', 53), ('Vincent Berkley', 53), ('Shawn Leigh', 53), ('Joy Cloud', 61), ('Hazel Persinger', 61), ('Billy Luu', 61), ('Michele Ford', 61), ('Darlene Bergman', 61), ('Travis Trull', 61), ('Susie Peoples', 61), ('Kenneth Lewellen', 61), ('Jeffery Bounds', 61), ('Arlene Wellington', 61), ('Philip Gauthier', 61), ('Shelia Sorrells', 61), ('Katherine Ryan', 61), ('Kurt Gettings', 62), ('Javier Weatherwax', 62), ('Willie Pence', 62), ('Roberto Munroe', 62), ('Yvette Scalise', 62), ('Jeanette Herbst', 62), ('Daniel Binkley', 62), ('Craig Bulter', 62), ('Matthew Beaman', 62), ('Rodney Toth', 62), ('Walter Sappington', 62), ('Brent Curtsinger', 62), ('Jorge Madry', 62), ('Ruby Traylor', 62), ('Valerie Jenson', 67), ('Jessie Mayville', 67), ('Clarence Batchelder', 67), ('Chad Maclean', 67), ('Neil Rosenbeck', 67), ('Audrey Buck', 67), ('Herman Appling', 67), ('Jose Flannery', 67), ('Christian Durall', 67), ('Nereida Viscardi', 67), ('Sarah Waterman', 67), ('Belen Sheerin', 67), ('Pamela French', 67), ('Dorothy Groves', 76), ('Eugene Gore', 76), ('Chris Reiner', 76), ('Dennis Donahue', 76), ('Darren Pedrosa', 76), ('Clara Huck', 76), ('Guy Winders', 76), ('Christa Monsour', 76), ('Brandon Saari', 76), ('Cody Russom', 76), ('Verla Pali', 76), ('Thomas Giffin', 76), ('Jacob Dandrea', 76), ('Amelia Jarrell', 76)]

Duration taken by QuickSort to sort 500 pickup requests in random order: 0.004800 seconds

500 pickup request list in nearly sorted order sorted

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Mckenna', 0), ('Teresa Booth', 0), ('Cassandra Creed', 0), ('Jose Gregg', 0), ('Johnny Turnbull', 0), ('Karen Pena', 0), ('Jocelyn Lynde', 0), ('Chad Steffens', 0), ('Eva Pursley', 0), ('Dustin Gadsden', 0), ('Larry Chapa', 0), ('Kelly Cheatam', 0), ('Chris Embry', 0), ('Scott Luevano', 0), ('Mike Rodarte', 0), ('Opal Tobias', 0), ('Steve Rushing', 0), ('Jessie Efaw', 0), ('William Backus', 0), ('Chad Doucet', 0), ('Justin Soloman', 0), ('Bruce Dailey', 0), ('Denise Herrington', 0), ('Erik Weinman', 0), ('Jimmy Pritchard', 0), ('Olivia Fall', 0), ('Sherry Winfrey', 0), ('Daniel Trimble', 0), ('Sally Soule', 0), ('Elsie Sowers', 0), ('Wayne Merkle', 0), ('Jeremy Catalano', 0), ('Leon Urbina', 0), ('Ronald Henning', 0), ('Harriett Zollinger', 0), ('Christopher Merwin', 0), ('Jessie Reta', 0), ('Timothy Vetter', 0), ('Stanley Littleton', 0), ('Clarence Keyser', 0), ('Norman Mcnulty', 0), ('Erma Creech', 0), ('Sonia Bass', 0), ('Nelson Guitierrez', 0), ('Jason Eskridge', 0), ('Albert Lockett', 0), ('Eleanor Fontenot', 0), ('Jeff Vanbuskirk', 0), ('Christopher Biles', 0), ('Manuel Wiser', 0), ('Douglas Dabney', 0), ('Allen Velarde', 0), ('Ronald Rushton', 0), ('Darren Hadaway', 0), ('Jonathan Gower', 0), ('Rick Brathwaite', 0), ('Carmella Vanwagenen', 0), ('Brandon Mask', 0), ('Vivian Milton', 0), ('Amy Boren', 0), ('Lena Judge', 0), ('Carl Campos', 0), ('Deanna Arndt', 0), ('Tonya Niemi', 0), ('Kurt Prowse', 0), ('George Manchester', 0), ('Jeremy Hyman', 0), ('Jacob Meeker', 0), ('Russell Oshea', 0), ('Stephen Fitz', 0), ('Shawn Rodriquez', 0), ('Gary Thorson', 0), ('Glenna Meaney', 0), ('Keisha Ragusa', 0), ('Chris Rahman', 0), ('Vernon Moultrie', 0), ('Stephanie Shanks', 0), ('Jodi Holm', 0), ('Daphne Renz', 0), ('Paige Lovitt', 0), ('Joshua Spooner', 0), ('Inez Yee', 0), ('Paul Albers', 0), ('Joseph Linville', 0), ('Russell Treat', 0), ('Rosemary Mcculley', 0), ('Larry Cantu', 0), ('Clayton Salls', 0), ('Sonia Turgeon', 0), ('Luis Lehmann', 0), ('Debbie Ruch', 0), ('Craig Avent', 0), ('Elinor Raco', 0), ('Candice Swisher', 0), ('Raymond Hynes', 0), ('Harry Norfleet', 0), ('James Urrutia', 0), ('Mabel Gist', 0), ('Antonio Braddock', 0), ('Ethel Reitz', 0), ('Juan Pink', 0), ('Matthew Massa', 0), ('Daisy Palladino', 0), ('Kelli Schulze', 0), ('Joyce Holtz', 0), ('Zachary Falgoust', 0), ('Candy Henn', 0), ('Dennis Housley', 0), ('Jack Kolb', 0), ('Vickie Giffin', 0), ('Jeff Singletary', 0), ('Lester Kealey', 0), ('Dennis Gault', 0), ('Edward Reader', 0), ('Arlene Bergmann', 0), ('Hector Horning', 0), ('Steven Ferrari', 0), ('Michele Valliere', 0), ('Keith Dykes', 0), ('Danny Corrigan', 0), ('Christine King', 0), ('Ingrid Mutter', 0), ('Orville Gillaspie', 0), ('Danielle Tincher', 0), ('Gerald Cousin', 0), ('Antonio Obryan', 0), ('Curtis Cordell', 0), ('Gary Quillen', 0), ('Jeffrey Marcum', 0), ('Lillie Frisby', 0), ('Gabriela Keaney', 0), ('Jorge Ventimiglia', 0), ('Denise Ferraro', 0), ('Alicia Womble', 0), ('Darren Suchan', 0), ('Patsy Finlay', 0), ('Earnestine Simo', 0), ('Kevin Schreiber', 0), ('Marjorie Tello', 0), ('Leonard Delrio', 0), ('Heather Laureano', 0), ('Scott Ring', 0), ('Yvette Dowdy', 0), ('Louise Lively', 0), ('Vernon Sermons', 0), ('Jerry Herring', 0), ('Tony Brogan', 0), ('Cindy Penney', 0), ('Christian Klemm', 0), ('Wesley Harm', 0), ('Fred Meeker', 0), ('Juan Vandyke', 0), ('Javier Kratochvil', 0), ('Ruby Avent', 0), ('Dorcas Abbitt', 0), ('Noemi Marcucci', 0), ('Darren Cotta', 0), ('Alice Knoll', 0), ('Patsy Horne', 0), ('Chad Pardo', 0), ('Harold Seaman', 0), ('Brandi Mcauliffe', 0), ('Jo Patino', 0), ('Wilma Rey', 0), ('Timothy Congdon', 0), ('Tyrone Mondor', 0), ('Richard Melvin', 0), ('Johnny Dozier', 0), ('Brian Martindale', 0), ('Claire Britton', 0), ('Elaine Elias', 0), ('Corinne Kocsis', 0), ('Shanna

Elsea', 0), ('Kenneth Casares', 0), ('Wilfred Mourning', 0), ('Arthur Larrabee', 0), ('Kayla Paynter', 0), ('Judith Lemay', 0), ('Tonya Canfield', 0), ('Ruby Phifer', 0), ('Max Kirkendall', 0), ('Howard Storm', 0), ('Geraldine Vargas', 0), ('Sherri Kinser', 0), ('Carolyn Kay', 0), ('Javier Hausman', 0), ('Edward Scalise', 0), ('Lonnie Neveu', 5), ('Mark Grimm', 5), ('Adam Bingham', 5), ('Jamie Sing', 5), ('Jason Word', 5), ('Keith Hyder', 5), ('Keith Hurt', 5), ('Geraldine Flora', 5), ('Dean Scioneaux', 5), ('Donald Parmer', 5), ('Tina Haugen', 5), ('Larry Mitchel', 5), ('Joshua Kingsbury', 5), ('Violet Felker', 5), ('Candice Brower', 5), ('Stephen Core', 5), ('Carmella Sheffey', 5), ('Bobby Faris', 5), ('Yvette Chenault', 5), ('Craig Dash', 5), ('Adam Feeney', 5), ('Harold Andre', 5), ('Sean Howerton', 5), ('Walter Chapa', 5), ('Earnestine Croswell', 5), ('Larry Lamp', 7), ('Amelia Mckeever', 8), ('Carole Stuckey', 8), ('Paul Gowen', 11), ('Lawrence Hecker', 11), ('Lois Sexton', 11), ('Ryan Giglio', 11), ('Pearlie Storie', 14), ('Jesse Sandstrom', 14), ('Nora Hashimoto', 14), ('Hillary Tall', 14), ('Doris Mays', 14), ('Roslyn Philipp', 14), ('Leonard Urbina', 14), ('Pauline Penton', 14), ('Gustavo Sysak', 14), ('Roberto Routon', 14), ('Cindy Motley', 14), ('Norman Wingfield', 14), ('Alan Harness', 14), ('Clarence Correll', 14), ('Juanita Wildman', 14), ('Louis Schreiber', 14), ('Larry Weatherspoon', 14), ('Nicole Ellingson', 14), ('Zelma Vallier', 14), ('Gerald Whitesell', 17), ('George Blow', 17), ('Jesse Horn', 17), ('Sharonda Remsberg', 17), ('Jeffery Ruiz', 17), ('Darlene Sandler', 17), ('Brandi Keil', 19), ('Yvonne Mickens', 19), ('Todd Huss', 19), ('Eugene Forman', 24), ('Daniel Deloach', 24), ('Corey Devane', 24), ('Anthony Pires', 24), ('Howard Hake', 24), ('Dianne Mixon', 24), ('Kristy Strom', 24), ('Curtis Rounds', 24), ('Max Madamba', 24), ('Douglas Sedillo', 24), ('Curtis Bowyer', 24), ('Amelia Meacham', 24), ('Lillie Dunton', 24), ('Violet Eden', 24), ('Kristine Cave', 29), ('Diana Welker', 29), ('Elizabeth Dismuke', 29), ('Fannie Hesse', 29), ('Craig Baker', 29), ('Lester Gallivan', 29), ('Bruce Rickert', 29), ('Lilia Rojero', 29), ('Christopher List', 29), ('Natalie Bynum', 29), ('Tanisha Gamino', 29), ('Isabel Peart', 29), ('Shelley Scherer', 29), ('Christina Driskell', 29), ('Lance Buono', 33), ('Hugo Vogelgesang', 33), ('Raven Dalgleish', 33), ('Esther Hale', 33), ('Tyrone Ottman', 33), ('Jeremy Trainor', 33), ('Teresa Samaniego', 33), ('Ted Yetman', 33), ('Stephen Mccool', 33), ('Loraine Montford', 33), ('Elnora Coppedge', 33), ('Walter Roussel', 33), ('Debbie Sapp', 33), ('Patrick Bahena', 33), ('Katy Tressler', 33), ('Selena Lenart', 33), ('Ingrid Devaul', 33), ('Fernando Sobers', 35), ('Louis Koss', 35), ('Anna Marble', 35), ('Guy Vanalstyne', 35), ('Velma Burnett', 35), ('Eugene Singh', 35), ('Nora Neel', 35), ('Hector Burdge', 35), ('Anthony Ramey', 35), ('Kenneth Hudgens', 35), ('Yvonne Mccune', 35), ('Gary Tate', 35), ('Lula Eskridge', 35), ('Nelson Coltrane', 35), ('Robert Cooks', 35), ('Beatrice Rose', 35), ('Gary Poole', 35), ('Scott Simons', 35), ('Lonnie Kitzmiller', 35), ('Asia Gucman', 35), ('Charlene Pennington', 35), ('Kyla Hillin', 36), ('Jill Dickerson', 36), ('Lorraine Ake', 38), ('Guy Gonsoulin', 38), ('Matthew Will', 38), ('Laura Atwater', 38), ('Veronica Voss', 38), ('Jeanne Twigg', 38), ('Douglas Perrotta', 38), ('Jesse Pinon', 38), ('Verna Basso', 38), ('Chandra Trowell', 38), ('Shane Dare', 38), ('Stanley Bybee', 38), ('Laura Flanigan', 38), ('Justin Kuehn', 38), ('Mathew Eslick', 38), ('Steve Ellingson', 38), ('Julio Naccarato', 38), ('Della Carlile', 38), ('Mark Tate', 41), ('Andrew Colunga', 41),

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Duration taken by QuickSort to sort 500 pickup requests in nearly sorted order: 0.004988 seconds

500 pickup request list in reversed order sorted

[('Alice Knoll', 0), ('Chad Pardo', 0), ('Tonya Niemi', 0), ('Harold Seaman', 0), ('Antonio Obryan', 0), ('Jack Kolb', 0), ('Carl Campos', 0), ('Eva Pursley', 0), ('Edward Scalise', 0), ('Fred Meeker', 0), ('Amy Boren', 0), ('Tony Brogan', 0), ('Danielle Tincher', 0), ('Yvette Dowdy', 0), ('Brandon Mask', 0), ('Marjorie Tello', 0), ('Noemi Marcucci', 0), ('Darren Suchan', 0), ('Rick Brathwaite', 0), ('Gabriela Keaney', 0), ('Ingrid Mutter', 0), ('Curtis Cordell', 0), ('Darren Hadaway', 0), ('Orville Gillaspie', 0), ('Corinne Kocsis', 0), ('Keith Dykes', 0), ('Allen Velarde', 0), ('Arlene Bergmann', 0), ('Geraldine Vargas', 0), ('Jeff Singletary', 0), ('Danny Corrigan', 0), ('Candy Henn', 0), ('Manuel Wiser', 0), ('Daisy Palladino', 0), ('Ruby Avent', 0), ('Antonio Braddock', 0), ('Jeff Vanbuskirk', 0), ('Raymond Hynes', 0), ('Michele Valliere', 0), ('Debbie Ruch', 0), ('Albert Lockett', 0), ('Larry Cantu', 0), ('Ruby Phifer', 0), ('Paul Albers', 0), ('Tonya Canfield', 0), ('Phyllis Beene', 0), ('William Backus', 0), ('Elaine Elias', 0), ('Hector Horning', 0), ('Chris Rahman', 0), ('Erma Creech', 0), ('Shawn Rodriquez', 0), ('Juan Vandyke', 0), ('Jeremy Hyman', 0), ('Clarence Keyser', 0), ('Deanna Arndt', 0), ('Edward Reader', 0), ('Vivian Milton', 0), ('Timothy Vetter', 0), ('Jonathan Gower', 0), ('Claire Britton', 0), ('Douglas Dabney', 0), ('Christopher Merwin', 0), ('Eleanor Fontenot', 0), ('Lester Kealey', 0), ('Sonia Bass', 0), ('Ronald Henning', 0), ('Stanley Littleton', 0), ('Wesley Harm', 0), ('Harriett Zollinger', 0), ('Jeremy Catalano', 0), ('Wayne Merkle', 0), ('Max Kirkendall', 0), ('Sherry Winfrey', 0), ('Vickie Giffin', 0), ('Denise Herrington', 0), ('Elsie Sowers', 0), ('Nelson Guitierrez', 0), ('Sherri Kinser', 0), ('Mike Rodarte', 0), ('Daniel Trimble', 0), ('Larry Chapa', 0), ('Richard Melvin', 0), ('Jocelyn Lynde', 0), ('Dennis Housley', 0), ('Cassandra Creed', 0), ('Olivia Fall', 0), ('Allen Mckenna', 0), ('Cindy Penney', 0), ('Adam Harden', 0), ('Erik Weinman', 0), ('Theresa Townsley', 0), ('Zachary Falgoust', 0), ('Dean Buchanon', 0), ('Bruce Dailey', 0), ('Wayne Crabb', 0), ('Johnny Dozier', 0), ('Norman Styles', 0), ('Chad Doucet', 0), ('Joseph Favors', 0), ('Kelli Schulze', 0), ('Amy Bedford', 0), ('Jessie Efaw', 0), ('Gary Burmeister', 0), ('Jerry Herring', 0), ('Nelson Tooker', 0), ('Opal Tobias', 0), ('Annie Markel', 0), ('Matthew Massa', 0), ('Martin Harbin', 0), ('Scott Luevano', 0), ('Gary Thorson', 0), ('Judith Lemay', 0), ('Stephen Fitz', 0), ('Kelly Cheatam', 0), ('Jacob Meeker', 0), ('Ethel Reitz', 0), ('George Manchester', 0), ('Dustin Gadsden', 0), ('Patsy Horne', 0), ('Louise Lively', 0), ('Javier Kratochvil', 0), ('Chad Steffens', 0), ('Vernon Sermons', 0), ('Mabel Gist', 0), ('Earnestine Simo', 0), ('Karen Pena', 0), ('Jeffrey Marcum', 0), ('Tyrone Mondor', 0), ('Christine King', 0), ('Jose Gregg', 0), ('Dennis Gault', 0), ('Harry Norfleet', 0), ('Joyce Holtz', 0), ('Teresa Booth', 0), ('James Urrutia', 0), ('Scott Ring', 0), ('Sonia Turgeon', 0), ('Jonathan Wagoner', 0), ('Joshua Spooner', 0), ('Candice Swisher', 0), ('Stephanie Shanks', 0), ('Dale Ormsby', 0), ('Russell Oshea', 0), ('Javier Hausman', 0), ('Lena Judge', 0), ('Manuel Hammack', 0), ('Ronald Rushton', 0), ('Craig Avent', 0), ('Jason Eskridge', 0), ('Russell Ashmore', 0), ('Jessie Reta', 0), ('Leonard Delrio', 0), ('Sally Soule', 0), ('Philip Hammonds', 0), ('Justin Soloman', 0), ('Luis Lehmann', 0), ('Chris Embry', 0), ('Emily Chadwick', 0), ('Johnny Turnbull', 0), ('Wilma Rey', 0), ('Justin Hutson', 0), ('Luz

Coomer', 0), ('Juana Greenleaf', 0), ('Clayton Salls', 0), ('Harriett Loveall', 0), ('Jessica Villegas', 0), ('Jeff Moten', 0), ('Kevin Schreiber', 0), ('Harriett Curatolo', 0), ('Eileen Bunch', 0), ('Alma Mcneal', 0), ('Rosemary Mcculley', 0), ('Lillie Frisby', 0), ('Harold Hamill', 0), ('Gary Quillen', 0), ('Arthur Larrabee', 0), ('Darren Cotta', 0), ('Noreen Durfey', 0), ('Heather Laureano', 0), ('Joseph Linville', 0), ('Gerald Cousin', 0), ('Marilyn Mickens', 0), ('Shanna Elsea', 0), ('Patsy Finlay', 0), ('Elinor Raco', 0), ('Peter Whitman', 0), ('Daphne Renz', 0), ('Inez Yee', 0), ('Kurt Prowse', 0), ('Benjamin Prado', 0), ('Christopher Biles', 0), ('Kayla Paynter', 0), ('Leon Urbina', 0), ('Michael Michel', 0), ('Steve Rushing', 0), ('Paige Lovitt', 0), ('Brian Martindale', 0), ('Bessie Minton', 0), ('Jennie Tuggle', 0), ('Alicia Womble', 0), ('Janice Granados', 0), ('Gregory Despain', 0), ('Jacob Ingle', 0), ('Wilfred Mourning', 0), ('Jo Patino', 0), ('Jodi Holm', 0), ('Denise Ferraro', 0), ('Ramona Stacy', 0), ('Juan Pink', 0), ('Howard Storm', 0), ('Glenna Meaney', 0), ('Maribel Earnest', 0), ('Norman Mcnulty', 0), ('Timothy Congdon', 0), ('Dorcas Abbitt', 0), ('Brandi Mcauliffe', 0), ('Michael Lefevre', 0), ('Vernon Moultrie', 0), ('Kenneth Casares', 0), ('Louise Bellamy', 0), ('Steven Ferrari', 0), ('Jorge Ventimiglia', 0), ('Carmella Vanwagenen', 0), ('Carolyn Kay', 0), ('Ellen Millican', 0), ('Cody Basta', 0), ('Christian Klemm', 0), ('Keisha Ragusa', 0), ('Jimmy Pritchard', 0), ('Noemi Widrick', 0), ('Russell Treat', 0), ('Becky Zinn', 0), ('Earnestine Croswell', 5), ('Craig Dash', 5), ('Harold Andre', 5), ('Lonnie Neveu', 5), ('Yvette Chenault', 5), ('Mark Grimm', 5), ('Bobby Faris', 5), ('Sean Howerton', 5), ('Walter Chapa', 5), ('Adam Feeney', 5), ('Carmella Sheffey', 5), ('Stephen Core', 5), ('Candice Brower', 5), ('Violet Felker', 5), ('Joshua Kingsbury', 5), ('Larry Mitchel', 5), ('Tina Haugen', 5), ('Donald Parmer', 5), ('Dean Scioneaux', 5), ('Geraldine Flora', 5), ('Keith Hurt', 5), ('Keith Hyder', 5), ('Jason Word', 5), ('Jamie Sing', 5), ('Adam Bingham', 5), ('Larry Lamp', 7), ('Carole Stuckey', 8), ('Amelia Mckeever', 8), ('Lois Sexton', 11), ('Ryan Giglio', 11), ('Paul Gowen', 11), ('Lawrence Hecker', 11), ('Larry Weatherspoon', 14), ('Juanita Wildman', 14), ('Clarence Correll', 14), ('Alan Harness', 14), ('Pearlie Storie', 14), ('Norman Wingfield', 14), ('Cindy Motley', 14), ('Roberto Routon', 14), ('Gustavo Sysak', 14), ('Pauline Penton', 14), ('Leonard Urbina', 14), ('Roslyn Philipp', 14), ('Doris Mays', 14), ('Zelma Vallier', 14), ('Hillary Tall', 14), ('Nicole Ellingson', 14), ('Nora Hashimoto', 14), ('Jesse Sandstrom', 14), ('Louis Schreiber', 14), ('Jeffery Ruiz', 17), ('Jesse Horn', 17), ('Gerald Whitesell', 17), ('George Blow', 17), ('Darlene Sandler', 17), ('Sharonda Remsberg', 17), ('Yvonne Mickens', 19), ('Brandi Keil', 19), ('Todd Huss', 19), ('Amelia Meacham', 24), ('Douglas Sedillo', 24), ('Max Madamba', 24), ('Curtis Rounds', 24), ('Kristy Strom', 24), ('Dianne Mixon', 24), ('Howard Hake', 24), ('Anthony Pires', 24), ('Corey Devane', 24), ('Daniel Deloach', 24), ('Lillie Dunton', 24), ('Violet Eden', 24), ('Eugene Forman', 24), ('Curtis Bowyer', 24), ('Shelley Scherer', 29), ('Tanisha Gamino', 29), ('Natalie Bynum', 29), ('Christopher List', 29), ('Lilia Rojero', 29), ('Bruce Rickert', 29), ('Lester Gallivan', 29), ('Craig Baker', 29), ('Fannie Hesse', 29), ('Elizabeth Dismuke', 29), ('Diana Welker', 29), ('Christina Driskell', 29), ('Kristine Cave', 29), ('Isabel Peart', 29), ('Ingrid Devaul', 33), ('Lance Buono', 33), ('Katy Tressler', 33), ('Patrick Bahena', 33), ('Debbie Sapp', 33), ('Walter Roussel', 33), ('Elnora Coppedge', 33), ('Loraine Montford',

33), ('Stephen Mccool', 33), ('Ted Yetman', 33), ('Teresa Samaniego', 33), ('Jeremy Trainor', 33), ('Tyrone Ottman', 33), ('Esther Hale', 33), ('Raven Dalgleish', 33), ('Hugo Vogelgesang', 33), ('Selena Lenart', 33), ('Scott Simons', 35), ('Kenneth Hudgens', 35), ('Robert Cooks', 35), ('Anthony Ramey', 35), ('Gary Poole', 35), ('Velma Burnett', 35), ('Lonnie Kitzmiller', 35), ('Nelson Coltrane', 35), ('Hector Burdge', 35), ('Louis Koss', 35), ('Beatrice Rose', 35), ('Gary Tate', 35), ('Asia Gucman', 35), ('Yvonne Mccune', 35), ('Eugene Singh', 35), ('Anna Marble', 35), ('Charlene Pennington', 35), ('Fernando Sobers', 35), ('Guy Vanalstyne', 35), ('Lula Eskridge', 35), ('Nora Neel', 35), ('Jill Dickerson', 36), ('Kyla Hillin', 36), ('Della Carlile', 38), ('Steve Ellingson', 38), ('Mathew Eslick', 38), ('Lorraine Ake', 38), ('Justin Kuehn', 38), ('Laura Flanigan', 38), ('Stanley Bybee', 38), ('Shane Dare', 38), ('Chandra Trowell', 38), ('Verna Basso', 38), ('Jesse Pinon', 38), ('Douglas Perrotta', 38), ('Jeanne Twigg', 38), ('Veronica Voss', 38), ('Laura Atwater', 38), ('Matthew Will', 38), ('Guy Gonsoulin', 38), ('Julio Naccarato', 38), ('Andrew Colunga', 41), ('Mark Tate', 41), ('Ellen Childress', 42), ('Donald Daly', 42), ('Roger Pellerin', 42), ('Kristin Keel', 42), ('Juan Hermes', 42), ('Roger Arriola', 42), ('Cora Mills', 42), ('Carolina Hamid', 42), ('Scott Prescott', 42), ('Tara Nissen', 42), ('Mathew Mercedes', 42), ('Darren Blau', 42), ('Cherie Mcguckin', 42), ('Marvin Hussey', 42), ('Terry Brush', 42), ('Lester Maya', 42), ('Allen Towns', 42), ('Adrienne Savoy', 44), ('Jose Burge', 44), ('Neil Hunton', 44), ('Steve Hix', 44), ('Chris Mcmorris', 44), ('Frank Dunkin', 44), ('Joyce Etter', 44), ('Dianne Alarcon', 44), ('Renee Field', 44), ('Terry Caudle', 44), ('Paul Vaillancourt', 51), ('Clara Wainwright', 51), ('Antonio Bischoff', 51), ('Dan Lain', 51), ('Dean Bergey', 51), ('Sabina Corsaut', 51), ('Norman Mastin', 51), ('Louis Marino', 51), ('Keith Banister', 51), ('Dale Charlton', 51), ('Terry Tweed', 51), ('Bryan Corcoran', 51), ('Gustavo Theall', 51), ('Laurie Fullmer', 51), ('Michael Scheffler', 51), ('Marguerite Mccabe', 51), ('Molly Wilkinson', 51), ('Karen Kinard', 51), ('Anthony Deaver', 52), ('Mike Schumann', 52), ('Eunice Barr', 52), ('Christine Prather', 52), ('Alice Reese', 52), ('Harry Skeen', 52), ('Jack Tso', 52), ('Gregory Dantzler', 52), ('Timothy Markel', 52), ('Arthur Runge', 52), ('Tanya Pickering', 52), ('Valerie Byrd', 53), ('Vincent Berkley', 53), ('Hilda Wilmoth', 53), ('Alberta Bradley', 53), ('Jimmy Munoz', 53), ('Joseph Audet', 53), ('Mark Dillman', 53), ('Billy Partridge', 53), ('Roger Cool', 53), ('Sofia Kelling', 53), ('Shawn Leigh', 53), ('Joseph Donohue', 53), ('Rita Witkowski', 53), ('Luz Hong', 53), ('Darlene Bergman', 61), ('Hazel Persinger', 61), ('Philip Gauthier', 61), ('Susie Peoples', 61), ('Billy Luu', 61), ('Katherine Ryan', 61), ('Kenneth Lewellen', 61), ('Jeffery Bounds', 61), ('Arlene Wellington', 61), ('Michele Ford', 61), ('Joy Cloud', 61), ('Shelia Sorrells', 61), ('Travis Trull', 61), ('Willie Pence', 62), ('Jeanette Herbst', 62), ('Roberto Munroe', 62), ('Rodney Toth', 62), ('Javier Weatherwax', 62), ('Yvette Scalise', 62), ('Matthew Beaman', 62), ('Jorge Madry', 62), ('Ruby Traylor', 62), ('Kurt Gettings', 62), ('Daniel Binkley', 62), ('Craig Bulter', 62), ('Brent Curtsinger', 62), ('Walter Sappington', 62), ('Nereida Viscardi', 67), ('Clarence Batchelder', 67), ('Jose Flannery', 67), ('Christian Durall', 67), ('Belen Sheerin', 67), ('Jessie Mayville', 67), ('Sarah Waterman', 67), ('Chad Maclean', 67), ('Neil Rosenbeck', 67), ('Audrey Buck', 67), ('Pamela French', 67), ('Herman

Appling', 67), ('Valerie Jenson', 67), ('Thomas Giffin', 76), ('Cody Russom', 76), ('Amelia Jarrell', 76), ('Eugene Gore', 76), ('Darren Pedrosa', 76), ('Guy Winders', 76), ('Verla Pali', 76), ('Christa Monsour', 76), ('Jacob Dandrea', 76), ('Dorothy Groves', 76), ('Brandon Saari', 76), ('Chris Reiner', 76), ('Clara Huck', 76), ('Dennis Donahue', 76)]

Duration taken by QuickSort to sort 500 pickup requests in reverse order: 0.005106 seconds

```
Duration taken by QuickSort to sort 500 pickup requests in reverse order: 0.005106 seconds

The Median of 3 QuickSort algorithm Reflection for different scenarios of 500 pickup requests:
Duration to sort 500 pickup requests in random order: 0.004800 seconds
Duration to sort 500 pickup requests in nearly sorted order: 0.004988 seconds
Duration to sort 500 pickup requests in reverse order: 0.005106 seconds

Looking at the durations for each scenario of 500 pickup request, QuickSort Median of 3 takes less time to
    sort the random list while more time is taken to sort the reverse list
Its time complexity is O(n log n) and its space complexity is Olog(n).
```

Figure 76: Short Reflection of using Quick Sort on different 500 pickup request lists

Dataset of 1000 pickup requests:

1000 pickup request list in random order:

Test List of 1000 pickup requests in random order

[('Fernando Mccardell', 61), ('Lance Rein', 61), ('Cecile Lafontaine', 33), ('Allen Mckinney', 53), ('Claire Britton', 0), ('Jorge Ventimiglia', 0), ('Darlene Sandler', 0), ('Timothy Congdon', 0), ('Marlene Berman', 14), ('Sofia Mcinturff', 33), ('Thomas Hardeman', 76), ('Eunice Barr', 0), ('Wilfred Mourning', 0), ('Jeremy Hyman', 0), ('Shelia Sorrells', 0), ('Lawrence Velarde', 62), ('Walter Sappington', 27), ('Tanya Pickering', 0), ('Gregory Dantzler', 0), ('Matthew Soper', 14), ('Jodi Derosier', 33), ('Justin Skelton', 42), ('Jeffery Bounds', 0), ('Bruce Rickert', 0), ('Selena Hoose', 14), ('Roger Sung', 5), ('Noemi Marcucci', 0), ('Roxie Underdahl', 61), ('Arthur Larrabee', 0), ('Charles Babich', 5), ('Elnora Coppedge', 0), ('Sofia Bonfiglio', 29), ('George Blow', 0), ('Patsy Horne', 0), ('Gary Tate', 0), ('Verna Basso', 0), ('Chris Rahman', 0), ('Max Kirkendall', 0), ('Keith Overturf', 76), ('Neil Brabant', 5), ('Danny Hartsfield', 38), ('Russell Ashmore', 0), ('Katie Plunkett', 35), ('Donna Pepin', 14), ('Lula Eskridge', 0), ('Dale Ormsby', 0), ('Valerie Jenson', 0), ('Stanley Hindman', 76), ('Maurice Stilwell', 42),

('Penelope Gayman', 24), ('Hugo Vogelgesang', 0), ('Joseph Donohue', 0), ('Stella Morrissey', 29), ('Latonya Norby', 51), ('Inez Basile', 35), ('Chad Steffens', 0), ('Sonia Pippin', 44), ('Dustin Gadsden', 0), ('Justin Soloman', 0), ('Asia Gucman', 0), ('Lillie Frisby', 0), ('David Crandell', 44), ('Ralph Cartagena', 33), ('Kelly Cheatam', 0), ('Rick Monday', 14), ('Cora Stauffer', 29), ('Javier Mungia', 0), ('Douglas Dabney', 0), ('Lynda Blum', 44), ('Jose Flannery', 0), ('Kelly Hilchey', 51), ('Bruce Dailey', 0), ('Juan Pink', 0), ('Terry Tweed', 11), ('Geraldine Macon', 52), ('Richard Melvin', 0), ('Candice Brower', 0), ('Louise Wendt', 76), ('Gustavo Sysak', 0), ('Travis Mojica', 42), ('Shane Dare', 0), ('Clarence Correll', 0), ('Mark Tate', 0), ('Christopher Merwin', 0), ('Raymond Hynes', 0), ('Douglas Perrotta', 8), ('Veronica Voss', 0), ('Philip Gauthier', 0), ('Shawn Rodriquez', 0), ('Erma Creech', 0), ('Vera Sargeant', 52), ('Maggie Molloy', 29), ('Hillary Tall', 0), ('William Backus', 0), ('Brandon Phelps', 61), ('Jimmy Fults', 67), ('Sean Holley', 14), ('Ruby Avent', 0), ('Laurie Coons', 62), ('Nicole Prevost', 29), ('Leonard Willett', 44), ('Fannie Tipton', 62), ('Nicole Ellingson', 0), ('Bessie Minton', 0), ('Yvette Scalise', 0), ('James Hayward', 76), ('Zelma Vallier', 0), ('Darren Hadaway', 0), ('Nicholas Applegate', 14), ('Noemi Vancamp', 52), ('Travis Trull', 0), ('Brandon Mask', 0), ('Debbie Northcutt', 38), ('Maricela Godard', 38), ('Tania Silvernail', 35), ('Lonnie Neveu', 0), ('Keith Hurt', 0), ('Nicholas Goins', 35), ('Samantha Giroux', 62), ('Mark Grimm', 0), ('Juana Hodgson', 29), ('Melinda Millard', 29), ('Nicole Robinson', 35), ('Carolina Hamid', 0), ('Violet Felker', 0), ('Craig Baker', 0), ('Claudia Sthilaire', 61), ('Paul Edgar', 42), ('Nancy Welker', 62), ('Shawn Leigh', 0), ('Joshua Spooner', 0), ('Christa Monsour', 36), ('Della Carlile', 0), ('Joseph Linville', 0), ('Jillian Mcdougle', 42), ('Harry Carney', 14), ('Heather Tanaka', 62), ('Clayton Salls', 0), ('Wanda Harry', 35), ('Raymond Benge', 44), ('Deanna Whalen', 24), ('Mike Mcewen', 5), ('Elizabeth Dismuke', 0), ('Frank Smalls', 33), ('Chad Doucet', 0), ('Erika Cortes', 52), ('Tony Bost', 67), ('Matthew Massa', 0), ('Aaron Dalessio', 24), ('Agnes Fillmore', 33), ('April Wigfall', 52), ('Adam Bingham', 0), ('Guy Loredo', 0), ('Larry Chapa', 0), ('Darren Cotta', 0), ('Elsie Sowers', 0), ('Tyrone Sturman', 14), ('Wesley Harm', 0), ('Hazel Persinger', 0), ('Keith Bowlin', 76), ('Jorge Madry', 0), ('Arthur Runge', 0), ('Hector Horning', 0), ('Tonya Canfield', 0), ('Albert Lockett', 0), ('Allen Velarde', 0), ('Albert Rohrer', 44), ('Violet Eden', 0), ('Hector Burdge', 0), ('Leonard Philpot', 38), ('Milagros Harbuck', 52), ('Danny Corrigan', 0), ('Edward Bliss', 67), ('Jose Gregg', 0), ('Roy Hurtado', 53), ('Gary Quillen', 0), ('Kenneth Casares', 0), ('Tyrone Mondor', 0), ('Ted Yetman', 0), ('Patsy Finlay', 0), ('Leonard Urbina', 0), ('Opal Royal', 24), ('Vincent Dionne', 42), ('Harriett Curatolo', 0), ('Roy Maes', 14), ('Lester Maya', 0), ('Bruce Waldman', 62), ('Hazel Staten', 35), ('Luz Coomer', 0), ('Anthony Deaver', 0), ('Douglas Bischoff', 38), ('Dorothy Groves', 0), ('Walter Roussel', 0), ('Glenna Meaney', 0), ('Teresa Samaniego', 0), ('Katy Tressler', 0), ('Judy Burress', 62), ('Susie Haley', 44), ('Leon Urbina', 0), ('Craig Bulter', 0), ('Juan Vandyke', 0), ('Chris Embry', 0), ('Karen Pena', 0), ('Tammy Riddle', 61), ('Tina Haugen', 0), ('Lena Judge', 0), ('Amelia Mckeever', 0), ('Helen Farina', 52), ('Guy Fear', 33), ('Larry Lamp', 0), ('Kenneth Lewellen', 0), ('Betty Chi', 61), ('Wayne Merkle', 0), ('Constance Berg', 67), ('Arlene Wellington', 0), ('Vernon Sermons', 0),

('Darlene Bergman', 0), ('Kristin Keel', 0), ('James Urrutia', 0), ('Rebecca Eversole', 52), ('Suzanne Hiles', 52), ('Shanna Elsea', 0), ('Lucy Pisano', 44), ('Allan Hornbuckle', 33), ('Katherine Ryan', 0), ('Christine King', 0), ('Billy Partridge', 0), ('Molly Wilkinson', 11), ('Charlene Mccormack', 61), ('Paul Albers', 0), ('Dean Bergey', 11), ('Geraldine Vargas', 0), ('Jeffery Blount', 24), ('Susan Roby', 29), ('Chris Reiner', 0), ('Renee Field', 17), ('Christian Klemm', 0), ('Yvonne Mickens', 0), ('Vincent Malloy', 14), ('Velma Burnett', 0), ('Kelli Schulze', 0), ('Anthony Demarco', 29), ('Gerald Whitesell', 0), ('Eugene Molina', 29), ('Ernest Doherty', 52), ('Tanisha Coppin', 33), ('Steve Rushing', 0), ('Michael Lefevre', 0), ('Clarence Batchelder', 0), ('Andrew Colunga', 0), ('Jeanne Twigg', 0), ('Antonio Bischoff', 11), ('Allen Towns', 0), ('Larry Gehrke', 24), ('Shawn Standley', 24), ('Victor Leffler', 52), ('Sean Howerton', 0), ('Judy Lucia', 61), ('Gustavo Theall', 11), ('Gary Poole', 0), ('Jeff Singletary', 0), ('Rodney Toth', 0), ('Eileen Fischer', 24), ('Carmella Sheffey', 0), ('Sabina Mcmikle', 24), ('Dorothea Ruckman', 33), ('Nathan Pound', 33), ('Howard Shinn', 5), ('Ryan Giglio', 0), ('Cody Basta', 0), ('Nelson Weyer', 35), ('Norman Grimaldo', 24), ('Terry Brush', 0), ('Carolyn Kay', 0), ('Howard Hake', 0), ('Anthony Farley', 61), ('Lawrence Hecker', 0), ('Ingrid Devaul', 0), ('Javier Kratochvil', 0), ('Hector Kravitz', 44), ('William Trapp', 14), ('Steve Hix', 17), ('Mark Dillman', 0), ('Mike Schumann', 0), ('Curtis Bowyer', 0), ('Justin Mccleary', 51), ('Joseph Favors', 0), ('Jennifer Hoffmann', 53), ('Sarah Waterman', 0), ('Daisy Palladino', 0), ('Lakisha Izquierdo', 51), ('Rosemary Mcculley', 0), ('Esther Hale', 0), ('Kimberly Harrison', 29), ('Paul Hagler', 67), ('Isabel Peart', 0), ('Pearlie Willams', 76), ('Doris Mays', 0), ('Thomas Rawlins', 38), ('Kristy Strom', 0), ('Christine Prather', 0), ('Christopher List', 0), ('Carl Campos', 0), ('Max Madamba', 0), ('Chad Maclean', 0), ('Yvette Chenault', 0), ('Darren Blau', 0), ('Tyrone Letchworth', 52), ('Juana Greenleaf', 0), ('Corinne Kocsis', 0), ('Clyde Vosburgh', 61), ('Geraldine Flora', 0), ('Laura Flanigan', 8), ('Donald Parmer', 0), ('Lance Buono', 24), ('Patrick Bahena', 0), ('Chris Omara', 35), ('Danny Steinberg', 44), ('Timothy Vetter', 0), ('Stephen Fitz', 0), ('Alice Reese', 0), ('Anna Marble', 0), ('Jeff Vanbuskirk', 0), ('Jessie Mayville', 0), ('Joyce Poole', 42), ('Candice Swisher', 0), ('Stephen Core', 0), ('Fred Meeker', 0), ('Edwina Fedor', 52), ('Joseph Desimone', 38), ('Edward Reader', 0), ('Clara Wainwright', 16), ('Yvonne Mccune', 0), ('Norman Styles', 0), ('Gregory Despain', 0), ('Jerry Herring', 0), ('Cherie Mcguckin', 0), ('Nita Sandor', 5), ('Lois Karl', 29), ('Valerie Byrd', 0), ('Clinton Escalona', 52), ('Tony Burnett', 24), ('Daniel Deloach', 0), ('Ronald Rushton', 0), ('Dale Adamson', 42), ('Adrienne Lasley', 29), ('Jacob Ingle', 0), ('Cora Thorson', 29), ('Corey Sloat', 14), ('Noreen Smyre', 5), ('Harry Skeen', 0), ('Brandi Keil', 0), ('George Manchester', 0), ('Faye Scribner', 62), ('Cathy Tubbs', 24), ('Herman Appling', 0), ('Theresa Townsley', 0), ('Kevin Omara', 52), ('Craig Dash', 0), ('Antonio Braddock', 0), ('Vincent Roark', 5), ('Curtis Rounds', 0), ('Deanna Arndt', 0), ('Terry Caudle', 17), ('Vincent Augustine', 52), ('Jimmy Doak', 67), ('Kurt Gettings', 0), ('Lance Newhard', 29), ('Justin Moye', 42), ('Clarence Lamar', 38), ('Ethel Glidden', 5), ('Harold Hamill', 0), ('Joshua Zack', 33), ('Vivian Milton', 0), ('Keith Hyder', 0), ('Kristen Pulliam', 35), ('Roslyn Philipp', 5), ('Yvette Dowdy', 0), ('Brent Bovee', 35), ('Johnny

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1000 pickup request list in nearly sorted order

[('Hillary Tall', 0), ('Cody Basta', 0), ('Elnora Coppedge', 0), ('Jeff Singletary', 0), ('Larry Weatherspoon', 0), ('Jorge Ventimiglia', 0), ('Stephen Fitz', 0), ('Timothy Congdon', 0), ('Doris Mays', 0), ('Lawrence Velarde', 62), ('Gary Thorson', 0), ('Jodi Holm', 0), ('Steve Ellingson', 0), ('Howard Storm', 0), ('Shelia Sorrells', 0), ('Gregory Despain', 0), ('Craig Bulter', 0), ('Bessie Minton', 0), ('Esther Hale', 0), ('Michael Michel', 0), ('Antonio Braddock',

0), ('Benjamin Prado', 0), ('Jeffery Bounds', 0), ('Peter Whitman', 0), ('Ingrid Devaul', 0), ('Marilyn Mickens', 0), ('Daniel Binkley', 0), ('Noreen Durfey', 0), ('Stanley Bybee', 0), ('Harold Hamill', 0), ('Jeremy Hyman', 0), ('Eileen Bunch', 0), ('Chris Reiner', 0), ('Jessica Villegas', 0), ('Matthew Will', 0), ('Luz Coomer', 0), ('Chris Rahman', 0), ('Philip Hammonds', 0), ('Willie Pence', 0), ('Emily Chadwick', 0), ('Ruby Traylor', 0), ('Russell Ashmore', 0), ('Pearlie Storie', 0), ('Manuel Hammack', 0), ('Jeanne Twigg', 0), ('Dale Ormsby', 0), ('Lorraine Ake', 0), ('Jonathan Wagoner', 0), ('Guy Gonsoulin', 0), ('Teresa Booth', 0), ('Hugo Vogelgesang', 0), ('Jose Gregg', 0), ('Teresa Samaniego', 0), ('Karen Pena', 0), ('Yvette Scalise', 0), ('Chad Steffens', 0), ('Stephen Mccool', 0), ('Dustin Gadsden', 0), ('Della Carlile', 0), ('Kelly Cheatam', 0), ('Paul Albers', 0), ('Opal Tobias', 0), ('Katy Tressler', 0), ('Scott Luevano', 0), ('Curtis Cordell', 0), ('Chad Doucet', 0), ('Javier Mungia', 0), ('Jessie Efaw', 0), ('Larry Cantu', 0), ('Erik Weinman', 0), ('Leonard Urbina', 0), ('Bruce Dailey', 0), ('Dennis Donahue', 0), ('Olivia Fall', 0), ('Amelia Jarrell', 0), ('Richard Melvin', 0), ('Jesse Sandstrom', 0), ('Sherri Kinser', 0), ('Gustavo Sysak', 0), ('Vickie Giffin', 0), ('Shane Dare', 0), ('Jeremy Catalano', 0), ('Denise Herrington', 0), ('Christopher Merwin', 0), ('Raymond Hynes', 0), ('Timothy Vetter', 0), ('Wilma Rey', 0), ('Ronald Henning', 0), ('Fred Meeker', 0), ('Erma Creech', 0), ('Cody Russom', 0), ('Clarence Keyser', 0), ('Clarence Correll', 0), ('William Backus', 0), ('Verna Basso', 0), ('Ruby Phifer', 0), ('Darren Suchan', 0), ('Ruby Avent', 0), ('Nora Hashimoto', 0), ('Michele Valliere', 0), ('Daisy Palladino', 0), ('Allen Velarde', 0), ('Nicole Ellingson', 0), ('Danny Corrigan', 0), ('Jo Patino', 0), ('Rick Brathwaite', 0), ('Jodi Derosier', 33), ('Darren Hadaway', 0), ('Candy Henn', 0), ('Amy Boren', 0), ('Travis Trull', 0), ('Brandon Mask', 0), ('Marjorie Tello', 0), ('Tonya Niemi', 0), ('Jonathan Gower', 0), ('George Manchester', 0), ('Chad Pardo', 0), ('Keisha Ragusa', 0), ('Carolyn Kay', 0), ('Louise Bellamy', 0), ('Rodney Toth', 0), ('Carl Campos', 0), ('Brent Curtsinger', 0), ('Maribel Earnest', 0), ('Alicia Womble', 0), ('Paige Lovitt', 0), ('Kayla Paynter', 0), ('Inez Yee', 0), ('Roger Barnwell', 24), ('Wilfred Mourning', 0), ('Arthur Larrabee', 0), ('Rosemary Mcculley', 0), ('Guy Loredo', 0), ('Joseph Linville', 0), ('Veronica Voss', 0), ('Jeffery Marchant', 62), ('Arlene Wellington', 0), ('Clayton Salls', 0), ('Raven Dalgleish', 0), ('Craig Avent', 0), ('Joy Cloud', 0), ('Marian Ruble', 44), ('Tyrone Mondor', 0), ('Mabel Gist', 0), ('Jeanette Herbst', 0), ('Harry Norfleet', 0), ('Judith Lemay', 0), ('Matthew Massa', 0), ('Jerry Herring', 0), ('Kelli Schulze', 0), ('Johnny Dozier', 0), ('Zachary Falgoust', 0), ('Michele Ford', 0), ('Ethel Reitz', 0), ('Daniel Trimble', 0), ('Elsie Sowers', 0), ('Max Kirkendall', 0), ('Wesley Harm', 0), ('Hazel Persinger', 0), ('Dennis Housley', 0), ('Jorge Madry', 0), ('Claire Britton', 0), ('Hector Horning', 0), ('Tonya Canfield', 0), ('Albert Lockett', 0), ('Carmella Rondon', 62), ('Jeremy Trainor', 0), ('Juan Vandyke', 0), ('Corinne Kocsis', 0), ('Ingrid Mutter', 0), ('Marguerite Mccabe', 11), ('Danielle Tincher', 0), ('Edward Scalise', 0), ('Antonio Obryan', 0), ('Alice Knoll', 0), ('Gary Quillen', 0), ('Kenneth Casares', 0), ('Lillie Frisby', 0), ('Ted Yetman', 0), ('Geraldine Vargas', 0), ('Eugene Gore', 0), ('Alma Mcneal', 0), ('Jacob Dandrea', 0), ('Harriett Curatolo', 0), ('Darren Pedrosa', 0), ('Jeff Moten', 0), ('Loraine Montford', 0), ('Harriett Loveall', 0), ('Ellen Millican', 0), ('Justin Hutson', 0), ('Brian Martindale', 0), ('Johnny Turnbull', 0), ('Walter

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('Javier Crichton', 14), ('Vincent Malloy', 14), ('Guy Cullinan', 14), ('Matthew Dowden', 14), ('Donald Mack', 14), ('Rick Monday', 14), ('Tyra Rhew', 14), ('Johnny Watt', 14), ('Matthew Soper', 14), ('Russell Parson', 14), ('Roy Maes', 14), ('Corey Sloat', 14), ('Zachary Carruthers', 14), ('Tyrone Sturman', 14), ('Dan Ulm', 14), ('Shantel Zevallos', 14), ('Henry Mckenna', 14), ('Kelli Sallee', 14), ('Ryan Bennet', 14), ('Daisy Nino', 14), ('Antonio Bischoff', 11), ('Donald Daly', 11), ('Bryan Corcoran', 11), ('Molly Wilkinson', 11), ('Terry Tweed', 11), ('Keith Banister', 11), ('Karen Kinard', 11), ('Dean Bergey', 11), ('Louis Marino', 11), ('Marguerite Mccabe', 11), ('Sabina Corsaut', 11), ('Liza Borchers', 11), ('Dale Charlton', 11), ('Norman Mastin', 11), ('Paul Vaillancourt', 11), ('Michael Scheffler', 11), ('Gustavo Theall', 11), ('Dan Lain', 11), ('Martha Holloway', 11), ('Laura Flanigan', 8), ('Laura Atwater', 8), ('Jesse Pinon', 8), ('Douglas Perrotta', 8), ('Chandra Trowell', 8), ('Julio Naccarato', 8), ('Roslyn Philipp', 5), ('Howard Shinn', 5), ('Sandra Lacey', 5), ('Cindy Motley', 5), ('Kelsey Levey', 5), ('Javier Swint', 5), ('Wesley Newquist', 5), ('Charles Babich', 5), ('Kurt Bridgett', 5), ('Tiffany Sain', 5), ('Ellen Wimmer', 5), ('Thomas Peckham', 5), ('Vincent Roark', 5), ('Katherine Martens', 5), ('Nita Sandor', 5), ('Norman Wingfield', 5), ('Alberta Howie', 5), ('Mike Mcewen', 5), ('Roger Sung', 5), ('Roberto Routon', 5), ('Clint Tsuchiya', 5), ('Shelia Roberts', 5), ('Ralph Parks', 5), ('David Gabbard', 5), ('Russell Hollar', 5), ('Caroline Bible', 5), ('Jenny Gold', 5), ('Shelia Deck', 5), ('Ethel Glidden', 5), ('Juanita Wildman', 5), ('Kenneth Frizzell', 5), ('Alan Harness', 5), ('Noreen Smyre', 5), ('Dolores Moses', 5), ('Scott Beckwith', 5), ('Louis Schreiber', 5), ('Jose Haight', 5), ('Pauline Penton', 5), ('Priscilla Ellis', 5), ('Roberta Stephens', 5), ('Neil Brabant', 5), ('Leonard Delrio', 0), ('Martin Harbin', 0), ('Brandi Mcauliffe', 0), ('Amy Bedford', 0), ('Stanley Littleton', 0), ('Joseph Donohue', 0), ('Antonio Braddock', 0), ('Patsy Finlay', 0), ('Sherry Winfrey', 0), ('Annie Markel', 0), ('Mike Rodarte', 0), ('Jacob Meeker', 0), ('Tony Brogan', 0), ('Javier Hausman', 0), ('Phyllis Beene', 0), ('Kevin Schreiber', 0), ('Orville Gillaspie', 0), ('Elaine Elias', 0), ('Ramona Stacy', 0), ('Brandi Keil', 0), ('Douglas Dabney', 0), ('Denise Ferraro', 0), ('Eleanor Fontenot', 0), ('Gerald Cousin', 0), ('Sonia Bass', 0), ('Carole Stuckey', 0), ('Harriett Zollinger', 0), ('Russell Treat', 0), ('Wayne Merkle', 0), ('Glenna Meaney', 0), ('Debbie Ruch', 0), ('Norman Mcnulty', 0), ('Gabriela Keaney', 0), ('Nelson Guitierrez', 0), ('Allen Mckenna', 0), ('Dorcas Abbitt', 0), ('Jack Kolb', 0), ('Larry Chapa', 0), ('Jocelyn Lynde', 0), ('Cassandra Creed', 0), ('Adam Harden', 0), ('Jennie Tuggle', 0), ('Theresa Townsley', 0), ('Michael Lefevre', 0), ('Dean Buchanon', 0), ('Janice Granados', 0), ('Keith Dykes', 0), ('Becky Zinn', 0), ('Wayne Crabb', 0), ('Harold Seaman', 0), ('Norman Styles', 0), ('Joseph Favors', 0), ('Jeremy Hyman', 0), ('Manuel Wiser', 0), ('Shawn Rodriquez', 0), ('Gary Burmeister', 0), ('Yvette Dowdy', 0), ('Nelson Tooker', 0), ('Arlene Bergmann', 0), ('Edward Reader', 0), ('Vivian Milton', 0), ('Lester Kealey', 0), ('Deanna Arndt', 0), ('Eva Pursley', 0), ('Keith Hyder', 0), ('Cindy Penney', 0), ('Thomas Giffin', 0), ('Kurt Gettings', 0), ('Todd Huss', 0), ('Louise Lively', 0), ('Douglas Sedillo', 0), ('Jack Tso', 0), ('Curtis Rounds', 0), ('Scott Ring', 0), ('Jacob Ingle', 0), ('Craig Dash', 0), ('Christian Durall', 0), ('Yvonne Mccune', 0), ('Herman Appling', 0), ('Robert Cooks', 0), ('Darren Blau', 0), ('Lonnie Neveu', 0), ('Guy Vanalstyne', 0), ('Harry

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0), ('Lillie Frisby', 0), ('Kenneth Casares', 0), ('Gary Quillen', 0), ('Alice Knoll', 0), ('Antonio Obryan', 0), ('Edward Scalise', 0), ('Danielle Tincher', 0), ('Noemi Marcucci', 0), ('Ingrid Mutter', 0), ('Corinne Kocsis', 0), ('Juan Vandyke', 0), ('Jeremy Trainor', 0), ('Jeff Vanbuskirk', 0), ('Albert Lockett', 0), ('Tonya Canfield', 0), ('Hector Horning', 0), ('Claire Britton', 0), ('Jorge Madry', 0), ('Dennis Housley', 0), ('Hazel Persinger', 0), ('Wesley Harm', 0), ('Max Kirkendall', 0), ('Elsie Sowers', 0), ('Daniel Trimble', 0), ('Ethel Reitz', 0), ('Michele Ford', 0), ('Zachary Falgoust', 0), ('Johnny Dozier', 0), ('Kelli Schulze', 0), ('Jerry Herring', 0), ('Matthew Massa', 0), ('Judith Lemay', 0), ('Harry Norfleet', 0), ('Jeanette Herbst', 0), ('Mabel Gist', 0), ('Tyrone Mondor', 0), ('Candice Swisher', 0), ('Joy Cloud', 0), ('Craig Avent', 0), ('Raven Dalgleish', 0), ('Clayton Salls', 0), ('Arlene Wellington', 0), ('Luis Lehmann', 0), ('Wilma Rey', 0), ('Joseph Linville', 0), ('Guy Loredo', 0), ('Rosemary Mcculley', 0), ('Arthur Larrabee', 0), ('Wilfred Mourning', 0), ('Dorothy Groves', 0), ('Inez Yee', 0), ('Kayla Paynter', 0), ('Paige Lovitt', 0), ('Alicia Womble', 0), ('Maribel Earnest', 0), ('Brent Curtsinger', 0), ('Carl Campos', 0), ('Rodney Toth', 0), ('Louise Bellamy', 0), ('Carolyn Kay', 0), ('Keisha Ragusa', 0), ('Chad Pardo', 0), ('George Manchester', 0), ('Jonathan Gower', 0), ('Tonya Niemi', 0), ('Marjorie Tello', 0), ('Brandon Mask', 0), ('Travis Trull', 0), ('Amy Boren', 0), ('Candy Henn', 0), ('Darren Hadaway', 0), ('Zelma Vallier', 0), ('Rick Brathwaite', 0), ('Jo Patino', 0), ('Danny Corrigan', 0), ('Nicole Ellingson', 0), ('Allen Velarde', 0), ('Daisy Palladino', 0), ('Michele Valliere', 0), ('Nora Hashimoto', 0), ('Ruby Avent', 0), ('Darren Suchan', 0), ('Ruby Phifer', 0), ('Verna Basso', 0), ('William Backus', 0), ('Clarence Correll', 0), ('Clarence Keyser', 0), ('Cody Russom', 0), ('Erma Creech', 0), ('Fred Meeker', 0), ('Ronald Henning', 0), ('Veronica Voss', 0), ('Timothy Vetter', 0), ('Raymond Hynes', 0), ('Christopher Merwin', 0), ('Denise Herrington', 0), ('Jeremy Catalano', 0), ('Shane Dare', 0), ('Vickie Giffin', 0), ('Gustavo Sysak', 0), ('Sherri Kinser', 0), ('Jesse Sandstrom', 0), ('Richard Melvin', 0), ('Amelia Jarrell', 0), ('Olivia Fall', 0), ('Dennis Donahue', 0), ('Bruce Dailey', 0), ('Leonard Urbina', 0), ('Erik Weinman', 0), ('Larry Cantu', 0), ('Jessie Efaw', 0), ('Javier Mungia', 0), ('Chad Doucet', 0), ('Curtis Cordell', 0), ('Scott Luevano', 0), ('Katy Tressler', 0), ('Opal Tobias', 0), ('Paul Albers', 0), ('Kelly Cheatam', 0), ('Della Carlile', 0), ('Dustin Gadsden', 0), ('Stephen Mccool', 0), ('Chad Steffens', 0), ('Yvette Scalise', 0), ('Karen Pena', 0), ('Teresa Samaniego', 0), ('Jose Gregg', 0), ('Hugo Vogelgesang', 0), ('Teresa Booth', 0), ('Guy Gonsoulin', 0), ('Jonathan Wagoner', 0), ('Lorraine Ake', 0), ('Dale Ormsby', 0), ('Jeanne Twigg', 0), ('Manuel Hammack', 0), ('Pearlie Storie', 0), ('Russell Ashmore', 0), ('Ruby Traylor', 0), ('Emily Chadwick', 0), ('Willie Pence', 0), ('Philip Hammonds', 0), ('Chris Rahman', 0), ('Luz Coomer', 0), ('Matthew Will', 0), ('Jessica Villegas', 0), ('Chris Reiner', 0), ('Eileen Bunch', 0), ('Matthew Beaman', 0), ('Harold Hamill', 0), ('Stanley Bybee', 0), ('Noreen Durfey', 0), ('Daniel Binkley', 0), ('Marilyn Mickens', 0), ('Ingrid Devaul', 0), ('Peter Whitman', 0), ('Jeffery Bounds', 0), ('Benjamin Prado', 0), ('Guy Winders', 0), ('Michael Michel', 0), ('Esther Hale', 0), ('Bessie Minton', 0), ('Craig Bulter', 0), ('Gregory Despain', 0), ('Shelia Sorrells', 0), ('Howard Storm', 0), ('Steve Ellingson', 0), ('Jodi Holm', 0), ('Gary Thorson', 0), ('Vernon Moultrie', 0), ('Doris Mays', 0), ('Timothy Congdon', 0), ('Stephen

Fitz', 0), ('Jorge Ventimiglia', 0), ('Larry Weatherspoon', 0), ('Jeff Singletary', 0), ('Elnora Coppedge', 0), ('Cody Basta', 0), ('Hillary Tall', 0)]

Merge Sort for different 1000 pickup request lists

1000 pickup request list in random order sorted

Note: The first names are not the same for the different test scenarios as the lists are sorted using only the pickup times not the names.

[('Claire Britton', 0), ('Jorge Ventimiglia', 0), ('Darlene Sandler', 0), ('Timothy Congdon', 0), ('Eunice Barr', 0), ('Wilfred Mourning', 0), ('Jeremy Hyman', 0), ('Shelia Sorrells', 0), ('Tanya Pickering', 0), ('Gregory Dantzler', 0), ('Jeffery Bounds', 0), ('Bruce Rickert', 0), ('Noemi Marcucci', 0), ('Arthur Larrabee', 0), ('Elnora Coppedge', 0), ('George Blow', 0), ('Patsy Horne', 0), ('Gary Tate', 0), ('Verna Basso', 0), ('Chris Rahman', 0), ('Max Kirkendall', 0), ('Russell Ashmore', 0), ('Lula Eskridge', 0), ('Dale Ormsby', 0), ('Valerie Jenson', 0), ('Hugo Vogelgesang', 0), ('Joseph Donohue', 0), ('Chad Steffens', 0), ('Dustin Gadsden', 0), ('Justin Soloman', 0), ('Asia Gucman', 0), ('Lillie Frisby', 0), ('Kelly Cheatam', 0), ('Javier Mungia', 0), ('Douglas Dabney', 0), ('Jose Flannery', 0), ('Bruce Dailey', 0), ('Juan Pink', 0), ('Richard Melvin', 0), ('Candice Brower', 0), ('Gustavo Sysak', 0), ('Shane Dare', 0), ('Clarence Correll', 0), ('Mark Tate', 0), ('Christopher Merwin', 0), ('Raymond Hynes', 0), ('Veronica Voss', 0), ('Philip Gauthier', 0), ('Shawn Rodriquez', 0), ('Erma Creech', 0), ('Hillary Tall', 0), ('William Backus', 0), ('Ruby Avent', 0), ('Nicole Ellingson', 0), ('Bessie Minton', 0), ('Yvette Scalise', 0), ('Zelma Vallier', 0), ('Darren Hadaway', 0), ('Travis Trull', 0), ('Brandon Mask', 0), ('Lonnie Neveu', 0), ('Keith Hurt', 0), ('Mark Grimm', 0), ('Carolina Hamid', 0), ('Violet Felker', 0), ('Craig Baker', 0), ('Shawn Leigh', 0), ('Joshua Spooner', 0), ('Della Carlile', 0), ('Joseph Linville', 0), ('Clayton Salls', 0), ('Elizabeth Dismuke', 0), ('Chad Doucet', 0), ('Matthew Massa', 0), ('Adam Bingham', 0), ('Guy Loredo', 0), ('Larry Chapa', 0), ('Darren Cotta', 0), ('Elsie Sowers', 0), ('Wesley Harm', 0), ('Hazel Persinger', 0), ('Jorge Madry', 0), ('Arthur Runge', 0), ('Hector Horning', 0), ('Tonya Canfield', 0), ('Albert Lockett', 0), ('Allen Velarde', 0), ('Violet Eden', 0), ('Hector Burdge', 0), ('Danny Corrigan', 0), ('Jose Gregg', 0), ('Gary Quillen', 0), ('Kenneth Casares', 0), ('Tyrone Mondor', 0), ('Ted Yetman', 0), ('Patsy Finlay', 0), ('Leonard Urbina', 0), ('Harriett Curatolo', 0), ('Lester Maya', 0), ('Luz Coomer', 0), ('Anthony Deaver', 0), ('Dorothy Groves', 0), ('Walter Roussel', 0), ('Glenna Meaney', 0), ('Teresa Samaniego', 0), ('Katy Tressler', 0), ('Leon Urbina', 0), ('Craig Bulter', 0), ('Juan Vandyke', 0), ('Chris Embry', 0), ('Karen Pena', 0), ('Tina Haugen', 0), ('Lena Judge', 0),

('Amelia Mckeever', 0), ('Larry Lamp', 0), ('Kenneth Lewellen', 0), ('Wayne Merkle', 0), ('Arlene Wellington', 0), ('Vernon Sermons', 0), ('Darlene Bergman', 0), ('Kristin Keel', 0), ('James Urrutia', 0), ('Shanna Elsea', 0), ('Katherine Ryan', 0), ('Christine King', 0), ('Billy Partridge', 0), ('Paul Albers', 0), ('Geraldine Vargas', 0), ('Chris Reiner', 0), ('Christian Klemm', 0), ('Yvonne Mickens', 0), ('Velma Burnett', 0), ('Kelli Schulze', 0), ('Gerald Whitesell', 0), ('Steve Rushing', 0), ('Michael Lefevre', 0), ('Clarence Batchelder', 0), ('Andrew Colunga', 0), ('Jeanne Twigg', 0), ('Allen Towns', 0), ('Sean Howerton', 0), ('Gary Poole', 0), ('Jeff Singletary', 0), ('Rodney Toth', 0), ('Carmella Sheffey', 0), ('Ryan Giglio', 0), ('Cody Basta', 0), ('Terry Brush', 0), ('Carolyn Kay', 0), ('Howard Hake', 0), ('Lawrence Hecker', 0), ('Ingrid Devaul', 0), ('Javier Kratochvil', 0), ('Mark Dillman', 0), ('Mike Schumann', 0), ('Curtis Bowyer', 0), ('Joseph Favors', 0), ('Sarah Waterman', 0), ('Daisy Palladino', 0), ('Rosemary Mcculley', 0), ('Esther Hale', 0), ('Isabel Peart', 0), ('Doris Mays', 0), ('Kristy Strom', 0), ('Christine Prather', 0), ('Christopher List', 0), ('Carl Campos', 0), ('Max Madamba', 0), ('Chad Maclean', 0), ('Yvette Chenault', 0), ('Darren Blau', 0), ('Juana Greenleaf', 0), ('Corinne Kocsis', 0), ('Geraldine Flora', 0), ('Donald Parmer', 0), ('Patrick Bahena', 0), ('Timothy Vetter', 0), ('Stephen Fitz', 0), ('Alice Reese', 0), ('Anna Marble', 0), ('Jeff Vanbuskirk', 0), ('Jessie Mayville', 0), ('Candice Swisher', 0), ('Stephen Core', 0), ('Fred Meeker', 0), ('Edward Reader', 0), ('Yvonne Mccune', 0), ('Norman Styles', 0), ('Gregory Despain', 0), ('Jerry Herring', 0), ('Cherie Mcguckin', 0), ('Valerie Byrd', 0), ('Daniel Deloach', 0), ('Ronald Rushton', 0), ('Jacob Ingle', 0), ('Harry Skeen', 0), ('Brandi Keil', 0), ('George Manchester', 0), ('Herman Appling', 0), ('Theresa Townsley', 0), ('Craig Dash', 0), ('Antonio Braddock', 0), ('Curtis Rounds', 0), ('Deanna Arndt', 0), ('Kurt Gettings', 0), ('Harold Hamill', 0), ('Vivian Milton', 0), ('Keith Hyder', 0), ('Yvette Dowdy', 0), ('Johnny Turnbull', 0), ('Vincent Berkley', 0), ('Debbie Ruch', 0), ('Lorraine Ake', 0), ('Arlene Bergmann', 0), ('Dean Buchanon', 0), ('Edward Scalise', 0), ('Cody Russom', 0), ('Curtis Cordell', 0), ('Jacob Dandrea', 0), ('Jeremy Catalano', 0), ('Ruby Traylor', 0), ('Norman Mcnulty', 0), ('Harold Seaman', 0), ('Daphne Renz', 0), ('Roger Cool', 0), ('Denise Herrington', 0), ('Eleanor Fontenot', 0), ('Denise Ferraro', 0), ('Elaine Elias', 0), ('Danielle Tincher', 0), ('Phyllis Beene', 0), ('Vernon Moultrie', 0), ('Tony Brogan', 0), ('Paul Gowen', 0), ('Sherry Winfrey', 0), ('Ellen Millican', 0), ('Stanley Littleton', 0), ('Peter Whitman', 0), ('Brandi Mcauliffe', 0), ('Shelley Scherer', 0), ('Alicia Womble', 0), ('Audrey Buck', 0), ('Wayne Crabb', 0), ('Jeremy Trainor', 0), ('Jeanette Herbst', 0), ('Scott Ring', 0), ('Wilma Rey', 0), ('Todd Huss', 0), ('Eugene Forman', 0), ('Lois Sexton', 0), ('Belen Sheerin', 0), ('Nelson Guitierrez', 0), ('Zachary Falgoust', 0), ('Jill Dickerson', 0), ('Anthony Ramey', 0), ('Kyla Hillin', 0), ('Adam Harden', 0), ('Russell Treat', 0), ('Jocelyn Lynde', 0), ('Guy Winders', 0), ('Howard Storm', 0), ('Dean Scioneaux', 0), ('Amy Bedford', 0), ('Manuel Wiser', 0), ('Fannie Hesse', 0), ('Philip Hammonds', 0), ('Johnny Dozier', 0), ('Javier Hausman', 0), ('Kayla Paynter', 0), ('Michele Ford', 0), ('Neil Rosenbeck', 0), ('Darren Suchan', 0), ('Candy Henn', 0), ('Ethel Reitz', 0), ('Scott Prescott', 0), ('Harry Norfleet', 0), ('Douglas Sedillo', 0), ('Nora Hashimoto', 0), ('Pamela French', 0), ('Sonia Bass', 0), ('Daniel Trimble', 0), ('Janice Granados', 0), ('Stephen

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('Roberto Routon', 5), ('Alberta Howie', 5), ('Ellen Wimmer', 5), ('Kurt Bridgett', 5), ('Sandra Lacey', 5), ('Scott Beckwith', 5), ('Louis Schreiber', 5), ('Katherine Martens', 5), ('Priscilla Ellis', 5), ('Tiffany Sain', 5), ('Dolores Moses', 5), ('Russell Hollar', 5), ('Kelsey Levey', 5), ('Cindy Motley', 5), ('Thomas Peckham', 5), ('Wesley Newquist', 5), ('Shelia Deck', 5), ('Jenny Gold', 5), ('Javier Swint', 5), ('Norman Wingfield', 5), ('David Gabbard', 5), ('Douglas Perrotta', 8), ('Laura Flanigan', 8), ('Chandra Trowell', 8), ('Jesse Pinon', 8), ('Laura Atwater', 8), ('Julio Naccarato', 8), ('Terry Tweed', 11), ('Molly Wilkinson', 11), ('Dean Bergey', 11), ('Antonio Bischoff', 11), ('Gustavo Theall', 11), ('Marguerite Mccabe', 11), ('Norman Mastin', 11), ('Michael Scheffler', 11), ('Martha Holloway', 11), ('Keith Banister', 11), ('Karen Kinard', 11), ('Bryan Corcoran', 11), ('Liza Borchers', 11), ('Dale Charlton', 11), ('Sabina Corsaut', 11), ('Donald Daly', 11), ('Dan Lain', 11), ('Paul Vaillancourt', 11), ('Louis Marino', 11), ('Marlene Berman', 14), ('Matthew Soper', 14), ('Selena Hoose', 14), ('Donna Pepin', 14), ('Rick Monday', 14), ('Sean Holley', 14), ('Nicholas Applegate', 14), ('Harry Carney', 14), ('Tyrone Sturman', 14), ('Roy Maes', 14), ('Vincent Malloy', 14), ('William Trapp', 14), ('Corey Sloat', 14), ('Zachary Carruthers', 14), ('Katherine Oswald', 14), ('Tyra Rhew', 14), ('Henry Mckenna', 14), ('Shantel Zevallos', 14), ('Russell Parson', 14), ('Guy Cullinan', 14), ('Joanna Nixon', 14), ('Jeffrey Pulliam', 14), ('Dean Harvill', 14), ('Genevieve Holston', 14), ('Janice Tong', 14), ('Daisy Nino', 14), ('Matthew Dowden', 14), ('Donald Mack', 14), ('Kelli Sallee', 14), ('Johnny Watt', 14), ('Ryan Bennet', 14), ('Patrick Booker', 14), ('Javier Crichton', 14), ('Dan Ulm', 14), ('Larry Elmer', 14), ('Amanda Woodward', 14), ('Guy Pressler', 14), ('Maxine Kyser', 14), ('Clara Wainwright', 16), ('Laurie Fullmer', 16), ('Renee Field', 17), ('Steve Hix', 17), ('Terry Caudle', 17), ('Joyce Etter', 17), ('Jose Burge', 17), ('Neil Hunton', 17), ('Dianne Alarcon', 17), ('Chris Mcmorris', 17), ('Frank Dunkin', 17), ('Nora Neel', 17), ('Eugene Singh', 17), ('Scott Simons', 17), ('Penelope Gayman', 24), ('Deanna Whalen', 24), ('Aaron Dalessio', 24), ('Opal Royal', 24), ('Jeffery Blount', 24), ('Larry Gehrke', 24), ('Shawn Standley', 24), ('Eileen Fischer', 24), ('Sabina Mcmikle', 24), ('Norman Grimaldo', 24), ('Lance Buono', 24), ('Tony Burnett', 24), ('Cathy Tubbs', 24), ('Tyrone Ottman', 24), ('Bernice Schafer', 24), ('Joseph Woodbury', 24), ('Pamela Funkhouser', 24), ('Ted Makuch', 24), ('Shelley Bollinger', 24), ('Ester Janis', 24), ('Jonathan Glasgow', 24), ('Debbie Sapp', 24), ('Roger Barnwell', 24), ('Benjamin Wendt', 24), ('Henry Mazur', 24), ('Gina Gooden', 24), ('Dale Oxendine', 24), ('Tim Daggett', 24), ('Bertha Makowski', 24), ('Megan Hargrave', 24), ('Victor Lucier', 24), ('Ricardo Maselli', 24), ('Julianne Cressman', 24), ('Toni Levi', 24), ('Albert Wardell', 24), ('Margery Erving', 24), ('Dollie Dignan', 24), ('Ruth Dressler', 24), ('Bobby Zito', 24), ('Elaine Santillan', 24), ('Dan Peranio', 24), ('Walter Sappington', 27), ('Roberto Munroe', 27), ('Sofia Bonfiglio', 29), ('Stella Morrissey', 29), ('Cora Stauffer', 29), ('Maggie Molloy', 29), ('Nicole Prevost', 29), ('Juana Hodgson', 29), ('Melinda Millard', 29), ('Susan Roby', 29), ('Anthony Demarco', 29), ('Eugene Molina', 29), ('Kimberly Harrison', 29), ('Lois Karl', 29), ('Adrienne Lasley', 29), ('Cora Thorson', 29), ('Lance Newhard', 29), ('Aaron Taveras', 29), ('Justin Kuehn', 29), ('Eunice Mason', 29), ('Mike Pitre', 29), ('Jessie Hanshaw', 29), ('Jeff Everson', 29), ('Pearl

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('Susie Haley', 44), ('Lucy Pisano', 44), ('Hector Kravitz', 44), ('Danny Steinberg', 44), ('Alan Matos', 44), ('Thomas Hooten', 44), ('Johnny Segovia', 44), ('Jill Borkowski', 44), ('Ramona Pence', 44), ('Lonnie Scheffel', 44), ('Emma Mcmorris', 44), ('Timothy Burkhalter', 44), ('Fred Cushman', 44), ('Walter Wolf', 44), ('Kathrine Lainez', 44), ('Timothy Delvalle', 44), ('Charles Kirkman', 44), ('Dorothy Lemmons', 44), ('Nicholas Chabot', 44), ('Nina Chaney', 44), ('Dewey Vashaw', 44), ('Clarence Lemoine', 44), ('Marian Ruble', 44), ('Glenn Irizarry', 44), ('Latonya Norby', 51), ('Kelly Hilchey', 51), ('Justin Mccleary', 51), ('Lakisha Izquierdo', 51), ('Judith Brauer', 51), ('Jessica Manuel', 51), ('Angelica Joe', 51), ('Harry Numbers', 51), ('Bobby Ericson', 51), ('Lillian Sproul', 51), ('Steven Salvador', 51), ('Jeannette Gooch', 51), ('Timothy Deguzman', 51), ('Bonnie Koski', 51), ('Amie Delcambre', 51), ('Dale Horne', 51), ('Monique Dearman', 51), ('Chad Tolliver', 51), ('Henry Eng', 51), ('Albert Roseberry', 51), ('Edward Jeffries', 51), ('Emilia Erikson', 51), ('Erik Calvi', 51), ('Raymond Oller', 51), ('Janie Julius', 51), ('Geraldine Macon', 52), ('Vera Sargeant', 52), ('Noemi Vancamp', 52), ('Erika Cortes', 52), ('April Wigfall', 52), ('Milagros Harbuck', 52), ('Helen Farina', 52), ('Rebecca Eversole', 52), ('Suzanne Hiles', 52), ('Ernest Doherty', 52), ('Victor Leffler', 52), ('Tyrone Letchworth', 52), ('Edwina Fedor', 52), ('Clinton Escalona', 52), ('Kevin Omara', 52), ('Vincent Augustine', 52), ('Billy Wellman', 52), ('Howard Ung', 52), ('Brittany Grande', 52), ('Sherry Edmondson', 52), ('Roger Willis', 52), ('Ronald Winkle', 52), ('Jesica Cammon', 52), ('Emma Kingery', 52), ('Roy Parke', 52), ('Ariana Mongelli', 52), ('Danny Ragan', 52), ('Robyn Tallent', 52), ('Dean Cambridge', 52), ('Marjorie Marek', 52), ('Fred Morehead', 52), ('Amber Raymond', 52), ('Olga Cimino', 52), ('Sheri Corona', 52), ('Clinton Velardo', 52), ('Russell Goodale', 52), ('Allen Mckinney', 53), ('Roy Hurtado', 53), ('Jennifer Hoffmann', 53), ('Debora Bolles', 53), ('Todd Pillow', 53), ('Lucy Halsey', 53), ('Jill Welch', 53), ('Lance Goodpaster', 53), ('Elinor Buff', 53), ('Carmella Calandra', 53), ('Eunice Cooney', 53), ('Darlene Foote', 53), ('Gerald Gosselin', 53), ('Arthur Gard', 53), ('Gregory Durant', 53), ('Willie Monroy', 53), ('Erika Hicks', 53), ('Gertrude Toms', 53), ('Christy Kendall', 53), ('Fernando Mccardell', 61), ('Lance Rein', 61), ('Roxie Underdahl', 61), ('Brandon Phelps', 61), ('Claudia Sthilaire', 61), ('Tammy Riddle', 61), ('Betty Chi', 61), ('Charlene Mccormack', 61), ('Judy Lucia', 61), ('Anthony Farley', 61), ('Clyde Vosburgh', 61), ('Carlos Pfeffer', 61), ('Misty Arzola', 61), ('Nora Therrien', 61), ('Katie Jeffery', 61), ('Jason Carney', 61), ('Dena Staff', 61), ('Herman Latch', 61), ('Clara Bender', 61), ('Cecilia Shank', 61), ('Marvin Gober', 61), ('Joe Weitzel', 61), ('Travis Dennard', 61), ('Louis Gribble', 61), ('Jessie Treiber', 61), ('Travis Hightower', 61), ('Lori Connor', 61), ('Maggie Mallon', 61), ('Agnes Carrell', 61), ('Martin Mcswain', 61), ('Herman Harnish', 61), ('Claudia Keane', 61), ('Roger Birdwell', 61), ('Shelia Simoneau', 61), ('Alice Gayton', 61), ('Lawrence Velarde', 62), ('Laurie Coons', 62), ('Fannie Tipton', 62), ('Samantha Giroux', 62), ('Nancy Welker', 62), ('Heather Tanaka', 62), ('Bruce Waldman', 62), ('Judy Burress', 62), ('Faye Scribner', 62), ('Terri Landon', 62), ('Kathleen Wray', 62), ('Rachel Holmgren', 62), ('Kay Shearer', 62), ('Clyde Baran', 62), ('Lorraine Tavarez', 62), ('Janie Counts', 62), ('Rose Hinson', 62), ('Vivian Mather', 62), ('Tyrone Awong', 62), ('Ronald Nettles', 62), ('Marguerite Carman', 62), ('Jeffery Marchant',

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Duration taken by mergesort to sort 1000 pickup requests in random order: 0.002335 seconds

1000 nearly sorted pickup request list sorted

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('Maggie Mallon', 61), ('Misty Arzola', 61), ('Herman Harnish', 61), ('Claudia Keane', 61), ('Nora Therrien', 61), ('Roger Birdwell', 61), ('Marvin Gober', 61), ('Cecilia Shank', 61), ('Alice Gayton', 61), ('Herman Latch', 61), ('Claudia Sthilaire', 61), ('Michael Rankins', 62), ('Carmella Rondon', 62), ('Fannie Tipton', 62), ('Nancy Welker', 62), ('Clara Asher', 62), ('Adrienne Velez', 62), ('Dorothy Mcginley', 62), ('Rachel Holmgren', 62), ('Terri Landon', 62), ('Travis Bliss', 62), ('Alana Glade', 62), ('Tyrone Awong', 62), ('Kenneth Lentz', 62), ('Rose Hinson', 62), ('Samantha Giroux', 62), ('Betty Kirkwood', 62), ('Clyde Baran', 62), ('Heather Tanaka', 62), ('Marguerite Carman', 62), ('Kathleen Wray', 62), ('Judy Burress', 62), ('Janie Counts', 62), ('Julio Dunkel', 62), ('Faye Scribner', 62), ('Patrick Downing', 62), ('Ronald Nettles', 62), ('Kay Shearer', 62), ('Henry Bertrand', 62), ('Laurie Coons', 62), ('Lawrence Velarde', 62), ('Trudy Shadrick', 62), ('Marsha Bump', 62), ('Vivian Mather', 62), ('Bruce Waldman', 62), ('Anna Harber', 62), ('Suzanne Stacy', 62), ('Lorraine Tavarez', 62), ('Jeffery Marchant', 62), ('Danny Carden', 62), ('Allen Olvera', 62), ('Edward Bliss', 67), ('Tony Bost', 67), ('Arlene Serrano', 67), ('Chad Cardin', 67), ('Carrie Ballinger', 67), ('Bruce Pfeffer', 67), ('Jeff Todd', 67), ('Chad Knight', 67), ('Guy Snipe', 67), ('Kelly Silliman', 67), ('Sofia Debolt', 67), ('Vickie Cathcart', 67), ('Constance Berg', 67), ('Della Condon', 67), ('Jimmy Doak', 67), ('Tania Agbayani', 67), ('Paul Hagler', 67), ('Jerry Lennon', 67), ('Jimmy Fults', 67), ('Nicholas Arrowood', 76), ('Louise Wendt', 76), ('Sean Scarberry', 76), ('Jeffery Neves', 76), ('Joseph Feeney', 76), ('Pauline Callender', 76), ('Martin Hagen', 76), ('Keith Overturf', 76), ('Pearlie Willams', 76), ('Thomas Hardeman', 76), ('Stephanie Mcintosh', 76), ('Liza Guyette', 76), ('Stanley Hindman', 76), ('Jonathan Calloway', 76), ('Daniel Birch', 76), ('Bruce Heard', 76), ('Norma Zack', 76), ('Keith Bowlin', 76), ('William Wingfield', 76), ('Russell Nino', 76), ('James Hayward', 76), ('Kristin Grantham', 76), ('John Tolbert', 76), ('Joseph Smith', 76), ('Peter Kinser', 76), ('Roger Easton', 76), ('Jeanne Fortin', 76)]

Duration taken by MergeSort to sort 1000 pickup requests in nearly sorted order: 0.009898 seconds

1000 reversed pickup requests list sorted

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('Sherry Edmondson', 52), ('Suzanne Hiles', 52), ('April Wigfall', 52), ('Ariana Mongelli', 52), ('Roy Parke', 52), ('Ronald Winkle', 52), ('Dean Cambridge', 52), ('Erika Cortes', 52), ('Geraldine Macon', 52), ('Gertrude Toms', 53), ('Willie Monroy', 53), ('Gregory Durant', 53), ('Jennifer Hoffmann', 53), ('Roy Hurtado', 53), ('Erika Hicks', 53), ('Allen Mckinney', 53), ('Christy Kendall', 53), ('Jill Welch', 53), ('Lucy Halsey', 53), ('Darlene Foote', 53), ('Arthur Gard', 53), ('Eunice Cooney', 53), ('Carmella Calandra', 53), ('Debora Bolles', 53), ('Todd Pillow', 53), ('Elinor Buff', 53), ('Lance Goodpaster', 53), ('Gerald Gosselin', 53), ('Maggie Mallon', 61), ('Martin Mcswain', 61), ('Joe Weitzel', 61), ('Brandon Phelps', 61), ('Clyde Vosburgh', 61), ('Cecilia Shank', 61), ('Shelia Simoneau', 61), ('Roger Birdwell', 61), ('Roxie Underdahl', 61), ('Claudia Keane', 61), ('Anthony Farley', 61), ('Misty Arzola', 61), ('Agnes Carrell', 61), ('Judy Lucia', 61), ('Louis Gribble', 61), ('Travis Dennard', 61), ('Dena Staff', 61), ('Fernando Mccardell', 61), ('Lori Connor', 61), ('Herman Latch', 61), ('Carlos Pfeffer', 61), ('Marvin Gober', 61), ('Travis Hightower', 61), ('Herman Harnish', 61), ('Lance Rein', 61), ('Jason Carney', 61), ('Tammy Riddle', 61), ('Clara Bender', 61), ('Jessie Treiber', 61), ('Nora Therrien', 61), ('Charlene Mccormack', 61), ('Katie Jeffery', 61), ('Betty Chi', 61), ('Claudia Sthilaire', 61), ('Alice Gayton', 61), ('Marguerite Carman', 62), ('Lorraine Tavarez', 62), ('Michael Rankins', 62), ('Trudy Shadrick', 62), ('Judy Burress', 62), ('Alana Glade', 62), ('Lawrence Velarde', 62), ('Carmella Rondon', 62), ('Clyde Baran', 62), ('Marsha Bump', 62), ('Samantha Giroux', 62), ('Bruce Waldman', 62), ('Kenneth Lentz', 62), ('Henry Bertrand', 62), ('Jeffery Marchant', 62), ('Ronald Nettles', 62), ('Terri Landon', 62), ('Faye Scribner', 62), ('Dorothy Mcginley', 62), ('Janie Counts', 62), ('Clara Asher', 62), ('Kathleen Wray', 62), ('Fannie Tipton', 62), ('Heather Tanaka', 62), ('Danny Carden', 62), ('Betty Kirkwood', 62), ('Anna Harber', 62), ('Rose Hinson', 62), ('Allen Olvera', 62), ('Tyrone Awong', 62), ('Vivian Mather', 62), ('Travis Bliss', 62), ('Suzanne Stacy', 62), ('Rachel Holmgren', 62), ('Kay Shearer', 62), ('Adrienne Velez', 62), ('Julio Dunkel', 62), ('Nancy Welker', 62), ('Patrick Downing', 62), ('Laurie Coons', 62), ('Kelly Silliman', 67), ('Paul Hagler', 67), ('Guy Snipe', 67), ('Tania Agbayani', 67), ('Chad Knight', 67), ('Tony Bost', 67), ('Jeff Todd', 67), ('Della Condon', 67), ('Bruce Pfeffer', 67), ('Vickie Cathcart', 67), ('Carrie Ballinger', 67), ('Jimmy Fults', 67), ('Chad Cardin', 67), ('Jimmy Doak', 67), ('Arlene Serrano', 67), ('Sofia Debolt', 67), ('Edward Bliss', 67), ('Constance Berg', 67), ('Jerry Lennon', 67), ('John Tolbert', 76), ('Sean Scarberry', 76), ('Louise Wendt', 76), ('Jeffery Neves', 76), ('Kristin Grantham', 76), ('Peter Kinser', 76), ('Nicholas Arrowood', 76), ('William Wingfield', 76), ('Jeanne Fortin', 76), ('Daniel Birch', 76), ('Liza Guyette', 76), ('Stephanie Mcintosh', 76), ('Russell Nino', 76), ('Martin Hagen', 76), ('Thomas Hardeman', 76), ('Roger Easton', 76), ('Joseph Smith', 76), ('James Hayward', 76), ('Jonathan Calloway', 76), ('Stanley Hindman', 76), ('Keith Bowlin', 76), ('Joseph Feeney', 76), ('Pauline Callender', 76), ('Norma Zack', 76), ('Keith Overturf', 76), ('Bruce Heard', 76), ('Pearlie Willams', 76)]

Duration taken by MergeSort to sort 1000 pickup requests in reverse order: 0.013226 seconds

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Duration taken by MergeSort to sort 1000 pickup requests in reverse order: 0.013226 seconds

The Top down MergeSort algorithm Reflection for different scenarios of 1000 pickup request:
Duration to sort 1000 pickup requests in random order: 0.002335 seconds
Duration to sort 1000 pickup requests in nearly sorted order: 0.009898 seconds
Duration to sort 1000 pickup requests in reverse order: 0.013226 seconds

Looking at the durations for each scenario of 1000 pickup request, top down MergeSort takes less time to
    sort the random list while more time is taken to sort the reverse list
Its time complexity is O(n log n) and its space complexity is Olog(n).
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Figure 77: Short Reflection on using Merge Sort to sort different 1000 pickup request list

Quick Sort for different 1000 pickup request lists

1000 pickup request list in random order sorted

Note: The first names are not the same for the different test scenarios as the lists are sorted using only the pickup times not the names.

[('Alicia Womble', 0), ('Jorge Ventimiglia', 0), ('Audrey Buck', 0), ('Claire Britton', 0), ('Wayne Crabb', 0), ('Timothy Congdon', 0), ('Jeremy Trainor', 0), ('Wilfred Mourning', 0), ('Jeanette Herbst', 0), ('Shelia Sorrells', 0), ('Scott Ring', 0), ('Gregory Dantzler', 0), ('Wilma Rey', 0), ('Bruce Rickert', 0), ('Todd Huss', 0), ('Arthur Larrabee', 0), ('Eugene Forman', 0), ('George Blow', 0), ('Lois Sexton', 0), ('Gary Tate', 0), ('Belen Sheerin', 0), ('Chris Rahman', 0), ('Nelson Guitierrez', 0), ('Russell Ashmore', 0), ('Zachary Falgoust', 0), ('Dale Ormsby', 0), ('Jill Dickerson', 0), ('Hugo Vogelgesang', 0), ('Anthony Ramey', 0), ('Chad Steffens', 0), ('Kyla Hillin', 0), ('Justin Soloman', 0), ('Adam Harden', 0), ('Lillie Frisby', 0), ('Russell Treat', 0), ('Javier Mungia', 0), ('Jocelyn Lynde', 0), ('Jose Flannery', 0), ('Guy Winders', 0), ('Juan Pink', 0), ('Howard Storm', 0), ('Candice Brower', 0), ('Dean Scioneaux', 0), ('Shane Dare', 0), ('Amy Bedford', 0), ('Mark Tate', 0), ('Manuel Wiser', 0), ('Raymond Hynes', 0), ('Fannie Hesse', 0), ('Philip Gauthier', 0), ('Philip Hammonds', 0), ('Erma Creech', 0), ('Johnny Dozier', 0), ('William Backus', 0), ('Javier Hausman', 0), ('Nicole Ellingson', 0), ('Kayla Paynter', 0), ('Yvette Scalise', 0), ('Michele Ford', 0), ('Darren Hadaway', 0), ('Neil

Rosenbeck', 0), ('Brandon Mask', 0), ('Darren Suchan', 0), ('Keith Hurt', 0), ('Candy Henn', 0), ('Carolina Hamid', 0), ('Ethel Reitz', 0), ('Craig Baker', 0), ('Scott Prescott', 0), ('Joshua Spooner', 0), ('Harry Norfleet', 0), ('Joseph Linville', 0), ('Douglas Sedillo', 0), ('Elizabeth Dismuke', 0), ('Nora Hashimoto', 0), ('Matthew Massa', 0), ('Pamela French', 0), ('Guy Loredo', 0), ('Sonia Bass', 0), ('Darren Cotta', 0), ('Daniel Trimble', 0), ('Wesley Harm', 0), ('Janice Granados', 0), ('Jorge Madry', 0), ('Stephen Mccool', 0), ('Hector Horning', 0), ('Gary Burmeister', 0), ('Albert Lockett', 0), ('Benjamin Prado', 0), ('Violet Eden', 0), ('Robert Cooks', 0), ('Danny Corrigan', 0), ('Jesse Sandstrom', 0), ('Gary Quillen', 0), ('Diana Welker', 0), ('Tyrone Mondor', 0), ('Jeff Moten', 0), ('Patsy Finlay', 0), ('Hilda Wilmoth', 0), ('Harriett Curatolo', 0), ('Jack Tso', 0), ('Luz Coomer', 0), ('Marilyn Mickens', 0), ('Dorothy Groves', 0), ('Matthew Beaman', 0), ('Glenna Meaney', 0), ('Thomas Giffin', 0), ('Katy Tressler', 0), ('Gabriela Keaney', 0), ('Craig Bulter', 0), ('Adam Feeney', 0), ('Chris Embry', 0), ('Martin Harbin', 0), ('Tina Haugen', 0), ('Marjorie Tello', 0), ('Amelia Mckeever', 0), ('Kevin Schreiber', 0), ('Kenneth Lewellen', 0), ('Eva Pursley', 0), ('Arlene Wellington', 0), ('Jeffrey Marcum', 0), ('Darlene Bergman', 0), ('Luz Hong', 0), ('James Urrutia', 0), ('Michael Michel', 0), ('Katherine Ryan', 0), ('Nereida Viscardi', 0), ('Billy Partridge', 0), ('Dennis Gault', 0), ('Geraldine Vargas', 0), ('Jason Word', 0), ('Christian Klemm', 0), ('Gerald Cousin', 0), ('Velma Burnett', 0), ('Pearlie Storie', 0), ('Gerald Whitesell', 0), ('Dianne Mixon', 0), ('Michael Lefevre', 0), ('Keith Dykes', 0), ('Andrew Colunga', 0), ('Jesse Horn', 0), ('Allen Towns', 0), ('Joyce Holtz', 0), ('Gary Poole', 0), ('Kurt Prowse', 0), ('Rodney Toth', 0), ('Raven Dalgleish', 0), ('Ryan Giglio', 0), ('Jessica Villegas', 0), ('Terry Brush', 0), ('Michele Valliere', 0), ('Howard Hake', 0), ('Annie Markel', 0), ('Ingrid Devaul', 0), ('Tonya Niemi', 0), ('Mark Dillman', 0), ('Cora Mills', 0), ('Curtis Bowyer', 0), ('Manuel Hammack', 0), ('Sarah Waterman', 0), ('Natalie Bynum', 0), ('Rosemary Mcculley', 0), ('Jason Eskridge', 0), ('Isabel Peart', 0), ('Brent Curtsinger', 0), ('Kristy Strom', 0), ('Guy Vanalstyne', 0), ('Christopher List', 0), ('Tanisha Gamino', 0), ('Max Madamba', 0), ('Inez Yee', 0), ('Yvette Chenault', 0), ('Dennis Housley', 0), ('Juana Greenleaf', 0), ('Matthew Will', 0), ('Geraldine Flora', 0), ('Anthony Pires', 0), ('Patrick Bahena', 0), ('Scott Luevano', 0), ('Stephen Fitz', 0), ('Jennie Tuggle', 0), ('Anna Marble', 0), ('Christian Durall', 0), ('Jessie Mayville', 0), ('Louise Lively', 0), ('Stephen Core', 0), ('Darren Pedrosa', 0), ('Edward Reader', 0), ('Rick Brathwaite', 0), ('Norman Styles', 0), ('Ingrid Mutter', 0), ('Jerry Herring', 0), ('Joy Cloud', 0), ('Valerie Byrd', 0), ('Jeffery Ruiz', 0), ('Ronald Rushton', 0), ('Eugene Gore', 0), ('Harry Skeen', 0), ('Billy Luu', 0), ('George Manchester', 0), ('Brian Martindale', 0), ('Theresa Townsley', 0), ('Harriett Zollinger', 0), ('Antonio Braddock', 0), ('Timothy Markel', 0), ('Deanna Arndt', 0), ('Noreen Durfey', 0), ('Harold Hamill', 0), ('Nelson Tooker', 0), ('Keith Hyder', 0), ('Jimmy Munoz', 0), ('Johnny Turnbull', 0), ('Carole Stuckey', 0), ('Debbie Ruch', 0), ('Leonard Delrio', 0), ('Arlene Bergmann', 0), ('Jodi Holm', 0), ('Edward Scalise', 0), ('Paige Lovitt', 0), ('Curtis Cordell', 0), ('Loraine Montford', 0), ('Jeremy Catalano', 0), ('Opal Tobias', 0), ('Norman Mcnulty', 0), ('Orville Gillaspie', 0), ('Daphne Renz', 0), ('Sally Soule', 0), ('Denise Herrington', 0), ('Lonnie Kitzmiller', 0), ('Denise Ferraro', 0), ('Louise Bellamy', 0),

('Danielle Tincher', 0), ('Carmella Vanwagenen', 0), ('Vernon Moultrie', 0), ('Sharonda Remsberg', 0), ('Paul Gowen', 0), ('Teresa Booth', 0), ('Ellen Millican', 0), ('Daniel Binkley', 0), ('Peter Whitman', 0), ('Keisha Ragusa', 0), ('Shelley Scherer', 0), ('Steve Ellingson', 0), ('Darlene Sandler', 0), ('Mike Rodarte', 0), ('Jeremy Hyman', 0), ('Jimmy Pritchard', 0), ('Jeffery Bounds', 0), ('Cindy Penney', 0), ('Elnora Coppedge', 0), ('Erik Weinman', 0), ('Verna Basso', 0), ('Emily Chadwick', 0), ('Lula Eskridge', 0), ('Amelia Jarrell', 0), ('Joseph Donohue', 0), ('Sonia Turgeon', 0), ('Asia Gucman', 0), ('Beatrice Rose', 0), ('Douglas Dabney', 0), ('Amelia Meacham', 0), ('Richard Melvin', 0), ('Steven Ferrari', 0), ('Clarence Correll', 0), ('Joseph Audet', 0), ('Veronica Voss', 0), ('Louis Koss', 0), ('Hillary Tall', 0), ('Vickie Giffin', 0), ('Bessie Minton', 0), ('Jonathan Wagoner', 0), ('Travis Trull', 0), ('Tara Nissen', 0), ('Mark Grimm', 0), ('Nelson Coltrane', 0), ('Shawn Leigh', 0), ('Stanley Bybee', 0), ('Clayton Salls', 0), ('Sherri Kinser', 0), ('Adam Bingham', 0), ('Noemi Widrick', 0), ('Elsie Sowers', 0), ('Justin Hutson', 0), ('Arthur Runge', 0), ('Dorcas Abbitt', 0), ('Allen Velarde', 0), ('Kenneth Hudgens', 0), ('Jose Gregg', 0), ('Lester Kealey', 0), ('Ted Yetman', 0), ('Becky Zinn', 0), ('Lester Maya', 0), ('Alberta Bradley', 0), ('Walter Roussel', 0), ('Larry Weatherspoon', 0), ('Leon Urbina', 0), ('Jonathan Gower', 0), ('Karen Pena', 0), ('Sofia Kelling', 0), ('Larry Lamp', 0), ('Christopher Biles', 0), ('Vernon Sermons', 0), ('Bobby Faris', 0), ('Shanna Elsea', 0), ('Susie Peoples', 0), ('Paul Albers', 0), ('Larry Mitchel', 0), ('Yvonne Mickens', 0), ('Jessie Efaw', 0), ('Steve Rushing', 0), ('Allen Mckenna', 0), ('Jeanne Twigg', 0), ('Charlene Pennington', 0), ('Jeff Singletary', 0), ('Larry Cantu', 0), ('Cody Basta', 0), ('Ruby Phifer', 0), ('Lawrence Hecker', 0), ('Kristine Cave', 0), ('Mike Schumann', 0), ('Mathew Mercedes', 0), ('Daisy Palladino', 0), ('Willie Pence', 0), ('Doris Mays', 0), ('Guy Gonsoulin', 0), ('Carl Campos', 0), ('Roger Arriola', 0), ('Darren Blau', 0), ('Luis Lehmann', 0), ('Donald Parmer', 0), ('Alma Mcneal', 0), ('Alice Reese', 0), ('Heather Laureano', 0), ('Candice Swisher', 0), ('Maribel Earnest', 0), ('Yvonne Mccune', 0), ('Antonio Obryan', 0), ('Cherie Mcguckin', 0), ('Christina Driskell', 0), ('Jacob Ingle', 0), ('Dennis Donahue', 0), ('Herman Appling', 0), ('Walter Chapa', 0), ('Curtis Rounds', 0), ('Jo Patino', 0), ('Vivian Milton', 0), ('Russell Oshea', 0), ('Vincent Berkley', 0), ('Chad Pardo', 0), ('Dean Buchanon', 0), ('Ronald Henning', 0), ('Jacob Dandrea', 0), ('Jessie Reta', 0), ('Harold Seaman', 0), ('Stephanie Shanks', 0), ('Eleanor Fontenot', 0), ('Roger Pellerin', 0), ('Phyllis Beene', 0), ('Jack Kolb', 0), ('Sherry Winfrey', 0), ('Craig Avent', 0), ('Brandi Mcauliffe', 0), ('Eileen Bunch', 0), ('Eunice Barr', 0), ('Judith Lemay', 0), ('Noemi Marcucci', 0), ('Clarence Keyser', 0), ('Max Kirkendall', 0), ('Alice Knoll', 0), ('Dustin Gadsden', 0), ('Gary Thorson', 0), ('Bruce Dailey', 0), ('Cassandra Creed', 0), ('Christopher Merwin', 0), ('Olivia Fall', 0), ('Ruby Avent', 0), ('Ramona Stacy', 0), ('Lonnie Neveu', 0), ('Jacob Meeker', 0), ('Della Carlile', 0), ('Mabel Gist', 0), ('Larry Chapa', 0), ('Amy Boren', 0), ('Tonya Canfield', 0), ('Harriett Loveall', 0), ('Kenneth Casares', 0), ('Rita Witkowski', 0), ('Anthony Deaver', 0), ('Lena Judge', 0), ('Wayne Merkle', 0), ('Kristin Keel', 0), ('Christine King', 0), ('Chris Reiner', 0), ('Kelli Schulze', 0), ('Clarence Batchelder', 0), ('Sean Howerton', 0), ('Carmella Sheffey', 0), ('Carolyn Kay', 0), ('Javier Kratochvil', 0), ('Joseph Favors', 0), ('Esther Hale', 0), ('Christine

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Duration taken by QuickSort to sort 1000 pickup requests in random order: 0.018282 seconds

1000 nearly sorted pickup request list sorted:

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Beaman', 0), ('Gary Thorson', 0), ('Jodi Holm', 0), ('Steve Ellingson', 0), ('Joseph Favors', 0), ('Shelia Sorrells', 0), ('Gregory Despain', 0), ('Craig Bulter', 0), ('Norman Styles', 0), ('Esther Hale', 0), ('Michael Michel', 0), ('Antonio Braddock', 0), ('Harold Seaman', 0), ('Jeffery Bounds', 0), ('Peter Whitman', 0), ('Ingrid Devaul', 0), ('Wayne Crabb', 0), ('Daniel Binkley', 0), ('Noreen Durfey', 0), ('Stanley Bybee', 0), ('Becky Zinn', 0), ('Jeremy Hyman', 0), ('Eileen Bunch', 0), ('Chris Reiner', 0), ('Keith Dykes', 0), ('Matthew Will', 0), ('Luz Coomer', 0), ('Chris Rahman', 0), ('Janice Granados', 0), ('Willie Pence', 0), ('Emily Chadwick', 0), ('Ruby Traylor', 0), ('Dean Buchanon', 0), ('Pearlie Storie', 0), ('Manuel Hammack', 0), ('Jeanne Twigg', 0), ('Michael Lefevre', 0), ('Lorraine Ake', 0), ('Jonathan Wagoner', 0), ('Guy Gonsoulin', 0), ('Theresa Townsley', 0), ('Hugo Vogelgesang', 0), ('Jose Gregg', 0), ('Teresa Samaniego', 0), ('Jennie Tuggle', 0), ('Yvette Scalise', 0), ('Chad Steffens', 0), ('Stephen Mccool', 0), ('Eugene Forman', 0), ('Della Carlile', 0), ('Kelly Cheatam', 0), ('Paul Albers', 0), ('Larry Chapa', 0), ('Katy Tressler', 0), ('Scott Luevano', 0), ('Curtis Cordell', 0), ('Jack Kolb', 0), ('Javier Mungia', 0), ('Jessie Efaw', 0), ('Larry Cantu', 0), ('Dorcas Abbitt', 0), ('Leonard Urbina', 0), ('Bruce Dailey', 0), ('Dennis Donahue', 0), ('Allen Mckenna', 0), ('Amelia Jarrell', 0), ('Richard Melvin', 0), ('Jesse Sandstrom', 0), ('Nelson Guitierrez', 0), ('Gustavo Sysak', 0), ('Vickie Giffin', 0), ('Shane Dare', 0), ('Gabriela Keaney', 0), ('Denise Herrington', 0), ('Christopher Merwin', 0), ('Raymond Hynes', 0), ('Norman Mcnulty', 0), ('Wilma Rey', 0), ('Ronald Henning', 0), ('Fred Meeker', 0), ('Debbie Ruch', 0), ('Cody Russom', 0), ('Clarence Keyser', 0), ('Clarence Correll', 0), ('Glenna Meaney', 0), ('Verna Basso', 0), ('Ruby Phifer', 0), ('Darren Suchan', 0), ('Russell Treat', 0), ('Nora Hashimoto', 0), ('Michele Valliere', 0), ('Daisy Palladino', 0), ('Harriett Zollinger', 0), ('Nicole Ellingson', 0), ('Danny Corrigan', 0), ('Jo Patino', 0), ('Carole Stuckey', 0), ('Darren Hadaway', 0), ('Candy Henn', 0), ('Amy Boren', 0), ('Sonia Bass', 0), ('Brandon Mask', 0), ('Marjorie Tello', 0), ('Tonya Niemi', 0), ('Gerald Cousin', 0), ('George Manchester', 0), ('Chad Pardo', 0), ('Keisha Ragusa', 0), ('Eleanor Fontenot', 0), ('Louise Bellamy', 0), ('Rodney Toth', 0), ('Carl Campos', 0), ('Denise Ferraro', 0), ('Maribel Earnest', 0), ('Alicia Womble', 0), ('Paige Lovitt', 0), ('Douglas Dabney', 0), ('Inez Yee', 0), ('Wilfred Mourning', 0), ('Arthur Larrabee', 0), ('Brandi Keil', 0), ('Guy Loredo', 0), ('Joseph Linville', 0), ('Veronica Voss', 0), ('Joseph Donohue', 0), ('Clayton Salls', 0), ('Raven Dalgleish', 0), ('Craig Avent', 0), ('Elaine Elias', 0), ('Tyrone Mondor', 0), ('Mabel Gist', 0), ('Jeanette Herbst', 0), ('Orville Gillaspie', 0), ('Judith Lemay', 0), ('Matthew Massa', 0), ('Jerry Herring', 0), ('Kevin Schreiber', 0), ('Johnny Dozier', 0), ('Zachary Falgoust', 0), ('Michele Ford', 0), ('Phyllis Beene', 0), ('Daniel Trimble', 0), ('Elsie Sowers', 0), ('Max Kirkendall', 0), ('Javier Hausman', 0), ('Hazel Persinger', 0), ('Dennis Housley', 0), ('Jorge Madry', 0), ('Tony Brogan', 0), ('Hector Horning', 0), ('Tonya Canfield', 0), ('Albert Lockett', 0), ('Jacob Meeker', 0), ('Juan Vandyke', 0), ('Corinne Kocsis', 0), ('Ingrid Mutter', 0), ('Mike Rodarte', 0), ('Edward Scalise', 0), ('Antonio Obryan', 0), ('Alice Knoll', 0), ('Annie Markel', 0), ('Kenneth Casares', 0), ('Lillie Frisby', 0), ('Ted Yetman', 0), ('Sherry Winfrey', 0), ('Eugene Gore', 0), ('Alma Mcneal', 0), ('Jacob Dandrea', 0), ('Guy Winders', 0), ('Darren Pedrosa', 0), ('Jeff Moten', 0), ('Loraine Montford', 0), ('Ramona Stacy', 0), ('Ellen Millican',

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Duration taken by QuickSort to sort 1000 pickup requests in nearly sorted order: 0.016255 seconds

1000 reversed pickup request list sorted

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('Curtis Cordell', 0), ('Jerry Herring', 0), ('Patsy Horne', 0), ('Matthew Massa', 0), ('Scott Luevano', 0), ('Judith Lemay', 0), ('Gerald Whitesell', 0), ('Harry Norfleet', 0), ('Katy Tressler', 0), ('Jeanette Herbst', 0), ('Jesse Horn', 0), ('Mabel Gist', 0), ('Opal Tobias', 0), ('Tyrone Mondor', 0), ('George Blow', 0), ('Candice Swisher', 0), ('Paul Albers', 0), ('Joy Cloud', 0), ('Billy Luu', 0), ('Craig Avent', 0), ('Kelly Cheatam', 0), ('Raven Dalgleish', 0), ('Katherine Ryan', 0), ('Clayton Salls', 0), ('Della Carlile', 0), ('Arlene Wellington', 0), ('Dennis Gault', 0), ('Luis Lehmann', 0), ('Dustin Gadsden', 0), ('Wilma Rey', 0), ('Martin Harbin', 0), ('Joseph Linville', 0), ('Stephen Mccool', 0), ('Guy Loredo', 0), ('Annie Markel', 0), ('Rosemary Mcculley', 0), ('Chad Steffens', 0), ('Arthur Larrabee', 0), ('Elaine Elias', 0), ('Wilfred Mourning', 0), ('Yvette Scalise', 0), ('Dorothy Groves', 0), ('Carole Stuckey', 0), ('Inez Yee', 0), ('Karen Pena', 0), ('Kayla Paynter', 0), ('Nelson Guitierrez', 0), ('Paige Lovitt', 0), ('Teresa Samaniego', 0), ('Alicia Womble', 0), ('Jennie Tuggle', 0), ('Maribel Earnest', 0), ('Jose Gregg', 0), ('Brent Curtsinger', 0), ('Harold Seaman', 0), ('Carl Campos', 0), ('Hugo Vogelgesang', 0), ('Rodney Toth', 0), ('Nelson Tooker', 0), ('Louise Bellamy', 0), ('Teresa Booth', 0), ('Carolyn Kay', 0), ('Cindy Penney', 0), ('Keisha Ragusa', 0), ('Guy Gonsoulin', 0), ('Chad Pardo', 0), ('Scott Ring', 0), ('George Manchester', 0), ('Jonathan Wagoner', 0), ('Jonathan Gower', 0), ('Lonnie Neveu', 0), ('Tonya Niemi', 0), ('Lorraine Ake', 0), ('Marjorie Tello', 0), ('Elizabeth Dismuke', 0), ('Brandon Mask', 0), ('Dale Ormsby', 0), ('Travis Trull', 0), ('Cherie Mcguckin', 0), ('Amy Boren', 0), ('Jeanne Twigg', 0), ('Candy Henn', 0), ('Adam Feeney', 0), ('Darren Hadaway', 0), ('Manuel Hammack', 0), ('Zelma Vallier', 0), ('Alice Reese', 0), ('Rick Brathwaite', 0), ('Pearlie Storie', 0), ('Jo Patino', 0), ('Neil Rosenbeck', 0), ('Danny Corrigan', 0), ('Russell Ashmore', 0), ('Nicole Ellingson', 0), ('Chad Maclean', 0), ('Allen Velarde', 0), ('Ruby Traylor', 0), ('Lois Sexton', 0), ('Christopher List', 0), ('Leonard Delrio', 0), ('Emily Chadwick', 0), ('Stanley Littleton', 0), ('Sarah Waterman', 0), ('Sherry Winfrey', 0), ('Willie Pence', 0), ('Tony Brogan', 0), ('Anthony Ramey', 0), ('Orville Gillaspie', 0), ('Philip Hammonds', 0), ('Douglas Dabney', 0), ('Terry Brush', 0), ('Sonia Bass', 0), ('Chris Rahman', 0), ('Wayne Merkle', 0), ('Craig Baker', 0), ('Gabriela Keaney', 0), ('Luz Coomer', 0), ('Jack Kolb', 0), ('Geraldine Flora', 0), ('Adam Harden', 0), ('Matthew Will', 0), ('Dean Buchanon', 0), ('Lawrence Hecker', 0), ('Wayne Crabb', 0), ('Jessica Villegas', 0), ('Jeremy Hyman', 0), ('Ryan Giglio', 0), ('Yvette Dowdy', 0), ('Chris Reiner', 0), ('Vivian Milton', 0), ('Eugene Forman', 0), ('Keith Hyder', 0), ('Eileen Bunch', 0), ('Todd Huss', 0), ('Steven Ferrari', 0), ('Curtis Rounds', 0), ('Matthew Beaman', 0), ('Christian Durall', 0), ('Joseph Donohue', 0), ('Darren Blau', 0), ('Harold Hamill', 0), ('Mark Grimm', 0), ('Denise Ferraro', 0), ('Sean Howerton', 0), ('Stanley Bybee', 0), ('Scott Prescott', 0), ('Larry Chapa', 0), ('Belen Sheerin', 0), ('Noreen Durfey', 0), ('Roger Cool', 0), ('Manuel Wiser', 0), ('Kristy Strom', 0), ('Daniel Binkley', 0), ('Beatrice Rose', 0), ('Louise Lively', 0), ('Anna Marble', 0), ('Marilyn Mickens', 0), ('Howard Hake', 0), ('Jessie Mayville', 0), ('Donald Parmer', 0), ('Ingrid Devaul', 0), ('Hilda Wilmoth', 0), ('Diana Welker', 0), ('Yvette Chenault', 0), ('Peter Whitman', 0), ('Pamela French', 0), ('Carolina Hamid', 0), ('Charlene Pennington', 0), ('Jeffery Bounds', 0), ('Eunice Barr', 0), ('Christine Prather', 0), ('Louis Koss', 0), ('Benjamin

Prado', 0), ('Kristine Cave', 0), ('Curtis Bowyer', 0), ('Kristin Keel', 0), ('Guy Winders', 0), ('Tanya Pickering', 0), ('Bruce Rickert', 0), ('Rita Witkowski', 0), ('Michael Michel', 0), ('Gregory Dantzler', 0), ('Andrew Colunga', 0), ('Valerie Jenson', 0), ('Esther Hale', 0), ('Walter Chapa', 0), ('Christian Klemm', 0), ('Audrey Buck', 0), ('Bessie Minton', 0), ('Lester Maya', 0), ('Joyce Holtz', 0), ('Natalie Bynum', 0), ('Craig Bulter', 0), ('Darren Cotta', 0), ('Glenna Meaney', 0), ('Elnora Coppedge', 0), ('Gregory Despain', 0), ('Yvonne Mickens', 0), ('Lester Kealey', 0), ('Heather Laureano', 0), ('Shelia Sorrells', 0), ('Jeffrey Marcum', 0), ('Allen Towns', 0), ('Christine King', 0), ('Howard Storm', 0), ('Shanna Elsea', 0), ('Jason Word', 0), ('Amy Bedford', 0), ('Steve Ellingson', 0), ('Jacob Meeker', 0), ('Nelson Coltrane', 0), ('Brandi Keil', 0), ('Jodi Holm', 0), ('Russell Treat', 0), ('Javier Kratochvil', 0), ('Dorcas Abbitt', 0), ('Gary Thorson', 0), ('Juan Pink', 0), ('Javier Hausman', 0), ('Antonio Braddock', 0), ('Vernon Moultrie', 0), ('Phyllis Beene', 0), ('Yvonne Mccune', 0), ('Eleanor Fontenot', 0), ('Doris Mays', 0), ('Debbie Ruch', 0), ('Cora Mills', 0), ('Jocelyn Lynde', 0), ('Timothy Congdon', 0), ('Keith Dykes', 0), ('Susie Peoples', 0), ('Shawn Rodriquez', 0), ('Stephen Fitz', 0), ('Deanna Arndt', 0), ('Shelley Scherer', 0), ('Douglas Sedillo', 0), ('Jorge Ventimiglia', 0), ('Herman Appling', 0), ('Janice Granados', 0), ('Shawn Leigh', 0), ('Alberta Bradley', 0), ('Valerie Byrd', 0), ('Tara Nissen', 0), ('Amelia Meacham', 0), ('Lula Eskridge', 0), ('Jose Flannery', 0), ('Keith Hurt', 0), ('Asia Gucman', 0), ('Max Madamba', 0), ('Billy Partridge', 0), ('Isabel Peart', 0), ('Roger Pellerin', 0), ('Adam Bingham', 0), ('Mike Schumann', 0), ('Bobby Faris', 0), ('Tanisha Gamino', 0), ('Kenneth Hudgens', 0), ('Dean Scioneaux', 0), ('Nereida Viscardi', 0), ('Gary Poole', 0), ('Joseph Audet', 0), ('Cody Basta', 0), ('Noemi Widrick', 0), ('Jeffery Ruiz', 0), ('Jeff Singletary', 0), ('Vernon Sermons', 0), ('Paul Gowen', 0), ('Kyla Hillin', 0), ('Darlene Sandler', 0), ('Larry Weatherspoon', 0), ('Patsy Finlay', 0), ('Kevin Schreiber', 0), ('Gerald Cousin', 0), ('Norman Mcnulty', 0), ('Norman Styles', 0), ('Arlene Bergmann', 0), ('Brandi Mcauliffe', 0), ('Thomas Giffin', 0), ('Ramona Stacy', 0), ('Jacob Ingle', 0), ('Allen Mckenna', 0), ('Guy Vanalstyne', 0), ('Roger Arriola', 0), ('Timothy Markel', 0), ('Harriett Zollinger', 0), ('Lonnie Kitzmiller', 0), ('Mike Rodarte', 0), ('Theresa Townsley', 0), ('Roberto Routon', 5), ('Dolores Moses', 5), ('David Gabbard', 5), ('Thomas Peckham', 5), ('Pauline Penton', 5), ('Noreen Smyre', 5), ('Russell Hollar', 5), ('Scott Beckwith', 5), ('Roslyn Philipp', 5), ('Jose Haight', 5), ('Caroline Bible', 5), ('Priscilla Ellis', 5), ('Javier Swint', 5), ('Neil Brabant', 5), ('Jenny Gold', 5), ('Sandra Lacey', 5), ('Tiffany Sain', 5), ('Kelsey Levey', 5), ('Shelia Deck', 5), ('Wesley Newquist', 5), ('Louis Schreiber', 5), ('Kurt Bridgett', 5), ('Ethel Glidden', 5), ('Ellen Wimmer', 5), ('Cindy Motley', 5), ('Howard Shinn', 5), ('Juanita Wildman', 5), ('Katherine Martens', 5), ('Alan Harness', 5), ('Norman Wingfield', 5), ('Kenneth Frizzell', 5), ('Mike Mcewen', 5), ('Charles Babich', 5), ('Shelia Roberts', 5), ('Roberta Stephens', 5), ('Clint Tsuchiya', 5), ('Nita Sandor', 5), ('Alberta Howie', 5), ('Roger Sung', 5), ('Ralph Parks', 5), ('Vincent Roark', 5), ('Laura Flanigan', 8), ('Jesse Pinon', 8), ('Laura Atwater', 8), ('Julio Naccarato', 8), ('Douglas Perrotta', 8), ('Chandra Trowell', 8), ('Paul Vaillancourt', 11), ('Molly Wilkinson', 11), ('Terry Tweed', 11), ('Norman Mastin', 11), ('Keith Banister', 11), ('Michael Scheffler', 11), ('Karen Kinard', 11), ('Dan Lain', 11),

('Dean Bergey', 11), ('Antonio Bischoff', 11), ('Louis Marino', 11), ('Dale Charlton', 11), ('Marguerite Mccabe', 11), ('Gustavo Theall', 11), ('Sabina Corsaut', 11), ('Bryan Corcoran', 11), ('Liza Borchers', 11), ('Martha Holloway', 11), ('Donald Daly', 11), ('Daisy Nino', 14), ('Kelli Sallee', 14), ('Joanna Nixon', 14), ('Shantel Zevallos', 14), ('Katherine Oswald', 14), ('Dan Ulm', 14), ('Maxine Kyser', 14), ('Nicholas Applegate', 14), ('Javier Crichton', 14), ('Janice Tong', 14), ('Vincent Malloy', 14), ('Larry Elmer', 14), ('Guy Cullinan', 14), ('Patrick Booker', 14), ('Matthew Dowden', 14), ('William Trapp', 14), ('Donald Mack', 14), ('Genevieve Holston', 14), ('Rick Monday', 14), ('Dean Harvill', 14), ('Tyra Rhew', 14), ('Henry Mckenna', 14), ('Johnny Watt', 14), ('Guy Pressler', 14), ('Matthew Soper', 14), ('Amanda Woodward', 14), ('Russell Parson', 14), ('Selena Hoose', 14), ('Roy Maes', 14), ('Jeffrey Pulliam', 14), ('Corey Sloat', 14), ('Marlene Berman', 14), ('Zachary Carruthers', 14), ('Harry Carney', 14), ('Tyrone Sturman', 14), ('Sean Holley', 14), ('Donna Pepin', 14), ('Ryan Bennet', 14), ('Clara Wainwright', 16), ('Laurie Fullmer', 16), ('Steve Hix', 17), ('Scott Simons', 17), ('Eugene Singh', 17), ('Jose Burge', 17), ('Dianne Alarcon', 17), ('Nora Neel', 17), ('Terry Caudle', 17), ('Joyce Etter', 17), ('Neil Hunton', 17), ('Renee Field', 17), ('Frank Dunkin', 17), ('Chris Mcmorris', 17), ('Aaron Dalessio', 24), ('Ricardo Maselli', 24), ('Dollie Dignan', 24), ('Gina Gooden', 24), ('Ted Makuch', 24), ('Henry Mazur', 24), ('Jonathan Glasgow', 24), ('Toni Levi', 24), ('Victor Lucier', 24), ('Benjamin Wendt', 24), ('Lance Buono', 24), ('Shelley Bollinger', 24), ('Shawn Standley', 24), ('Cathy Tubbs', 24), ('Pamela Funkhouser', 24), ('Norman Grimaldo', 24), ('Debbie Sapp', 24), ('Roger Barnwell', 24), ('Tim Daggett', 24), ('Albert Wardell', 24), ('Ester Janis', 24), ('Margery Erving', 24), ('Opal Royal', 24), ('Elaine Santillan', 24), ('Deanna Whalen', 24), ('Tyrone Ottman', 24), ('Bernice Schafer', 24), ('Eileen Fischer', 24), ('Joseph Woodbury', 24), ('Sabina Mcmikle', 24), ('Bertha Makowski', 24), ('Tony Burnett', 24), ('Ruth Dressler', 24), ('Bobby Zito', 24), ('Dan Peranio', 24), ('Julianne Cressman', 24), ('Larry Gehrke', 24), ('Jeffery Blount', 24), ('Megan Hargrave', 24), ('Penelope Gayman', 24), ('Dale Oxendine', 24), ('Walter Sappington', 27), ('Roberto Munroe', 27), ('Zachary Warne', 29), ('Lois Karl', 29), ('Carl Sperling', 29), ('Stella Morrissey', 29), ('Susan Roby', 29), ('Sofia Bonfiglio', 29), ('Travis Sweat', 29), ('Terry Macleod', 29), ('Nicole Prevost', 29), ('Pearl Moffett', 29), ('Mike Pitre', 29), ('Maggie Molloy', 29), ('Robyn Olin', 29), ('Anthony Demarco', 29), ('Justin Kuehn', 29), ('Kimberly Harrison', 29), ('Aaron Taveras', 29), ('Lance Newhard', 29), ('Roxanne Marte', 29), ('Juana Hodgson', 29), ('Eugene Molina', 29), ('Keisha Chancy', 29), ('George Jason', 29), ('Charlotte Catlin', 29), ('Jessie Hanshaw', 29), ('Adrienne Lasley', 29), ('Vickie Dorsett', 29), ('Eunice Mason', 29), ('Craig Mcdavid', 29), ('Vincent Kingsbury', 29), ('Stephen Delrosario', 29), ('Melinda Millard', 29), ('Elsie Fortenberry', 29), ('Roma Roskelley', 29), ('Kathryn Mcnulty', 29), ('Billy Beckwith', 29), ('Erik Cadiz', 29), ('Cora Stauffer', 29), ('Cora Thorson', 29), ('Jeff Everson', 29), ('Allan Hornbuckle', 33), ('Mattie Nichols', 33), ('Constance Liggett', 33), ('Barbara Vanover', 33), ('Sofia Mcinturff', 33), ('Tanisha Coppin', 33), ('Dorothea Ruckman', 33), ('Peter Lajoie', 33), ('Benjamin Weisman', 33), ('Ralph Cartagena', 33), ('Jacquelyn Michael', 33), ('Philip Rossman', 33), ('Dewey Swickard', 33), ('Holly Kibler',

33), ('Cassandra Hein', 33), ('Eugene Manis', 33), ('Patrick Snow', 33), ('Frank Smalls', 33), ('Agnes Fillmore', 33), ('Victor Pifer', 33), ('Nathan Pound', 33), ('Jodi Derosier', 33), ('Guy Fear', 33), ('Glenn Shade', 33), ('Darlene Piper', 33), ('Dennis Mckellar', 33), ('Cecile Lafontaine', 33), ('Bruce Roper', 33), ('Heidi Mcadoo', 33), ('Wendy Seda', 33), ('Belinda Beamer', 33), ('Lorrie Mungo', 33), ('Joshua Zack', 33), ('Marvin Fredrickson', 33), ('Edward Kozak', 33), ('Lucille Whiteley', 35), ('Clayton Rieder', 35), ('Chris Omara', 35), ('Inez Basile', 35), ('Thomas Holifield', 35), ('Nelson Weyer', 35), ('Nicole Robinson', 35), ('Kristen Pulliam', 35), ('Katie Plunkett', 35), ('Tania Silvernail', 35), ('Tony Garretson', 35), ('Ana Bridges', 35), ('Brent Bovee', 35), ('Rosemary Latour', 35), ('Maggie Bissell', 35), ('Alan Nino', 35), ('Nina Faria', 35), ('Vincent Howe', 35), ('Mamie Bazemore', 35), ('Lillian Crampton', 35), ('Floy Fruits', 35), ('Hazel Staten', 35), ('Yvette Silverman', 35), ('Patrick Mckenzie', 35), ('Jason Dull', 35), ('Pedro Sebring', 35), ('Sofia Cambridge', 35), ('Terri Lamm', 35), ('Gerald Totten', 35), ('Nicholas Goins', 35), ('Megan Miles', 35), ('Kristy Dewberry', 35), ('Wanda Harry', 35), ('Gordon Bross', 35), ('Brian Winchell', 35), ('Clara Huck', 36), ('Christa Monsour', 36), ('Eugene Abraham', 38), ('Lillian Starling', 38), ('Shantel Vagott', 38), ('Rose Mcroberts', 38), ('Clarence Lamar', 38), ('Eugene Steinke', 38), ('Edwina Hohn', 38), ('Rachel Fernandez', 38), ('Henry Quesada', 38), ('Marlyn Michalowski', 38), ('Thomas Rawlins', 38), ('Alan Simpson', 38), ('Russell Foust', 38), ('Maricela Godard', 38), ('Debbie Northcutt', 38), ('Elsie Alanis', 38), ('Leonard Philpot', 38), ('Jimmy Schaaf', 38), ('Holly Marlin', 38), ('Brian Aucoin', 38), ('Ronald Bryan', 38), ('Jesse Swiger', 38), ('John Peach', 38), ('Paul Leister', 38), ('Alan Clausen', 38), ('Danny Hartsfield', 38), ('Julio Kerbs', 38), ('Douglas Bischoff', 38), ('Joseph Desimone', 38), ('Blanca Kile', 38), ('Roxanne Turnage', 38), ('Charles Vaughan', 38), ('Leonard Fredericks', 38), ('Cathy Terrazas', 38), ('Brandon Saari', 41), ('Verla Pali', 41), ('Michael Minard', 42), ('Johnny Torrence', 42), ('Audrey Medrano', 42), ('Paul Edgar', 42), ('Vincent Dionne', 42), ('Angelica Ayala', 42), ('Natasha Scribner', 42), ('Maurice Stilwell', 42), ('Roy Gwinn', 42), ('Megan Fullerton', 42), ('Saundra Gow', 42), ('Dale Adamson', 42), ('Allen Rondeau', 42), ('Benjamin Maya', 42), ('Bruce Canady', 42), ('Lenore Decola', 42), ('Jillian Mcdougle', 42), ('Travis Mojica', 42), ('Carlene Buechner', 42), ('Joyce Poole', 42), ('Raymond Jeter', 42), ('Justin Skelton', 42), ('Justin Moye', 42), ('Rosemary Keating', 42), ('Terry Meyers', 42), ('Guy Fagundes', 42), ('Susie Haley', 44), ('Nina Chaney', 44), ('Sonia Pippin', 44), ('Albert Rohrer', 44), ('Clarence Lemoine', 44), ('Lucy Pisano', 44), ('Kathrine Lainez', 44), ('Alan Matos', 44), ('Walter Wolf', 44), ('Raymond Benge', 44), ('Thomas Hooten', 44), ('Jill Borkowski', 44), ('Dewey Vashaw', 44), ('Lynda Blum', 44), ('David Crandell', 44), ('Marian Ruble', 44), ('Lonnie Scheffel', 44), ('Emma Mcmorris', 44), ('Timothy Burkhalter', 44), ('Dorothy Lemmons', 44), ('Hector Kravitz', 44), ('Danny Steinberg', 44), ('Ramona Pence', 44), ('Fred Cushman', 44), ('Leonard Willett', 44), ('Charles Kirkman', 44), ('Johnny Segovia', 44), ('Nicholas Chabot', 44), ('Glenn Irizarry', 44), ('Timothy Delvalle', 44), ('Henry Eng', 51), ('Chad Tolliver', 51), ('Emilia Erikson', 51), ('Jeannette Gooch', 51), ('Latonya Norby', 51), ('Lillian Sproul', 51), ('Bobby Ericson', 51), ('Janie Julius', 51), ('Albert Roseberry', 51), ('Amie Delcambre', 51),

('Harry Numbers', 51), ('Lakisha Izquierdo', 51), ('Erik Calvi', 51), ('Angelica Joe', 51), ('Edward Jeffries', 51), ('Dale Horne', 51), ('Judith Brauer', 51), ('Timothy Deguzman', 51), ('Jessica Manuel', 51), ('Justin Mccleary', 51), ('Steven Salvador', 51), ('Monique Dearman', 51), ('Kelly Hilchey', 51), ('Raymond Oller', 51), ('Bonnie Koski', 51), ('Roger Willis', 52), ('Marjorie Marek', 52), ('Jesica Cammon', 52), ('Olga Cimino', 52), ('Russell Goodale', 52), ('Victor Leffler', 52), ('Tyrone Letchworth', 52), ('Kevin Omara', 52), ('Sheri Corona', 52), ('Milagros Harbuck', 52), ('Helen Farina', 52), ('Billy Wellman', 52), ('Emma Kingery', 52), ('Howard Ung', 52), ('Robyn Tallent', 52), ('Amber Raymond', 52), ('Ernest Doherty', 52), ('Sherry Edmondson', 52), ('Vincent Augustine', 52), ('Suzanne Hiles', 52), ('Rebecca Eversole', 52), ('April Wigfall', 52), ('Noemi Vancamp', 52), ('Ariana Mongelli', 52), ('Clinton Velardo', 52), ('Roy Parke', 52), ('Danny Ragan', 52), ('Ronald Winkle', 52), ('Edwina Fedor', 52), ('Dean Cambridge', 52), ('Brittany Grande', 52), ('Erika Cortes', 52), ('Fred Morehead', 52), ('Geraldine Macon', 52), ('Clinton Escalona', 52), ('Vera Sargeant', 52), ('Debora Bolles', 53), ('Willie Monroy', 53), ('Gregory Durant', 53), ('Jennifer Hoffmann', 53), ('Roy Hurtado', 53), ('Erika Hicks', 53), ('Allen Mckinney', 53), ('Christy Kendall', 53), ('Jill Welch', 53), ('Lucy Halsey', 53), ('Darlene Foote', 53), ('Arthur Gard', 53), ('Eunice Cooney', 53), ('Carmella Calandra', 53), ('Todd Pillow', 53), ('Elinor Buff', 53), ('Lance Goodpaster', 53), ('Gerald Gosselin', 53), ('Gertrude Toms', 53), ('Fernando Mccardell', 61), ('Lori Connor', 61), ('Herman Latch', 61), ('Joe Weitzel', 61), ('Carlos Pfeffer', 61), ('Clyde Vosburgh', 61), ('Marvin Gober', 61), ('Shelia Simoneau', 61), ('Travis Hightower', 61), ('Roxie Underdahl', 61), ('Herman Harnish', 61), ('Anthony Farley', 61), ('Lance Rein', 61), ('Agnes Carrell', 61), ('Jason Carney', 61), ('Louis Gribble', 61), ('Tammy Riddle', 61), ('Dena Staff', 61), ('Clara Bender', 61), ('Martin Mcswain', 61), ('Jessie Treiber', 61), ('Cecilia Shank', 61), ('Nora Therrien', 61), ('Claudia Keane', 61), ('Charlene Mccormack', 61), ('Judy Lucia', 61), ('Katie Jeffery', 61), ('Maggie Mallon', 61), ('Betty Chi', 61), ('Roger Birdwell', 61), ('Claudia Sthilaire', 61), ('Travis Dennard', 61), ('Alice Gayton', 61), ('Misty Arzola', 61), ('Brandon Phelps', 61), ('Fannie Tipton', 62), ('Lorraine Tavarez', 62), ('Michael Rankins', 62), ('Trudy Shadrick', 62), ('Judy Burress', 62), ('Alana Glade', 62), ('Lawrence Velarde', 62), ('Carmella Rondon', 62), ('Clyde Baran', 62), ('Marsha Bump', 62), ('Samantha Giroux', 62), ('Bruce Waldman', 62), ('Kenneth Lentz', 62), ('Henry Bertrand', 62), ('Jeffery Marchant', 62), ('Ronald Nettles', 62), ('Terri Landon', 62), ('Faye Scribner', 62), ('Dorothy Mcginley', 62), ('Janie Counts', 62), ('Clara Asher', 62), ('Kathleen Wray', 62), ('Travis Bliss', 62), ('Suzanne Stacy', 62), ('Heather Tanaka', 62), ('Rachel Holmgren', 62), ('Betty Kirkwood', 62), ('Kay Shearer', 62), ('Rose Hinson', 62), ('Adrienne Velez', 62), ('Tyrone Awong', 62), ('Julio Dunkel', 62), ('Marguerite Carman', 62), ('Nancy Welker', 62), ('Anna Harber', 62), ('Patrick Downing', 62), ('Vivian Mather', 62), ('Laurie Coons', 62), ('Allen Olvera', 62), ('Danny Carden', 62), ('Jimmy Doak', 67), ('Paul Hagler', 67), ('Guy Snipe', 67), ('Tania Agbayani', 67), ('Chad Knight', 67), ('Tony Bost', 67), ('Jeff Todd', 67), ('Della Condon', 67), ('Bruce Pfeffer', 67), ('Vickie Cathcart', 67), ('Carrie Ballinger', 67), ('Jimmy Fults', 67), ('Chad Cardin', 67), ('Edward Bliss', 67), ('Constance Berg', 67), ('Arlene Serrano', 67), ('Jerry

Lennon', 67), ('Kelly Silliman', 67), ('Sofia Debolt', 67), ('Liza Guyette', 76), ('Martin Hagen', 76), ('Thomas Hardeman', 76), ('Sean Scarberry', 76), ('Roger Easton', 76), ('Jeffery Neves', 76), ('Joseph Smith', 76), ('Peter Kinser', 76), ('James Hayward', 76), ('William Wingfield', 76), ('Jonathan Calloway', 76), ('Daniel Birch', 76), ('Stanley Hindman', 76), ('Stephanie Mcintosh', 76), ('Keith Bowlin', 76), ('John Tolbert', 76), ('Joseph Feeney', 76), ('Kristin Grantham', 76), ('Pauline Callender', 76), ('Jeanne Fortin', 76), ('Norma Zack', 76), ('Russell Nino', 76), ('Keith Overturf', 76), ('Nicholas Arrowood', 76), ('Bruce Heard', 76), ('Louise Wendt', 76), ('Pearlie Willams', 76)]

Duration taken by QuickSort to sort 1000 pickup requests in reverse order: 0.012246 seconds

```
The Median of 3 QuickSort algorithm Reflection for different scenarios of 1000 pickup request:
Duration to sort 1000 pickup requests in random order: 0.018282 seconds
Duration to sort 1000 pickup requests in nearly sorted order: 0.016255 seconds
Duration to sort 1000 pickup requests in reverse order: 0.012246 seconds

Looking at the durations for each scenario of 1000 pickup request, QuickSort Median of 3 takes less time to
    sort the reverse list while more time is taken to sort the random list
Its time complexity is O(n log n) and its space complexity is Olog(n).
```

Figure 78: Short reflection on using Quick Sort to sort different 1000 pickup lists

```
********************************************************************************
TIMING TABLE
********************************************************************************

MergeSort
Dataset Size         Random     Nearly Sorted        Reverse
100                  0.000325   0.000287             0.000298
500                  0.001276   0.00197              0.002072
1000                 0.002335   0.009898             0.013226

Quicksort
Dataset Size         Random     Nearly Sorted        Reverse
100                  0.000167   0.000254             0.00025
500                  0.0048     0.004988             0.005106
1000                 0.018282   0.016255             0.012246
```

Figure 79: Timing Table

Runtime Analysis of module 4

Dataset in random order

It represents the most likely scenario of allocation of estimated pickup time to pickup requests. A realistic insight is given on the time taken to sort random lists of 100, 500 and 1000 pickup requests.

Random	Merge Sort	Quick Sort
100	0.000325	0.000167
500	0.001276	0.0048
1000	0.002335	0.018282

Analysis: It is clear to see that Quick Sort sorts the smallest random list more quickly than Merge Sort and Merge Sort sorts the largest dataset marginally faster.

Dataset in nearly sorted order

This scenario is less likely to happen in a ride-hailing company unless data is already nearly sorted.

Nearly Sorted	Merge Sort	Quick Sort
100	0.000287	0.000254
500	0.00197	0.004988
1000	0.009898	0.016255

Analysis: Here, Quick Sort performs better for small datasets and Merge Sort sorts the largest dataset marginally faster.

Dataset in reversed order

The reversed dataset is the worst-case scenario possible for both the Merge Sort and Quick Sort.

Reversed	Merge Sort	Quick Sort
100	0.000298	0.00025
500	0.002072	0.005106
1000	0.013226	0.012246

Analysis: Here, Quick Sort performs better for the smallest and largest dataset. Merge Sort sorts the 500 pickup requests dataset faster than Quick Sort.

To conclude for the runtime analysis of module 4, the median of three Quick Sort is faster than Merge Sort due to its use of partitioning and not using temporary arrays to sort. Moreover, it does not use too much of cache compared to Merge Sort which uses temporary arrays to sort. Theoretically, Quick Sort is faster than Merge Sort to sort large datasets. The runtimes obtained when using Quick Sort to sort the reversed lists showed it is quite robust as its runtime was lower than Merge Sort's. Both Merge Sort and Quick Sort have the same average time complexity O(n log n). Quick Sort has a time complexity of $O(n^2)$ for worst-case scenario. But Merge Sort is more efficient than Quick Sort when it comes to best-case time complexity (Ada Computer Science, n.d.).

Assumptions

1. For this assignment, it has been assumed that the urban area where ZipRide Dispatch System operates has only the following locations: Church, Harbor, Mall, Park, School, Suburbs, Supermarket, University, Zoo, Bookstore, Cinema.

Limitations

1. Further testing could have been for all four modules to demonstrate proper implementation such black box testing and white box testing.
2. Moreover, deepened runtime analysis could have been done using matplotlib to plot the runtimes for all different datasets.
3. Interactive menus for each module could have been implemented to help user understand how each module work.

Conclusion

The ZipRide Dispatch assignment deepened my understanding of data structures and algorithms (DSAs) and gave me an idea of how they can be used to identify and solve problems in real life. Thanks to the DSAs explained during the lectures, I was able to combine the use of four modules inside of a single main file to simulate how a ride-hailing company operates. This assignment was done keeping in mind the interest of the public who, in this case, are the passengers by using Dijkstra algorithm to find the shortest path and shortest pickup time to minimise their waiting time. This also ticks the box of Enhancement of Quality of life, which forms part of the Australian Computer Society (ACS) Code of Ethics.

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